Game Theory and Social Choice

Lecture Notes

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Preface

These are the lecture notes for PHIL 444, Game Theory and Social Choice, at the University of Michigan.

They cover the game theory half of the course, and they stop where Sen starts. The second half works through Amartya Sen's *Collective Choice and Social Welfare* (2017), and I'll do some notes for these, but I'm not going to do a book-like summary of a book..

Giacomo Bonanno's *Game Theory* (2024) is freely available online and is worth having open alongside these notes. It covers most of the same ground in a different voice, and it has 165 solved exercises, which these notes do not. Bonanno is by profession an economist not a philosopher (though he has published in philosophy journals, and is philosophically sophisticated), so he's naturally got a different emphasis at different points. And occasionally we disagree about the force of some consideration or other. But apart from differences in emphasis, the books mostly agree, and if you prefer his way of putting things (and over the years I've heard lots of students and GSIs say they learned a lot from his book), you should totally use it.

The notes began life in an older course, so the fit with what we are doing now is not exact. Where a section of the book is not on the schedule, it is there as background rather than as something you are expected to have read. (And the title of the book came from when it was meant to cover the whole of a course like this, not the first half.)

A PDF of the whole book is linked in the sidebar for anyone who would rather read it in one piece.

These notes arose from notes for various different courses (mostly at Michigan, but also at Rutgers and at St Andrews) over the years. I've received a lot of useful feedback on them, especially from faculty sitting in on the course at St Andrews, which has been incorporated into the notes. Before this term I ran the whole thing through Claude, to get advice on the prose and to see how it all fit together. It had many suggestions, both about the organisation and the prose, and about what needed to be added. I accepted a few of these suggestions in total, and many of them in part. It also chased down many of the references that I'd been cavalier about; I think I've double checked all of these, but I might have missed some.

The book is produced in Quarto, using HTML and PDF templates that Claude built.

1 Basics of Game Theory

1.1 Nature of Games

We'll think of a game as any situation where multiple players have to make some decisions, and the outcome they get is a function of (perhaps among other things) the decisions that each make. So understood, that's a very large class of situations. Indeed, game theory has been used to analyse any number of real-world interactions, as you might expect from a definition so broad.

We're going to start our focus on a special class of games, two player simultaneous move games. The term 'two player' probably obvious; it means that we're looking at games where just two players are involved. Simultaneous move games are where each player only gets to make one decision, or move. Each player makes that decision in ignorance of what the other players have decided. And once the decisions are made, the payoffs for each player can be revealed.

This feels like a very narrow class of games. After all, most interactions involve multiple moves by at least one player, or at least players making moves at different times. But there is a good sense in which this is not a particularly tight restriction. Think of a game where the players make multiple moves, like chess. We can imagine a game between two chess super-computers that have planned, in advance, what they will do at every possible stage of the game. Call such plans **strategies**. If two such computers played, they would in a sense each just have to make one, simultaneous, move, namely to announce their strategies. Then it would be relatively trivial to work out the result of the game from the strategies announced.

In practice, this isn't really possible with current technology. There are too many possible states of the board, even in a restricted game like chess, for a computer to have planned for every possible eventuality. But we can imagine such complete strategies for simple games like tic-tac-toe, and even for moderately complicated games like checkers.

For these reasons, when we are discussing simultaneous move games, we will often refer to the players' options as *strategies*. The representation of the game as an $n \times m$ matrix, where one player (represented on rows) has n options, and the other player (represented on columns) has m options, is sometimes called a **normal form** representation of a game and sometimes called a **strategic form** representation. And we'll start off with these matrix representations. When we complicate things by explicitly representing time in the game, we'll call the representation the **extensive form** representation of the game.

1.2 Utility

Here is a representation of an important kind of two player simultaneous move game.

Table 1.1: Prisoners' Dilemma

		Col	
		X	Y
Row	X	3, 3	0, 5
	Y	5, 0	1, 1

Each player, Row and Col, has two options: X and Y. The payoffs, given their moves, are as follows:

- If both play X, both get 3.
- If both play Y, both get 1.
- If one plays X and the other Y, then the one who plays X gets 0, and

the one who plays Y gets 5.

What does it mean to say that they 'get' 0, 1, 3 or 5? It means that the **utility** they get from the payout is that amount. For any given game, the results for the players can be specified in two related, but importantly distinct, ways.

- We can specify their **payouts**. These are the concrete things the players get.
- Alternatively, we can specify the **utility** each player gets from their payout.

Utility is a somewhat technical, somewhat theoretical, notion.

The idea is to numerically measure how good an outcome is to an agent. It is meant to sum up every aspect of outcomes that agents care about. For instance, if an agent cares a lot about their friend's payout, then their utility in an outcome will be a function not just of their own payout, but of their friend's. So if her payout is $\$x$ and her friend's payout is $\$y$, then her utility might equal $x + 0.5y$. This would represent that she cares about her friend's payout, but cares a bit more (well, twice as much) about her own outcome.

The substantive assumption here is that we can sensibly map the goodness of outcomes onto the real numbers. This assumes that goodness is, in a sense, one-dimensional. For any two numbers x, y, x is either greater than, less than, or equal to y. So one assumption we're making is that any two outcomes are comparable in this way. We're also assuming that we can compare the distance between pairs of outcomes. For any numbers x_1, y_1, x_2, y_2, we can ask which of $x_1 - y_1$ and $x_2 - y_2$ is greater. So if we can map outcomes onto numbers, then for any four outcomes, we can ask if the difference between the first two in goodness is more than, less than, or the same as, the difference in goodness between the last two. This isn't an absurd assumption, though it can seem a little arbitrary in some cases.

Although we assume that the goodness to an agent of outcomes can be mapped onto numbers, we do not assume that there is a unique way to do

this mapping. In fact there are two degrees of freedom in choosing a mapping. The outcome that gets utility zero can be any point is convenient to use. In practice, it is often convenient to give the status quo utility zero. But note this is arbitrary, we could use anything. The scale of the utility function is also arbitrary; we can multiply the utilities by a positive constant and it won't change their meaning.

Technically, if one utility function correctly represents an agent, then so will any other function generated from it by a positive affine transformation. So if u_1 is a utility function that correctly represents the agent, then so will the function u_2 defined as follows.

$$u_2(X) = au_1(X) + b, \text{ where } a \text{ is positive.}$$

What does it mean to say that a function u correctly represents the agent? First, it means that the agent prefers X to Y just in case $u(X) > u(Y)$. Second, it means that for any X, Y, Z the agent prefers Y to a gamble that returns X with probability p and Z with probability $1 - p$ iff $u(Y) > pu(X) + (1 - p)u(Z)$.

When it isn't stated otherwise, we will describe the outcomes of games in terms of the utility each agent gets. Sometimes we will have to just state what the payout is each agent gets, and the utility function will be either unknown, or will have to be figured out. But the default is that we will list the utility of each outcome.

1.3 Four Examples of Games

We'll start with a very simple class of games: 2 player simultaneous move games where each player has 2 choices for their move. As we'll see, there is plenty of complexity within even this class.

1.4 Prisoners' Dilemma

We've already seen this game, it was Table 1.1, but it's worthwhile having on the table again.

		Col	
		X	Y
Row	X	3, 3	0, 5
	Y	5, 0	1, 1

The general form of Prisoners' Dilemma is given in Table 1.2, assuming $a > b > c > d$, and $2b > a + d$.

Table 1.2: The general form of a Prisoners' Dilemma game

		Col	
		X	Y
Row	X	b, b	d, a
	Y	a, d	c, c

One characteristic of the game is that for each player, playing Y is better than playing X no matter what the other person does. That is, playing Y **dominates** playing X; it is better no matter what. But, and this is what gives the game its distinctive character, it is better for both players if they both play X than if they both play Y. (And in a repeated game it is better to always play X than to alternate between the outcomes where players make opposite moves.) It is an interesting way to think about modeling situations where there are rewards to cooperation, but incentives to individualist behavior.

1.5 Stag Hunt

The next game we'll look at is called Stag Hunt. We'll first do a version with commonly used numbers (in Table 1.3), then the general version.

Table 1.3: A familiar version of Stag Hunt

		Col	
		X	Y
Row	X	5, 5	0, 4
	Y	4, 0	2, 2

The general version, shown in Table 1.4, has $a > b \geqslant c > d$ and $b + c > a + d$. The case where $b = c$ is allowed; that's the case where the rabbit hunter's return doesn't depend on what the other hunter is doing.

Table 1.4: The general form of Stag Hunt

		Col	
		X	Y
Row	X	a, a	d, b
	Y	b, d	c, c

In Stag Hunt, the best outcome for all parties is the cooperative one. But if a player has no idea what the other player will do, and thinks it is 50/50 whether they will do the cooperative thing, then their best bet is to not cooperate.

The cute story behind this game comes from the eighteenth century political philosopher Jean-Jacques Rousseau (Rousseau [1755] 2004). Imagine two people are setting out on the day's hunt, and are going to hunt stag or rabbits. Catching a stag will lead to more meat, but it is impossible with just one person. One person can catch a lot of rabbits, especially if no one else is hunting rabbits. In this case, X is joining the stag hunt, and Y

is setting off on a solo rabbit hunt. It isn't as hard to get cooperative outcomes in this game as in Prisoners' Dilemma, but it takes some background trust.

1.6 Battle of the Sexes

I don't particularly like the name of this game, but it is common enough that you probably should know it. Table 1.5 gives a commonly used version of the game.

Table 1.5: Battle of the Sexes

		Col	
		X	Y
Row	X	2, 1	0, 0
	Y	0, 0	1, 2

Table 1.6 shows the general version, with $a > b > c$,

Table 1.6: The abstract form of Battle of the Sexes

		Col	
		X	Y
Row	X	a, b	c, c
	Y	c, c	b, a

The thought is that both parties prefer coordination - making the same pick - to doing different things. But they differ on what they would like to coordinate on; Row would like to coordinate on X, and Column would like to coordinate on Y. The name comes from the version of the game where Row and Column are husband and wife, and X and Y are stereotypically male/female things to prefer doing. You can probably fill in your own example here.

1.7 Chicken

Our last game, shown in Table 1.7, is not for the faint of heart.

Table 1.7: A familiar form of Chicken

		Col	
		X	Y
Row	X	-100, -100	1, -1
	Y	-1, 1	0, 0

The idea comes from a James Dean movie. Two cars are lined up facing each other at the end of a long straight. They start driving at each other. The drivers can either stay on the road (that's X), or swerve to the right (that's Y). If they both stay on the road, they die in a fiery inferno. If one swerves and the other stays, the one who stays wins the game, and the other loses. If they both swerve, the game is a draw.

Unlike the previous games we've looked at, there is no pure **equilibrium** to this game. That is, there is no point in the game where no one regrets their choice on finding out what the other players choose. There is an equilibrium to the game, it turns out, but getting to it requires a bit of background.

1.8 Types of Games

Let $u(x, o)$ be the utility that player x gets in outcome o. Then a two player game, with players p_1, p_2 is **zero-sum** if there is some constant c such that for any outcome o, $u(p_1, o) + u(p_2, o) = c$. That is, however good an outcome is for p_1, it is equally bad for p_2, and vice versa. Pure competitions are zero-sum, though real zero-sum interactions are rare in the real world.

By convention, when a game is zero-sum, we rescale the utilities so the constant c is 0. We always can do this - remember any positive affine transformation of the utility function doesn't change its meaning. That's why we can talk about zero-sum games, and not just constant sum games.

You might worry that whether a game is zero-sum or not depends on completely arbitrary choices about how to scale the utility functions. Strictly speaking, a game is zero-sum if there is *any* pair of utility functions that correctly represents the agents, and which makes the utility each gets from any outcome sum to a constant.

A game is **cooperative** if in any outcome, each player gets the same utility. As with the definition of zero-sum, it doesn't matter whether the game looks cooperative on every way of representing the utilities of the players, just if it looks cooperative on some representation.

1.9 Dominant Strategies

Say that one player has available to them strategies $s_1, \ldots, s_n$, and the other players have available to them collectively strategies $v_1, \ldots, v_m$. In the case of a two player game, the strategies available to other players will just be strategies. But in games with more than two players, we'll think of the alternatives available to other players as *vectors*, with each dimension being a choice available to one of the other players. So v_17 might be that player 2 plays X, and player 3 plays Y, and player 4 plays Z, and player 5 also plays Z. Every one of these vectors specifies the choices of all other players in the game.

Let $u_x(s_i, v_k)$ be the utility that x, the player we're focussing on, gets if she plays s_i and the others play v_k. Then strategy s_i **strongly dominates** strategy s_j if and only if the following condition is satisfied.

$$\forall m : u_x(s_1, v_m) > u_x(s_2, v_m)$$

That is, no matter what the other players end up playing, x will do better playing s_1 than playing s_2.

We'll say that strategy s_i **weakly dominates** strategy s_j if and only if the following condition is satisfied.

$$\forall m : u_x(s_1, v_m) \geq u_x(s_2, v_m) \& \exists m : u_x(s_1, v_m) > u_x(s_2, v_m)$$

That is, playing s_1 can't do any worse than playing s_2, no matter what the others do, and it could do better.

1.10 Dominance and Solving Games

Here is one simple way to make progress on a game.

1. Assume that no player will play a strongly dominated strategy.
2. Assume that every player will make a rational choice given that no player will play a dominated strategy.

The first step here lets us solve Prisoners' Dilemma. Since Y dominates X, neither player will play X. So the solution to the game is that both players play Y. Table 1.8 is another game that isn't solved by the first step, but is solved by the second step.

Table 1.8: An example of dominance reasoning

		Col	
		X	Y
Row	X	2, 1	3, 0
	Y	0, 0	1, 1

In Table 1.8 Column has no dominant strategy; what's best for them to play is a function of what Row plays. But Row does have a dominant strategy. Whatever Column does, they are better off playing Y. So we should assume they will play Y. And given they play Y, Column should also play Y. So the solution to the game again is that both players play Y.

In later chapters, we'll look at what happens when we iterate this idea. But first we need an introduction to probability theory.

2 Probability and Decision Theory

2.1 Probability

We'll start by going over the notion of probability that is used in decision theory. The best way to think about probability functions is by thinking about **measure** functions. Intuitively, a measure function is a non-negative **additive** function. That is, it is a function which only takes non-negative values, and where the value of a whole is equal to the sum of the value of the (non-overlapping) parts. In principle the domain of a measure function could be the size of any set, but we'll only consider functions with finite domains for now.

Such functions are frequently used in the real world. Consider, for instance, the functions from regions to their population and land mass. Table 2.1 lists the (very approximate) population and land mass for England, Scotland, Wales and Northern Ireland.

Table 2.1: Some statistics about the United Kingdom

	Population (Millions)	Land Mass (Thousands of mi^2)
England	50	50
Scotland	5	30
Wales	3	8
Northern Ireland	2	5

The nice thing about tables like Table 2.1 is that they give you a lot of information implicitly. You don't need a separate table to tell you that the combined population of England and Scotland is 55 million, or that the

combined land mass of England and Wales is 58,000 square miles. Those (approximate) facts follow from the facts listed on the table, as well as the fact that populations and land masses are additive. Once you know the population of the parts, you know the population of the whole.

The next notion we need is that of a **normalised** measure function. A normalised measure is a measure where every function takes the value assigned to the largest whole is 1. We often use this notion when talking about proportions. Table 2.2 is a table listing, again approximately, the portion of the UK population and land mass in each of the four countries.

Table 2.2: A normalised verison of Table 2.1.

	Population (Proportion)	Land Mass (Proportion)
England	0.84	0.54
Scotland	0.08	0.32
Wales	0.05	0.09
Northern Ireland	0.03	0.05

The way we turn the regular measure function into a normalised measure is by dividing each value in Table 2.1 by the value of whole. So the land mass of England is (about) 50,000 mi^2, and the land mass of the UK is (about) 93,000 mi^2, and $^{50,000}/_{93,000}$ is about 0.54. So that's why we right 0.54 in the top right cell of the normalised measure table.

The last notion we need is that of a possibility space, or more formally, a **field** of propositions. Although these terms might be unfamiliar, the notion should be very familiar; it is just what we represent with a truth table. We will identify, for these purposes, propositions with sets of possible worlds. Then a field of propositions is a set of propositions such that whenever A and B are in the set, so are $\neg A$ and $A \wedge B$. This is, of course, just what we see with truth tables. Whenever you can represent two propositions on a truth table, you can represent their conjunction and each of their negations. Just like with truth-tables, we'll only consider finite fields. It's

possible to extend this to cover any set-sized partition of possibility space, but we'll ignore such complications for now.

A **Probability function** is just a normalised measure on a possibility space. That is, it is a function satisfying these two conditions, where A and B are any propositions, and T is a logical truth.

- $0 \leq \Pr(A) \leq 1 = \Pr(T)$
- If A and B can't both be true, then $\Pr(A \vee B) = \Pr(A) + \Pr(B)$.

We normally suppose that any rational agent has **credences** over the space of (salient) epistemic possibilities, and these credences form a probability function.

In any finite field of propositions, there will be some **atoms**. These are propositions such that no stronger consistent proposition is in the field. (Some, but not all, infinite fields are atomic.) We will use these atoms a lot. I'll use the variable w to range over atoms, reminding us that for practical purposes, atoms behave a lot like *worlds*. They aren't worlds, of course; they only settle the truth values of a handful of propositions. But they are divisions of possibility space that are maximally precise for present purposes. That is, they are being treated as worlds. On a pragmatist approach to modality, there's no more to being a world than to be treated as a world in the right circumstances, and thinking about probability encourages, I think, such a pragmatist approach.

2.2 Conditional Probability

So far we've talked simply about the probability of various propositions. But sometimes we're not interested in the absolute probability of a proposition, we're interested in its **conditional** probability. That is, we're interested in the probability of the proposition *assuming* or *conditional on* some other proposition obtaining.

It isn't too hard to visualise how conditional probability works if we think of a possibility space simply as a truth table, so a measure over a possibility space is simply a measure over on the truth table. Table 2.3 how we might represent a simple probability function, defined over the Boolean combinations of three propositions, if we're thinking about things this way.

Table 2.3: A truth table, with measures

p	q	r	
T	T	T	0.0008
T	T	F	0.008
T	F	T	0.08
T	F	F	0.8
F	T	T	0.0002
F	T	F	0.001
F	F	T	0.01
F	F	F	0.1

The way to read Table 2.3 is that each row defines a world. The characteristics of the world are just the truth values of three salient propositions: p, q, r. If T is shown underneath the name of a proposition, that means the proposition is true in the world, and if an F is shown, it mans the proposition is false. So the third line is the world where p and r are true, and q is false. To the right of the vertical line is the probability of that world. We can work out the probability of any proposition that is true at multiple worlds by adding up the probabilities of the worlds where it is true. So $\Pr(q) = 0.0008 + 0.008 + 0.0002 + 0.001 = 0.01$.

Now let's see what happens when we look at conditional probabilities, i.e., probabilities you get from assuming some particular proposition is true. If we assume that B is true, then we should 'zero out', i.e. assign probability 0, to all the possibilities where B doesn't obtain. We're now left with a measure over only the B-possibilities. The problem is that it isn't a

normalised measure. The values will only sum to $\Pr(B)$, not to 1. We need to renormalise. So we divide by $\Pr(B)$ and we get a probability back. In a formula, we're left with

$$Pr(A|B) = \frac{\Pr(A \wedge B)}{\Pr(B)}$$

Assume now that we're trying to find the conditional probability of p given q in the above table. We could do this in two different ways. First, we could set the probability of any line where q is false to 0. So we will get Table 2.4.

Table 2.4: What happens to Table 2.3 if we 0 out $\neg q$

p	q	r	
T	T	T	0.0008
T	T	F	0.008
T	F	T	0
T	F	F	0
F	T	T	0.0002
F	T	F	0.001
F	F	T	0
F	F	F	0

The numbers don't sum to 1 any more. They sum to 0.01. So we need to divide everything by 0.01. It's sometimes easier to conceptualise this as multiplying by $1/\Pr(q)$, i.e. by multiplying by 100. Then we'll end up with Table 2.5.

Table 2.5: What happens to Table 2.4 if renormalise

p	q	r	
T	T	T	0.08
T	T	F	0.8
T	F	T	0
T	F	F	0
F	T	T	0.02
F	T	F	0.1
F	F	T	0
F	F	F	0

And since p is true on the top two lines of Table 2.5, the 'new' probability of p is 0.88. That is, the conditional probability of p given q is 0.88. As we were writing things above, $\Pr(p|q) = 0.88$.

Alternatively we could just use the formula given above. Just adding up rows gives us the following numbers.

$$\Pr(p \wedge q) = 0.0008 + 0.008 = 0.0088$$
$$\Pr(q) = 0.0008 + 0.008 + 0.0002 + 0.001 = 0.01$$

Then we can apply the formula.

$$\begin{aligned}\Pr(p|q) &= \frac{\Pr(p \wedge q)}{\Pr(q)} \\ &= \frac{0.0088}{0.01} \\ &= 0.88\end{aligned}$$

2.3 Bayes Theorem

This is a bit of a digression from what we're really focussing on in game theory, but it's important for understanding how conditional probabilities work. It is often easier to calculate conditional probabilities in the 'inverse' direction to what we are interested in. That is, if we want to know $\Pr(A|B)$, it might be much easier to discover $\Pr(B|A)$. In these cases, we use Bayes Theorem to get the right result.[1] I'll state Bayes Theorem in two distinct ways, then show that the two ways are ultimately equivalent.

$$\begin{aligned}\Pr(A|B) &= \frac{\Pr(B|A)\Pr(A)}{\Pr(B)} \\ &= \frac{\Pr(B|A)\Pr(A)}{\Pr(B|A)Pr(A) + \Pr(B|\neg A)Pr(\neg A)}\end{aligned}$$

These are equivalent because $\Pr(B) = \Pr(B|A)Pr(A) + \Pr(B|\neg A)\Pr(\neg A)$. Since this is an independently interesting result, it's worth going through the proof of it. First note that

$$\begin{aligned}\Pr(B|A)\Pr(A) &= \frac{\Pr(A \wedge B)}{\Pr(A)}\Pr(A) \\ &= \Pr(A \wedge B)\end{aligned}$$

$$\begin{aligned}\Pr(B|\neg A)\Pr(\neg A) &= \frac{\Pr(\neg A \wedge B)}{\Pr \neg(A)}\Pr(\neg A) \\ &= \Pr(\neg A \wedge B)\end{aligned}$$

Adding those two together we get

[1] Bayes Theory is named after the Reverend Thomas Bayes, an 18th century statistician and nonconformist minister.

$$\begin{aligned}\Pr(B|A)\Pr(A) + \Pr(B|\neg A)\Pr(\neg A) &= \Pr(A \wedge B) + \Pr(\neg A \wedge B) \\ &= \Pr((A \wedge B) \vee (\neg A \wedge B)) \\ &= \Pr(B)\end{aligned}$$

The second line uses the fact that $A \wedge B$ and $\neg A \wedge B$ are inconsistent. And the third line uses the fact that $(A \wedge B) \vee (\neg A \wedge B)$ is equivalent to A. So we get a nice result, one that we'll have occasion to use a bit in what follows.

$$\Pr(B) = \Pr(B|A)\Pr(A) + \Pr(B|\neg A)\Pr(\neg A)$$

So the two forms of Bayes Theorem are the same. We'll often find ourselves in a position to use the second form.

One kind of case where we have occasion to use Bayes Theorem is when we want to know how significant a test finding is. So imagine we're trying to decide whether the patient has disease D, and we're interested in how probable it is that the patient has the disease conditional on them returning a test that's positive for the disease. We also know the following background facts.

- In the relevant demographic group, 5% of patients have the disease.
- When a patient has the disease, the test returns a position result 80% of the time
- When a patient does not have the disease, the test returns a negative result 90% of the time

So in some sense, the test is fairly reliable. It usually returns a positive result when applied to disease carriers. And it usually returns a negative result when applied to non-carriers. But as we'll see when we apply Bayes Theorem, it is very unreliable in another sense. So let A be that the patient has the disease, and B be that the patient returns a positive test. We can use the

above data to generate some 'prior' probabilities, i.e. probabilities that we use prior to getting information about the test.

- $\Pr(A) = 0.05$, and hence $\Pr(\neg A) = 0.95$
- $\Pr(B|A) = 0.8$
- $\Pr(B|\neg A) = 0.1$

Now we can apply Bayes theorem in its second form.

$$\begin{aligned}
\Pr(A|B) &= \frac{\Pr(B|A)\Pr(A)}{\Pr(B|A)\Pr(A) + \Pr(B|\neg A)\Pr(\neg A)} \\
&= \frac{0.8 \times 0.05}{0.08 \times 0.05 + 0.1 \times 0.95} \\
&= \frac{0.04}{0.04 + 0.095} \\
&= \frac{0.04}{0.135} \\
&\approx 0.296
\end{aligned}$$

So in fact the probability of having the disease, conditional on having a positive test, is less than 0.3. So in that sense the test is quite unreliable.

This is actually a quite important point. The fact that the probability of B given A is quite high does not mean that the probability of A given B is equally high. By tweaking the percentages in the example I gave you, you can come up with cases where the probability of B given A is arbitrarily high, even 1, while the probability of A given B is arbitrarily low.

Confusing these two conditional probabilities is sometimes referred to as the *Prosecutors' fallacy*, though it's not clear how many actual prosecutors are guilty of it. The thought is that some prosecutors start with the premise that the probability of the defendant's blood (or DNA or whatever) matching the blood at the crime scene, conditional on the defendant being innocent, is 1 in a billion (or whatever it exactly is). They conclude that the probability of the defendant being innocent, conditional on their blood matching the crime scene, is about 1 in a billion. Because of derivations like the one we just saw, that is a clearly invalid move.

A more striking real world example comes from the analysis of soccer carried out by Charles Reep and Charles Hughes. (Reep and Hughes worked separately, but they reached similar conclusions by similar means.) They watched many games, and tracked each of the possessions (meaning a period of time where one team has control of the ball). They counted how many passes took place in each possession, and whether the possession resulted in a goal. They noted that the vast majority of the goals came from very short possessions, ones involving three or fewer passes; Reep put the figure at around 80% (Wilson 2008, 141). His own account of the study is Reep and Benjamin (1968). From this they concluded that soccer teams should be direct; kicking the ball to forwards as soon as possible to try to score from a short possession. Hughes became technical director of the Football Association in England, where he spread this idea of how to play soccer across the country.

You may be able to spot the error in their reasoning already. The figure that makes it vivid is one Reep did not dwell on: in the games he recorded, 91.5% of moves involved three passes or fewer (Wilson 2008, 141). So the share of goals that came from short possessions was *lower* than the share of possessions that were short, which is the opposite of what you would expect if short possessions were the productive ones. To see why that matters, work through the problem with some numbers. (Those in Table 2.6 are invented, and the split between medium and long is there to show a trend rather than to report one, but the overall proportions are close to what Reep found.) Divide all possessions into short (1-3 passes), medium (4-7 passes) and long (8+ passes). Imagine that we observed a bunch of games, and recorded the data in Table 2.6.

Table 2.6: Some imagined data for how soccer goals are scored

	Goal	No Goal
Short	78	822
Medium	12	48
Long	10	30

As you can see, most goals come from short possessions. But you shouldn't get excited by finding your team has had a short possession; over 90% of short possessions go nowhere. On the other hand, medium to long possessions often end in goals. In Table 2.7 I've added another column to Table 2.6 to make this clear.

Table 2.7: A version of Table 2.6 with success probability added.

	Goal	No Goal	Success %
Short	78	822	8.7
Medium	12	48	20
Long	10	30	25

The longer a possession is, the higher the frequency that it ends in a goal. I'm not sure in real soccer it is as dramatic as the numbers I gave you. (And in fact at high levels of soccer not nearly this high a percentage of possessions end in goals.) But I am confident that (at least at high levels, at most times in the history of soccer) the broad pattern I've outlined here is correct. Most goals come from short possessions, but short possessions lead to fewer goals, on average, than long possessions. Anyone who has taken a probability class should know that this suggests aiming for longer possessions rather than shorter ones, but apparently the England hierarchy didn't take that lesson for several decades.

2.4 Expected Value

One very convenient notion to define using a probability function is that of an **expected value**. Say that a **random variable** is any function from possibilities to real numbers. We usually designate random variables with capital letters. So expressions like $X = x$, which you'll often see in probability textbooks, means that the random variable X takes real value x.

The expected value of a random variable is calculated using the following formula.

$$E(X) = \sum_{w} \Pr(w)X(w)$$

Remember that w is a variable that ranges over atoms. So the summation here is over the atoms in the field. $\Pr(w)$ is the probability of that atom obtaining, and $X(w)$ is the value of the random variable at w. Here's a worked example to make it clearer.

Let our random variable X be the number of children of the next person to walk through the door. Let's assume that the department contains 10 people; 2 of whom have three children, 3 of whom have two children, 3 of whom have one child, and 2 are childless. Each member of the department is equally likely to walk through the door, and no one else is going to. So the probability of any given member of the department walking through the door is $1/10$. Then the expected value of X, i.e., $E(X)$, is:

$$\begin{aligned} 1/10 \times 3 + 1/10 \times 3+ \\ 1/10 \times 2 + 1/10 \times 2 + 1/10 \times 2+ \\ 1/10 \times 1 + 1/10 \times 1 + 1/10 \times 1+ \\ 1/10 \times 0 + 1/10 \times 0 = 1.5 \end{aligned}$$

Note that the expected value of X is *not* the value we 'expect', in the ordinary sense, X to take. We can be pretty sure that X's value will be an integer, so it won't be 1.5, for instance! That is to say, 'expected value' is a term of art.

As well as 'absolute' expected values, we can also define conditional expected values, as follows:

$$E(X|p) = \sum_{w} \Pr(w|p)X(w)$$

I won't go through the proof of this, but it turns out that conditional expected values satisfy a nice theorem. Let $\{p_1, \ldots, p_n\}$ be a partition of possibility space. Then the following holds.

$$E(X) = \sum_{i=1}^{n} \Pr(p_i) E(X|p_i)$$

Here are a few of immediate corrollaries of that result.

$$\begin{aligned} E(X) - E(Y) &= E(X - Y) \\ &= \sum_{i=1}^{n} \Pr(p_i)(E(X|p_i) - E(Y|p_i)) \\ &= \sum_{i=1}^{n} \Pr(p_i)(E((X - Y)|p_i)) \end{aligned}$$

Often we know more about the difference between two expected values than we do about the absolute value of either. For instance, if we are comparing the expected values of taking and declining a small bet, we can figure out which action has a higher expected value without figuring out the expected value of either, which presumably depends on all sorts of facts about how our life might (probably) go. But as long as we can work out the difference in expected value between the two acts conditional on each p_i, and the probability of each p_i, we can work out at least which has a higher expected value, and by how much.

2.5 Orthodox Decision Theory

Given all that, it is fairly easy to summarise orthodox decision theory.

- Rational agents have a credence function over salient alternatives that is identical to some probability function Pr.
- Rational agents also have a real-valued utility function U over salient

outcomes.

- The rational action for such an agent to do is the action that maximises **expected** utility, i.e., maximises the expected value of U relative to that function Pr.

2.6 Money Is Not Utility

Chapter 1 separated a player's payout from their utility, and said that we would normally specify games using the latter. This section is about why that separation matters, because the difference between money and utility plays several different explanatory roles.

Getting \$2x is not twice as good as getting \$x. Getting \$2x is like getting \$x, and then getting another \$x when you are already \$x richer, and the second \$x does you less good than the first did. This what is known as the **declining marginal utility** of money. The **marginal** utility of money is the margin between the utility you had before and after getting that money. A core thought of economic theory since the 1870s is that marginal utility is the core explanation of action; absolute utility is either non-existent or unimportant.[2] And the marginal utility of money falls, i.e., declines, the more of it you have.

Two claims that sound quite different both fall out of this. You are better off taking a million dollars than a fifty-fifty shot at two million. And a million dollars changes your life more than it changes a billionaire's. The orthodox view (and we'll see reasons that not everyone buys this when we get to Section 2.7) is that these are just two features of the same thing, that money has a declining marginal utility.

[2] Like many things, the importance of marginal utility seems to have been independently discovered by many people at once. The philosophy connection here is that one of the independent discoverers was William Stanley Jevons (1871). I'd like to think he came up with the idea while working in Australia in the 1850s, which would make this one of the earliest Australian contributions to Anglophone philosophy. But as far as I know, there is literally zero evidence for that timeline.

If we graph utility against wealth, declining marginal utility fixes two things about the shape. The curve slopes upward, because more money is better than less. And it bends downward; it is concave.

Here is a crude model. Let utility be the square root of net wealth in dollars, and take three equal steps of $300.

- Utility at $100 is $\sqrt{100} = 10$.
- Utility at $400 is $\sqrt{400} = 20$.
- Utility at $700 is $\sqrt{700} \approx 26.46$.
- Utility at $1000 is $\sqrt{1000} \approx 31.62$.

The same $300 buys 10 units of utility, then 6.46, then 5.17. Put the other wayround, equal gains in utility take larger and larger amounts of money.

For wealth above a thousand dollars or so, a logarithm is a more plausible model than a square root. Let utility be $\log_{10}$ of net wealth.

- Utility at $10,000 is 4.
- Utility at $100,000 is 5.
- Utility at $1,000,000 is 6.

Multiplying your wealth by ten *adds* one to your utility. The relation between dollars and utility is not just nonlinear, it is dramatically so.

That has consequences that are easy to miss. Take someone whose net wealth is $100,000, so their utility is 5, and offer them a bet that pays $900,000 if it wins and costs $90,000 if it loses, on the flip of a fair coin.

$$\begin{aligned} E(U(\text{bet})) &= \tfrac{1}{2} \times \log_{10}(1{,}000{,}000) + \tfrac{1}{2} \times \log_{10}(10{,}000) \\ &= \tfrac{1}{2} \times 6 + \tfrac{1}{2} \times 4 \\ &= 5 \end{aligned}$$

The bet has exactly the expected utility of doing nothing, so they should be indifferent, even though its expected dollar return is $405,000.

Now shrink the prize to $800,000, leaving everything else alone.

$$\begin{aligned} E(U(\text{bet})) &= 1/2 \times \log_{10}(900{,}000) + 1/2 \times \log_{10}(10{,}000) \\ &\approx 1/2 \times 5.954 + 1/2 \times 4 \\ &\approx 4.977 \end{aligned}$$

That is below the utility of the status quo, so they should decline, and the expected dollar return is still $355,000. Expected dollars and expected utility do not merely differ in size. They can point in opposite directions. This fact explains why there is an insurance industry; an insurance contract (from the consumer's perspective) typically has a negative expected dollar return, and a positive expected utility return.

2.7 The Allais Paradox

Some people have the intuition that expected utility theory doesn't sufficiently account for the preference people have for avoiding risk. After all, we routinely turn down bets with a positive expected dollar return. The orthodox reply (which I think is completely right, for what it's worth) is that these preferences over individual bets (when they are rational) can be explained by the declining marginal utility of money, plus the view that rationality requires maximising expected utility. No matter what bets people turn down, if we want to rationalise them, we can by positing a steeper declining marginal utility.

It turns out, as Maurice Allais (1953) showed[3], that the theory does have some striking predictions not for individual betting dispositions, but for *pairs* of betting dispositions. He came up with a pair of binary choices, and showed that there was no way for expected utility theory to explain why one attractive pair of dispositions to choose could be rationalised.

[3] Allais was a French economist. He won the 1988 Nobel, though not primarily for the result that he's most famous for in the Anglophone world, i.e., this paradox.

Here is the first choice.

- **1A**: $1,000,000, guaranteed.
- **1B**: a 1% chance of $0, an 89% chance of $1,000,000, and a 10% chance of $5,000,000.

Many people, I suspect most, take **1A**.[4] A guaranteed million is not worth risking for a one in ten shot at five, when the risk is walking away with nothing. That doesn't maximise expected dollars, but it's fine; as long as the marginal utility of the first million is sufficiently greater than the marginal utility of the next four million, **1A** could maximise expected utility.

Here is the second.

- **2A**: an 89% chance of $0, and an 11% chance of $1,000,000.
- **2B**: a 90% chance of $0, and a 10% chance of $5,000,000.

Now it seems fine to take **2B**, and many people do.[5] Crucially, many people who take **1A** take it. Why take **2B**? Well, you will almost certainly get nothing either way, so the extra 1% of risk looks cheap for the chance at five times the money.

Those two choices are inconsistent with expected utility theory. To see why, imagine the outcomes are settled by drawing one of a hundred numbered tickets, as shown in Table 2.8.

[4] The empirical evidence on the Allais paradox is actually a bit of a mess; see Blavatskyy et al. (2022) for a good overview. As I understand it, in these high stakes cases, this part at least is very standard. Of course, these are hypothetical choices; we don't run real experiments where we give away millions of dollars.

[5] Notably, as Blavatskyy et al. (2022) point out, it matters here whether **2A** is framed like it is here, as a 'flat' choice, or whether it is described as a compound lottery.

Table 2.8: The Allais choices, laid out over a hundred tickets

	Ticket 1	**Tickets 2-11**	**Tickets 12-100**
1A	\$1M	\$1M	\$1M
1B	\$0	\$5M	\$1M
2A	\$1M	\$1M	\$0
2B	\$0	\$5M	\$0

Look at the last column. Within each pair, the two options pay the same thing if a ticket from 12 to 100 comes up. So that column cannot bear on either choice. Delete it, and what is left of **1A** against **1B** is *exactly the same* what is left of **2A** against **2B**.

The algebra says the same thing. Write x for $U(\$5\text{M})$, y for $U(\$1\text{M})$ and z for $U(\$0)$. Preferring **1A** to **1B** means

$$y > 0.01z + 0.89y + 0.10x$$

which rearranges to

$$0.11y - 0.10x - 0.01z > 0$$

And the difference between the two options in the second choice is

$$\begin{aligned} E(U(2A)) - E(U(2B)) &= (0.11y + 0.89z) - (0.10x + 0.90z) \\ &= 0.11y - 0.10x - 0.01z \end{aligned}$$

which is the same expression. So anyone who prefers **1A** to **1B** is committed to preferring **2A** to **2B**, whatever their utilities for money happen to be. Taking **1A** and **2B** is not a sign of unusual attitudes to risk. It is inconsistent with expected utility theory. If you think **1A** and **2B** are rational, you think that expected utility theory is wrong.

2.8 Ellsberg

Around the same time, a similar challenge to expected utility theory was formulated by Daniel Ellsberg. Ellsberg is a fascinating figure; his Wikipedia page lists seven *films* that his story has a non-trivial part in. We're going to discuss something he discovered quite early in life, before the work that made him famous. How related it is to that later work is an interesting question; Ellsberg himself always said it was.

Like Allais, Ellsberg (1961) discussed a pair of choices. Just as before, Ellsberg showed that while an expected utility theorist could explain either choice that the typical subject made, they could not explain both. The choices concern an urn that has 90 marbles in it. The subject is told that 30 marbles are red, and the other 60 are a mix of black and yellow. But they are not told what the mixture is. They will be offered a choice between two bets, which have the same return, say $10 if you win, and $0 if you lose. All the bets turn on which marble is drawn from the urn at its next draw.

Here is the first choice.

- **1A**: You win if the marble is red, and lose otherwise.
- **1B**: You win if the marble is black, and lose otherwise.

Many people, I suspect most, take **1A**. They prefer knowing they have a 1/3 chance to not knowing what their odds are. That's actually fine; theory doesn't disagree.

Here is the second.

- **2A**: You win if the marble is red or yellow, and lose otherwise.
- **2B**: You win if the marble is black or yellow, and lose otherwise.

Now most people prefer **2B**. They prefer knowing that they have a 2/3 chance of winning, to the unknown odds with **2A**. Ellsberg's argument is that there is no way for standard expected utility theory to explain both preferences.

At some level, the problem is the same as in the Allais paradox. Table 2.9 is how the table looks if we draw the two of them together.

Table 2.9: The Ellsberg choices, laid out together

	Red	**Black**	**Yellow**
1A	Win	Lose	Lose
1B	Lose	Win	Lose
2A	Win	Lose	Win
2B	$0	Win	Win

Here's what both choices have in common. In both choices, there's a column where you get the same result whether you choose **A** or **B**. That's the right hand column in each. The pair **1A**/**2B** violates this principle.

Value of Irrelevant Alternatives

Consider any choice between lotteries **A** and **B** where there is some possibility where the two options return the same amount, and you prefer **A** to **B**. Changing the lottery simply by changing that common amount, without making any other changes, shouldn't change your preferences. All that matters it that you get the same amount in that outcome, not what that 'same amount' is.

This is a special case of what is known as the **Sure Thing Principle**. Both of these are consequences of the expected utility framework.[6] And both the Allais and Ellsberg paradoxes show that what I'm calling **Value of Irrelevant Alternatives** is sometimes counterintuitive.

But while the two paradoxes target the same principle of expected utility theory, they suggest different modifications to it. The Allais paradox

[6]This arguably understates things. One prominent argument for the expected utility framework takes the Sure Thing Principle as a premise, and argues that anyone who satisfies it, and some other principles, is in effect an expected utility maximiser. We're not going to go into this, but if you're interested, look up work on *representation theorems*.

suggests that we are wrong to say that all attitudes to risk are accounted for by declining marginal utility. The Ellsberg paradox suggests that we are wrong to act as if everyone has a probability function in their head. Think about how you'd do a version of the algebraic argument that **1A**/**2B** is inconsistent in the Ellsberg case.

Here's the most natural way to do this. By assumption, everyone has a probability for every important possibility in their head. Assume the chooser assigns probability y to the drawn marble being yellow. The probability of **1A** winning is 1/3, and the probability of **1B** winning is 2/3-y, so if they prefer **1A** to **1B**, it follows that y is greater than 1/3. The probability of **2A** winning is 1/3+y, and the probability of **2B** winning is 2/3. So y is less than 1/3, contradicting the claim that it is greater.

The response that Ellsberg and others gave to this argument is not to question the algebra, or to question the appeal to expected utility. It's to question the very assumption that there is such a thing as y. Or, in some versions, to question the assumption that it is a *number*. In Chapter 6, we'll come back to some game theoretic reasoning that also presupposes that things like *the probability that the marble is yellow* exist, and whether we might want to question that assumption.

2.9 For Next Time

Next time we'll start with an application of those ideas that leads to the most central concept in game theory.

3 Mixed Strategies and Nash Equilibrium

3.1 Mixed Strategies

In any game, if player x has strategies s_1, s_2 available, then we'll also assume they have available the **mixed strategy** of playing s_1 with probability p and s_2 with probability $1 - p$ (for arbitrary $p \in (0, 1)$). Such strategies have an expected value given strategies for the other players in the game, and we'll think of that expected value as the return to the player for playing the strategy.

Mixed strategies are contrasted with **pure strategies**. These are the strategies that are usually listed on the game table. They are the only strategies that we have considered so far.

I'm not sure whether there is a standard terminology for denoting mixed strategies. We'll write $\langle s_1, p; s_2, 1-p\rangle$ for the mixed strategy of playing s_1 with probability p and s_2 with probability $1 - p$. It is possible for mixed strategies to involve more than two pure strategies; in those cases the list of pure strategies and their probabilities will have to be longer.

If the agent has more than two pure strategies available to them, and plays $\langle s_1, p; s_2, 1 - p\rangle$, it wouldn't make any difference if they instead played $\langle s_1, p; s_2, 1 - p; s_3, 0; \ldots; s_n, 0\rangle$. We can treat any mixed strategy as a mixture of all the pure strategies, just that some of the strategies have probability 0 of being played. Indeed, we can treat any strategy as a mixed strategy that perhaps gives some pure strategies probability 0. So we no longer need to talk about pure strategies at all; they are just a limit case of mixed strategies

where one pure strategy gets probability 1, and the others all get probability zero.

Still, even if every mixed strategy can be represented as a mixture of all pure strategies, there is an important distinction between the pure strategies that get positive probability, and those that get zero probability. We'll call the set of pure strategies that get positive probability the **support** of the mixed strategy. What we used to call a pure strategy is a mixed strategy whose support is a singleton.

Now it's a somewhat delicate philosophical question just what this assumption amounts to. While the assumption is more or less universally made by game theorists, there is less agreement about just what it comes to. There are three natural interpretations, which we might call the objective chance, epistemic and frequentist readings. We'll leave them there for now; the question of which is right doesn't bite until we can see what a mixed equilibrium asks of a player, so we'll come back to it at the end of the chapter.

3.2 Best Responses

A strategy s is a **best response** for player x iff there is some probability distribution Pr over the possible strategies of other players such that no strategy has a higher expected payoff than s does, given Pr.

Let's look at an example of this. So far we've used the same names for the options that Row and Column have available. But we'll start to look at asymmetric games, so it helps to use asymmetric terminology. In Table 3.1 we'll use U, for Up, M for Middle, and D, for Down, for Row's options, and L for Left, and R for Right, for Column's options.

Table 3.1: An example of best responses.

		Col	
		L	R
	U	3, 0	0, 0
Row	M	2, 0	2, 0
	D	0, 0	3, 0

We will just look at things from the perspective of Row. What we want to show is that all three of the possible moves here are best responses. We will write $E(X) = x$ to mean that the expected value of strategy X is x.

It is clear that U is a best response. Set $\Pr(L) = 1, \Pr(R) = 0$. Then $E(U) = 3, E(M) = 2, E(D) = 0$, so U is best. It is also clear that D is a best response. Set $\Pr(L) = 0, \Pr(R) = 1$. Then $E(U) = 0, E(M) = 2, E(D) = 3$, so D is best.

The striking thing is that M can also be a best response. Set $\Pr(L) = \Pr(R) = 0.5$. Then $E(U) = E(D) = 1.5$. But $E(M) = 2$, which is greater than 1.5. So if Row thinks it is equally likely that Col will play either L or R, then she maximizes expected utility by playing M. Of course, she doesn't maximize actual utility. Maximizing actual utility requires making a gamble on which choice C will make. That isn't always wise; it might be best to take the safe option.

It's very plausible that agents should play best responses. If a move is not a best response, then there is no way that it maximizes expected utility, no matter what one's views of the other players are. And one should maximize expected utility.

Any strongly dominated option will never be a best response. That's because whatever strategy strongly dominates it will have a higher expected return no matter what the probabilities are. But a weakly dominated response can be a best response. That's because we've defined best responses to allow that something can be a best response when something else ties it. So in Table 3.2, D is weakly dominated by U, but it is a best response given $\Pr(R) = 1$.

Table 3.2: An example of weak dominance

		Col	
		L	*R*
Row	*U*	1, 0	0, 0
	D	0, 0	0, 0

What about the other direction? If something is not dominated, does that mean it is a best response? It turns out the answer to this question is a little more complicated. Start with Table 3.3.

Table 3.3: An example where something is dominated by a mixture.

		Col	
		L	*R*
	U	3, 0	0, 0
Row	*M*	1, 0	1, 0
	D	0, 0	3, 0

In this game, M is not dominated by either U or D. After all, if Col plays L, then M beats D. And if Col plays R then M beats D.

On the other hand, M is not a best response. The possible views Row could have about Col are all characterised by a single variable x, where x is the probability (according to Row) of Col playing L. This variable x can take any value in $[0, 1]$. But whatever value it takes, M is not the best option. Note that the expected return of L is $3x$, and the expected return of R is $3(1 - x)$. From that, we can easily compute that:

- If $0 \leq x < 0.5$ then D is the best response.
- If $x = 0.5$ then U and D are tied for best response.
- If $0.5 < x \leq 1$ then U is the best response.

So in no circumstance is M best. It is not a best response.

But on second thought, perhaps it is dominated. Let's add a row to the table for the mixed strategy $\langle U, 0.5; D, 0.5\rangle$. In Table 3.4, we'll add this to the table, and note its expected value.

Table 3.4: A version of Table 3.3 with the mixed strategy made explicit.

		Col	
		L	R
Row	U	3, 0	0, 0
	M	1, 0	1, 0
	D	0, 0	3, 0
	$\langle U, 0.5; D, 0.5\rangle$	1.5, 0	1.5, 0

And note that $\langle U, 0.5; D, 0.5\rangle$ dominates M; whatever Col plays, that mixed strategy has a higher expected return than M.

If we change the payout to M so that it isn't dominated, it becomes a best response. So recall the version of the game where the payout from M from a guaranteed 1 to a guaranteed 2. I've reprinted that as Table 3.5 with the mixed strategy made explicit.

Table 3.5: A version of Table 3.1 with the mixed strategy added.

		Col	
		L	R
Row	U	3, 0	0, 0
	M	2, 0	2, 0
	D	0, 0	3, 0
	$\langle U, 0.5; D, 0.5\rangle$	1.5, 0	1.5, 0

Now M is not dominated. Indeed, it dominates $\langle U, 0.5; D, 0.5\rangle$. And as we saw above it is also a best response. If Row thinks it is 50/50 whether Col plays L or R, then M has the highest expected return.

3.3 Iterated Best Responses

It seems very plausible that one should only play best responses. But not all best responses are worth playing. Some best responses are pretty hard to justify when playing against a rational opponent. Consider Table 3.6 from Row's perspective.

Table 3.6: A dominance example

		Col	
		L	*R*
Row	*U*	5, 5	0, -5
	D	0, 5	2, -5

In this game, *D* is a best response. It does best if Col chooses *R*. But why would Col do that? Col gets 5 for sure if she chooses *L*, and -5 for sure if she chooses *R*. Assuming the weakest possible rationality constraint on Col, she won't choose a sure loss over a sure gain. So given that, Row should choose *U*.

Now we could have shown that with just considering domination; indeed domination by pure strategies. Table 3.7 is a little more complex.

Table 3.7: A mixed strategy dominance example

		Col		
		L	*C*	*R*
Row	*U*	1, 3	0, 1	1, 0
	D	0, 0	1, 1	0, 3

In Table 3.7 neither player has a dominated move, as long as we restrict ourselves to pure strategies. And, from Row's perspective, thinking about best responses doesn't help either. *U* is a best response if Col is going to play *L* or *R* with probability at least 0.5, and *D* is best response if Col is going to play *m* with probability at least 0.5.

But note that *C* is not a best response for Col. One of the outer options has a higher expected value, whatever probability she gives to Row playing *U*. Now let's assume, when thinking from Row's perspective, that Col will play a best response. That is, we're assuming Col will play either *L* or *R*. Given that, the best thing for Row to do is to play *U*.

More carefully, if Row knows that Col is rational, and if Row knows that rational agents always play best responses, then Row has a compelling reason to play *U*. This suggests an important normative status. Call a **BRBR** strategy one that is a **B**est **R**esponse to a **B**est **R**esponse.

But we can go further than that. What should Col play, given what we just concluded about Row? She should play *L*. After all, she knows that Row will play *U*, and given that *L* has the best return.

The idea that players should play BRBR strategies isn't strong enough to get us that result. After all, *D* is a best response, and *R* is a best response to *D*. So *R* is BRBR. What it isn't is a best response to something that is a best response to a best response. Rather than keep counting layers of best responses, let's collapse this into one notion of being a best response to a best response to a best response to etc., to infinity. Call any strategy which has this feature an **IBR**, or **I**terated **B**est **R**esponse.

3.4 Nash Equilibrium

In a two-player game, for a strategy s_0 to be IBR, it must be the best response to some strategy s_1, which is the best response to some strategy s_2, which is the best response to some strategy s_3, etc. But we are assuming that there are only finitely many strategy choices available. So how can we get such an infinite chain going?

The answer, of course, is to repeat ourselves. As long as we get some kind of loop, we can extend such a chain forever, by keeping on circling around the loop. And the simplest loop will have just two steps in it. So consider any pair of strategies $\langle s_0, s_1 \rangle$ such that s_0 is the best response to s_1, and s_1 is the best response to s_0. In that case, each strategy will be BRBRI,

since we can run around that two-step 'loop' forever, with each stop on the loop being a strategy which is a best response to the strategy we previous stopped at.

When a pair of strategies fit together nicely like this, we say they are a **Nash Equilibrium**. More generally, in an n-player game, we use the following definition.

NE Some strategies $s_1, ..., s_n$ in an n-player game form a **N**ash **E**quilibrium iff for each i, s_i is a best response to the strategies played by the other $n-1$ players. That is, iff player i cannot do better by playing any alternative strategy to s_i, given that the other players are playing what they actually do.

The 'Nash' in Nash Equilibrium is in honour of John Nash, who developed the concept, and proved some striking mathematical results concerning it (Nash 1951).

3.5 Nash Equilibrium in Simultaneous Move Games

Consider Table 3.8:

Table 3.8: A game with no pure strategy equilibria

		Col	
		L	*R*
Row	*U*	2, 0	0, 2
	D	0, 4	1, 0

Any of the pure strategies is a best response. Indeed, any of them are iterated best responses. So is all we can say about the game that anything goes?

Most game theorists disagree with that claim. They say that once we consider mixed strategies, there is a distinctive solution to the game. Consider the following pair of strategies:

- Row plays $\langle U, 2/3; D, 1/3 \rangle$
- Col plays $\langle L, 1/3; R, 2/3 \rangle$

Given that Row has that strategy, let's work out the expected return (to Col) of Col's two possible plays.

$$\begin{aligned} E(L) &= 0 \times \frac{2}{3} + 4 \times \frac{1}{3} \\ &= 0 + \frac{4}{3} \\ &= \frac{4}{3} \\ E(R) &= 2 \times \frac{2}{3} + 0 \times \frac{1}{3} \\ &= \frac{4}{3} + 0 \\ &= \frac{4}{3} \end{aligned}$$

And we can do the same calculation for the expected return (to Row) of Row's possible strategies.

$$\begin{aligned} E(U) &= 2 \times \frac{1}{3} + 0 \times \frac{2}{3} \\ &= \frac{2}{3} + 0 \\ &= \frac{2}{3} \\ E(D) &= 0 \times \frac{1}{3} + 1 \times \frac{2}{3} \\ &= 0 + \frac{2}{3} \\ &= \frac{2}{3} \end{aligned}$$

Both players would be, if they knew what the other player was doing, indifferent between the two possible strategies that are available to them.

The expected return of Col's playing the mixed strategy $\langle L, x; R, 1-x\rangle$ is $xE(L) + (1-x)E(R)$. So if $E(L) = E(R) = 4/3$, then for all $x \in [0, 1]$, the expected return of $\langle L, x; R, 1-x\rangle$ is also $4/3$. So all strategies are equally good in response to Row playing $\langle U, 2/3; D, 1/3\rangle$. That is, they are all best responses. By similar reasoning, all strategies Row could play in response to Col playing $\langle L, 1/3; R, 2/3\rangle$ are equally good, so they are all best responses.

So the two strategies $\langle U, 2/3; D, 1/3\rangle$ and $\langle L, 1/3; R, 2/3\rangle$ are best responses to each other; they form a Nash Equilibrium. Indeed, this is the only Nash Equilibrium of the game.

We aren't going to prove this, but for any game with bounded payouts, and finite options for each player, there will be some Nash Equilibrium. This equilibria will often be a mixed strategy equilibria, but it is guaranteed to exist.

3.6 Finding Nash Equilibria

It isn't easy in general to find Nash Equilibria. But let's start with some general approaches we can use.

3.6.1 Iterated Dominance

Strongly dominated strategies can't be part of an equilibrium. (I'll just use 'equilibrium' for 'Nash Equilibrium' in this section, as is common in game theory texts.) So strategies that only make sense if they other player plays a dominated strategy can't be part of an equilibrium either. This idea can help us find the equilibrium in some games, such as Table 3.9.

Table 3.9: A game that can be solved with dominance

		Col	
		L	*R*
Row	*U*	1, 2	5, 0
	D	0, 0	3, 6

Neither of Col's strategies is dominated. Col wants to choose whatever Row chooses. But look at the game from Row's perspective. Here there is a strongly dominating strategy, namely choosing U. So D cannot be part of any equilbria. Indeed, if we assume that Row is rational, and so won't play strongly dominated strategies, it is really as if the strategy isn't even there. From that perspective, we can rewrite Table 3.9 as Table 3.10.

Table 3.10: The result of deleting a dominated strategy from Table 3.9

		Col	
		L	*R*
Row	*U*	1, 2	5, 0

Now in this game there is a dominating strategy for Col, namely L. So the only equilibrium of the game is $\langle U, L \rangle$. And it seems that those are the only rational strategies for the players, assuming they each know the other to be rational.

We can use the idea of removing dominating strategies to illustrate some puzzling features of a couple of other games. I won't do tables for these games, because they'd be much too big. The first is a location game that has many applications.

> Two trucks have to choose where they will sell ice-cream on a particular beach. There are 11 locations to choose from, which we'll number 0, 1, ..., 9, 10. Spot 0 is at the left end of the beach, Spot 10 is at the right end of the beach, and the other spots are equally spaced in between. There are 10

> people at each location. Each of them will buy ice-cream. If one truck is closer, they will buy ice-cream from that truck. If two trucks are equally close, then 5 of them will buy ice-cream from one truck, and 5 from the other. Each truck aims to maximise the amount of ice-cream it sells. Where should the trucks end up?

Let's start by looking, in Table 3.11, at a fragment of the payoff matrix. The payoffs are numbers of ice-creams sold. We'll call the players Row and Col, as usual, and just use the number n for the strategy of choosing location n.

Table 3.11: The top-left corner of the ice cream game

		Col					
		0	1	2	3	4	...
Row	0	55, 55	10, 100	15, 95	20, 90	25, 85	...
	1	100, 10	55, 55	20, 90	25, 95	30, 80	...
...							

I'll leave it as an exercise to confirm that these numbers are indeed correct. Note that no matter what Col selects, Row is better off picking 1 than 0. If Col picks 0 or 1, she is a lot better off; she sells 45 more ice-creams. And if Col picks a higher number, she is a bit better off; she sells 5 more ice-creams. So picking 1 dominates picking 0.

Let's look, in Table 3.12 at the opposite corner of the matrix.

Table 3.12: The bottom-right corner of the ice cream game

			Col				
		...	6	7	8	9	10
...							
Row	9	...	30, 80	25, 85	20, 90	55, 55	100, 10
	10	...	25, 85	20, 80	15, 95	10, 100	55, 55

Again, there is be a pattern. No matter what Col does, Row is better off picking 9 than 10. In most cases, this leads to selling 5 more ice-creams. If Col also picks 9 or 10, then picking 9 gives Row a big advantage. The argument here was obviously symmetric to the argument about picking 0 or picking 1, so I'll stop concentrating on what happens when both players select high numbers, and focus from here on the low numbers.

So there is a clear conclusion to be drawn from what we've said so far.

- Spot 0 is dominated by Spot 1, and Spot 10 is dominated by Spot 9. So if Row is rational, she won't pick either of those spots. Since the game is symmetric, if Col is rational, she won't pick either of those spots either. Moreover, neither 0 nor 10 can be part of any equilibrium.

Now let's turn, in Table 3.13, to the comparison between Spot 1 and Spot 2, after we conclude that Row won't play 0. Again, we'll just look at a fragment of the matrix.

Table 3.13: The top-left corner of the ice cream game, after deleting 0 for Row

		Col						
		0	1	2	3	4	5	...
...								
Row	1	100, 10	55, 55	20, 90	25, 95	30, 80	35, 75	...
	2	95, 15	90, 20	55, 55	30, 80	35, 75	40, 70	...
...								

Again, there's a pattern. If Col selects any number above 2, then Row sells 5 more ice-creams by picking 2 rather than 1. If Col selects either 1 or 2, then Row sells 35 more ice-creams by picking 2 rather than 1. But if Col selects 0, then Row sells 5 *fewer* ice-creams by picking 2 rather than 1. So picking 2 does *not* dominate picking 1.

But note the only circumstance when picking 2 is worse than picking 1 is if Col picks 0. And picking 0 is, for Col, a strongly dominated strategy. So picking 2 is sure to do better than picking 1 if Col does not play a strongly dominated strategy. So let's delete that dominated strategy. Table 3.14 is what the top left corner of the game matrix looks like when we do that.

Table 3.14: The top-left corner of the ice cream game, after deleting 0 for both players

		Col					
		1	2	3	4	5	...
Row	1	55, 55	20, 90	25, 95	30, 80	35, 75	...
	2	90, 20	55, 55	30, 80	35, 75	40, 70	...
...							

And now it looks like picking 1 is dominated. This teaches us another lesson about the game.

- If we 'delete' the option of picking either 0 or 10 for Col, then picking 2 dominates picking 1 for Row. In other words, if Row knows Col is rational, and hence won't pick a dominated option, then picking 2 dominates picking 1 for Row, relative to the space of epistemically possible moves in the game. For similar reasons, if Row knows Col is rational, then picking 8 dominates picking 9. And if Col knows Row is rational, then picking either 1 or 9 is dominated (by 2 and 8 respectively). And for similar reasons, neither 1 nor 9 can be part of an equilibrium strategy.

Summarising what we know so far,

- If the players are rational, they won't pick 0 or 10.
- If the players know the other player is rational, they also won't pick 1 or 9.

Let's continue down the matrix, now with Table 3.15. For simplicity, we'll leave off columns 0 and 10, since they are dominated, and we have deleted those as possible options.

Table 3.15: The top-left corner after deleting 0 and 1 for Row, and 0 for Column

		Col					
		1	2	3	4	5	...
Row	2	90, 20	55, 55	30, 80	35, 75	40, 70	...
	3	85, 25	80, 30	55, 55	40, 70	45, 65	...
...							

Picking 3 doesn't *quite* dominate picking 2. In most circumstances, Row does better by picking 3 rather than 2. But she does a little worse if Col picks 1. But wait! We just had an argument that Col shouldn't pick 1. Or, at least, if Col knows that Row is rational, she shouldn't pick 1. Let's assume that Col does know that Row is rational, and Row in turn knows that fact, so she can use it in reasoning. That means she knows Col won't pick 1. So she can delete it from consideration too. And once she does, picking 3 dominates picking 2, relative to the reduced space of epistemically possible outcomes.

I haven't belaboured the point as much as in the previous paragraphs, but hopefully the following conclusion is clear.

- If the players know that the players know that the other player is rational, they won't pick 2 or 8. So neither 2 nor 8 can be part of an equilibrium.

And you can see where this is going. Once we rule out each player picking 2 or 8, then picking 4 dominates picking 3, so picking 3 should be ruled out. And picking 7 should be ruled out for symmetric reasons. But once 3 and 7 are ruled out, picking 5 dominates picking either 4 or 6. So both players will end up picking 5.

And that is, in the standard economic textbooks, a nice explanation of why we see so much 'clustering' of shops. For instance, starting with work by Harold Hotelling (1929), this has been used to explain why there are several shops for one kind of good in one neighborhood, rather than spread over town. Of course the full explanation of clustering is more complicated, but it is nice to see such a simple model deliver such a strong outcome.

This process is called the **Iterative Deletion of Dominated Strategies**. More precisely, what we've used here is the strategy of iteratively deleting **strongly** dominated strategies. This is a powerful technique for solving games that don't, at first glance, admit of any easy solution. If only one strategy pair remains after using this technique, that is the only equilibrium of the game.

We'll return a bit to this point in what follows, but it's worth noting that in a one-shot version of this game, to get to the conclusion that each player should play 5, we had to use a number of very strong assumptions. We had to assume that each player is rational, that they know the other is rational, that they know that the other knows they are rational, and so on through six iterations. In principle, to justify iteratively deleting dominated strategies, we have to assume that there are arbitrarily many iterations of this idea that each player knows that the other player knows etc that they are rational.

3.6.2 Finding Best Responses

It can be useful to mark explicitly what the best responses are to each possible play by the other player. In a two-player game, that can sometimes lead to turning up Nash equilibria. Even when it does not, it can give you a hint about what the support for a mixed strategy equilibria could be. To see an example of this, consider Table 3.16.

Table 3.16: A complicated game

		Col				
		a	*b*	*c*	*d*	*e*
	A	9,9	5,1	6,4	3,3	1,8
	B	3,1	7,3	1,7	3,9	8,2
Row	*C*	4,5	8,7	2,4	4,1	2,9
	D	3,2	1,5	3,7	7,7	9,1
	E	5,3	8,3	7,6	3,5	8,4

We'll start by finding the best response for Row to each of Col's plays. We'll bold the payoff for Row that does best, assuming Col makes that choice. Table 3.17 shows what it looks like after we do that.

Table 3.17: A version of Table 3.16 with Row's best responses bolded.

		Col				
		a	*b*	*c*	*d*	*e*
	A	**9**,9	5,1	6,4	3,3	1,8
	B	3,1	7,3	1,7	3,9	8,2
Row	*C*	4,5	**8**,7	2,4	4,1	2,9
	D	3,2	1,5	3,7	**7**,7	**9**,1
	E	5,3	**8**,3	**7**,6	3,5	8,4

Note that under *b*, we bolded two possible payoffs. That's because they are tied for best, and best response is compatible with ties. In Table 3.18 we also bold Col's best responses.

Table 3.18: A version of Table 3.16 with best responses for both players bolded.

		Col				
		a	*b*	*c*	*d*	*e*
	A	**9**,**9**	5,1	6,4	3,3	1,8
	B	3,1	7,3	1,7	3,**9**	8,2
Row	*C*	4,5	**8**,7	2,4	4,1	2,**9**
	D	3,2	1,5	3,7	7,7	**9**,1
	E	5,3	**8**,3	**7**,**6**	3,5	8,4

Any cell where both values are bolded is an equilibrium; neither player can improve their lot by unilaterally changing their play. And we can see there are three pure strategy equilibria here: $\langle Aa \rangle$, $\langle Dd \rangle$, $\langle Ec \rangle$. I'll leave finding the mixed strategy equilibria as an exercise for the interested reader.

3.6.3 Finding Mixed Strategy Equilibria

It isn't easy in general to find mixed strategy equilibria. But we can start by dividing the problem into two parts.

1. Finding the support for each player's part of the equilibria.
2. Finding the probability with which they play each part of their support.

And it turns out, a bit surprisingly, that it is the first part of the problem that is harder in general. (Recall that the support of a mixed strategy is the set of pure strategies that have a positive probability of being played given that mixed strategy.) And while that isn't an easy problem to solve in general, it isn't something we need to worry about in very simple games. In games with just two options per player, then either there will be a pure strategy equilibria, or the mixed strategy Nash Equilibria will have all strategies in its support. Even if there are three strategies available per player, then

it usually isn't too hard to find the support for the mixed strategy equilibria by just noting that strongly dominated strategies cannot be part of an equilibria.

We'll start by considering the very abstractly defined game, as in Table 3.19.

Table 3.19: An abstract game with no pure strategy equilibrium.

		Col	
		L	*R*
Row	*U*	**a**,b	c,**d**
	D	e,**f**	**0**,0

Don't make too much of the fact that I've put 0 for both players in the bottom right. Since utility scales are only defined up to affine transformation, adding a constant to each possible payout a player gets literally doesn't change the game. So any game representation can be changed into an equivalent representation with 0 for each player in an arbitrarily chosen cell. We'll make that cell be the bottom right, but it could be anywhere.

But do notice the bolding. That has meaning. In particular, it means that all of the following must hold.

- $a > e$, since a is bolded and e is not, and those are the choices if Col chooses L.
- $0 > c$, since 0 is bolded (for Row) and c is not, and those are the choices if Col chooses R.
- $d > b$, since d is bolded and b is not, and those are the choices if Row chooses U.
- $f > 0$, since f is bolded and 0 is not (for Col), and those are the choices if Row chooses D.

Since there is no cell where both are bolded, there is no pure strategy equilibrium. So the equilibria must include at least one mixed strategy.

Recall how we calculated the expected return $E(M)$ of the mixed strategy $\langle x, L; 1 - x, R\rangle$, given the expected returns of L and R. It is

$$E(M) = xE(L) + (1 - x)E(R)$$

If we let $k = E(R) - E(L)$, we can rewrite this as

$$\begin{aligned} E(M) &= xE(L) + (1 - x)E(R) \\ &= xE(L) + (1 - x)(E(L) + k) \\ &= xE(L) + (1 - x)E(L) + k(1 - x) \\ &= (x + (1 - x))E(L) + k(1 - x) \\ &= E(L) + k(1 - x) \end{aligned}$$

A similar bit of reasoning shows that $E(M) = E(R) + kx$. Now if the strategy is genuinely mixed, then both x and $1 - x$ will be greater than 0. And so if $k \neq 0$, then one of kx and $k(1 - x)$ will be positive. And if one of them is positive, then $E(M)$ will be less than one of $E(L)$ and $E(R)$, so it won't be a best response.

So the only time when $\langle x, L; 1 - x, R\rangle$ is a best response for Col is when $E(L) = E(R)$. In that circumstance, all responses are best responses. In any other circumstance, one of the pure responses is best. So,

- If the equilibrium has Col playing a mixed strategy, then Row must play a strategy that makes Col's expected returns from L and R equal.
- And that can only obtain if Row herself plays a mixed strategy.

I've done all this from Col's perspective, but we could redo the argument from Row's perspective to get a similar result; if a mixed strategy is to be best for Row, then Col must play the mixed strategy that makes each of Row's pure strategies equally good.

So now we have a recipe, or at least something like one, for finding the mixed strategy equilibria. Each player must choose a mixed strategy that

makes the expected return of the other player's pure strategies equal. So let's assume that

- Row plays $\langle U, x; D, 1-x\rangle$; and
- Col plays $\langle L, y; R, 1-y\rangle$

Then we need x to be such that $E(L) = E(R)$, and y to be such that $E(U) = E(D)$. So we can solve for each of these variables in turn.

$$\begin{aligned} E(L) &= E(R) \\ xb + (1-x)f &= xd \\ xb + f - xf &= xd \\ f &= xd + xf - xb \\ f &= x(d + f - b) \\ \frac{f}{d+f-b} &= x \end{aligned}$$

So Row must play the mixed strategy $\langle U, \frac{f}{d+f-b}; D, \frac{d-b}{d+f-b}\rangle$ if there is to be an equilibrium. If she does, then Col's expected return from playing either L, or R, or any mixture of them, will be $\frac{df-bf}{d+f-b}$.

Let's now solve for y using the same technique.

$$\begin{aligned} E(U) &= E(D) \\ ya + (1-y)c &= ye \\ ya + c - yc &= ye \\ c &= ye + yc - ya \\ c &= y(e + c - a) \\ \frac{c}{e+c-a} &= y \end{aligned}$$

So Col must play the mixed strategy $\langle L, \frac{c}{e+c-a}; R, \frac{e-a}{e+c-a} \rangle$ if there is to be an equilibrium. If she does, then Row's expected return from playing either U, or D, or any mixture of them, will be $\frac{ec}{e+c-a}$.

There is something puzzling about this result. Imagine a version of the game where Col is a driver, and Row a police officer. Col doesn't care much about the law, or the safety of other users of the road. Col just wants to get where they are going as fast as possible, but really doesn't want a speeding ticket. Row wants to catch speeding motorists, but it is costly to engage in detection work. Col has a choice, Speed or Not Speed. Row has a different choice, Detect or Not Detect. The payoffs are in Table 3.20.

Table 3.20: The speeding game

		Col	
		S	*NS*
Row	*D*	10, -10	-1, 0
	ND	0, 1	0, 0

The equilibrium of this game is that Col speeds with probability $\frac{1}{11}$, and Row detects with probability $\frac{1}{11}$, and each has an expected payoff of 0.

Now what happens if the legislature decides that there is too much speeding, and doubles the fine for speeding. Now the payoff table looks like Table 3.21.

Table 3.21: The speeding game, with raised fines

		Col	
		S	*NS*
Row	*D*	10, -20	-1, 0
	ND	0, 1	0, 0

If nothing else apart from the penalty for speeding, the rational response by police department Row is to reduce monitoring of speeding, and spend time on other crimes. After all, the new equilibrium is that Row detects

with probability $\frac{1}{21}$, and Col still speeds with probability $\frac{1}{11}$. If there is a game in a mixed strategy equilibrium, and the result of any change to the payoffs will be to move it to a new equilibrium, then the only way to change Col's behavior is to either create a new dominant strategy, or to change Row's payoffs. Perhaps what the legislature should do here is to not just increase speeding fines, but to give the money to police departments - that would change the equilibrium of the game in a hurry.

3.6.4 Combining Strategies

We'll work out the equilibria for Table 3.22.

Table 3.22: The example for Section 3.6.4

		Col		
		a	*b*	*c*
	A	5,0	0,1	1,-1
Row	*B*	0,2	1,0	2,1
	C	3,4	-1,2	6,3

As a first step, we'll bold the best responses, and see if any equilibria turn up that way.

Table 3.23: A version of Table 3.22 with best responses bolded

		Col		
		a	*b*	*c*
	A	**5**,0	0,**1**	1,-1
Row	*B*	0,**2**	**1**,0	2,1
	C	3,**4**	-1,2	**6**,3

The bolding in Table 3.23 doesn't look like it helps. Row doesn't have any dominated strategies either. But Col does have a dominated strategy, since *a* dominates *c*. It doesn't matter that *c* might outperform *b*; there is no

reason to play c since a will always do 1 point better. So let's delete that, as in Table 3.24.

Table 3.24: A version of Table 3.23 with a dominated strategy deleted

		Col	
		a	*b*
Row	*A*	**5**,0	0,**1**
	B	0,**2**	**1**,0
	C	3,**4**	-1,2

We don't have to change the bolding, because a dominated strategy is never a best response.

Now compare options A and C. If we rule out Col playing c, it turns out there is never a reason for Row to play C over A. Option A does better than C whether Col plays a or b. So let's delete option C as well.

Table 3.25: A version of Table 3.24 with a strategy deleted by iterated deletion

		Col	
		a	*b*
Row	*A*	**5**,0	0,**1**
	B	0,**2**	**1**,0

Now we have a familiar kind of game. Each player has two options, and there is no pure strategy equilibrium. The problem has reduced to the form of the problem we tackled in the previous subsection. The only equilibrium will be a mixed strategy equilibrium. And we find it by making each player leave the other indifferent between their options. I won't repeat the algebra for finding it, but you can verify yourself that the following works.

- Row plays $\langle A, \frac{2}{3}; B, \frac{1}{3}\rangle$.
- Col plays $\langle a, \frac{1}{6}; b, \frac{5}{6}\rangle$.

If you prefer to have the mixed strategies range over all of the options that were originally available, with the strategies outside the support getting probability 0, we can write the equilibrium of the game as.

- Row plays $\langle A, \frac{2}{3}; B, \frac{1}{3}; C, 0\rangle$.
- Col plays $\langle a, \frac{1}{6}; b, \frac{5}{6}; c, 0\rangle$.

3.7 Why Care about Equilibria?

There is a philosophical question about why we should care about Nash equilibria. Thinking about the mixed strategy equilibria can make these questions particularly vivid.

Imagine I'm Col in one of the driving games from the last section. Why should I ever play a mixed strategy? Here is an argument that I should not.

- If police Row is playing the mixed strategy that brings the game into equilibrium, then my expected return from playing either Speed or Not Speed is 0. So I don't have any incentive to play the particular mixed strategy that brings the game into equilibrium.
- Figuring out how to make it that I speed with probability $\frac{1}{11}$ seems like hard work, when it will have no positive return.
- If police Row is not playing the mixed strategy that brings the game into equilibrium, then one of the pure strategies open to me will have a higher expected return than playing the mixed strategy.

So either playing the mixed strategy will involve work for no gain, or it will actually lower my expected return compared to an available alternative. This isn't seeming very promising. But we can imagine some circumstances when the mixed strategy would be more appealing. Here are three related circumstances where it seems useful.

1. Imagine that I'm too incompetent to make decisions for myself in games like this. You have to advise me. But the advice will be public;

police Row will see it as well. Then the only advice you can rationally offer is that I should play the mixed strategy. Any other advice will backfire because the police will respond to it in ways that make it worthless.

2. Imagine that my strategy will leak to the police, no matter what. That could be because I'm so incompetent that I need to rely on public advice, or it could be because the police have a way of reading my moves. And imagine that the police are in the same boat; their strategy will leak to me. Then we should each play the equilibrium strategy; it is immune to doing worse in light of leaks. That's an implausible situation in most cases, but there is one special case worth considering.
3. Imagine the game will be repeated so many times that eventually our strategies will become visible enough just from the record of our play. Then we are (eventually) back into the 'leaked strategy' situation. And in that situation, we should play equilibrium strategies.

Personally, I think it's the last possibility that makes the concept of an equilibrium play so important. Equilibria (in one shot games) are interesting because they alone are sensible plays in repeated games.

This matches up nicely with our intuitive judgments about games like the driving game. What would happen, intuitively, if one player deviated dramatically from the equilibrium in a one-shot game? The answer is that that player would probably do fine. If Col speeds, then they'll probably get away with it. But think about what would happen (again intuitively) if a player repeated non-equilibrium play over a long period of time. I think it would end up being counterproductive. If Col speeds all the time, Row will adjust and catch them. If Row detects all the time, Col will slow down, but Row will end up wasting detecting resources. It's in repeated games, that non-equilibrium play leads to disaster.

Recall that in Section 3.1 we said that it is hard to sometimes make sense of what it even means to play a mixed strategy outside of the context of a repeated game. The reasoning of the last paragraph is why I don't find that

worrying. Just like it is hard to define a mixed strategy in a genuinely one-shot game, it is hard to motivate such a strategy. Mixed strategies become easier to define in repeated games, and they become easier to motivate in those games too.

3.8 What Is a Mixed Strategy?

Back in Section 3.1 we assumed that players have mixed strategies available, and noted that it is a tricky question what that assumption amounts to. We can now make that note much more precise.

This is the feature of mixed equilibria that generates the difficulty. At a mixed equilibrium, every pure strategy in the support has the same expected return as the mixture itself. We use that fact to discover equilibria. The trick is to find the probabilities that make the *other* player indifferent. That means that at the equilibrium, a player has no reason at all to play the mixture rather than any of the strategies in its support. Playing U for certain does exactly as well as playing $\langle U, 0.5; D, 0.5\rangle$, given what the other player is doing.

For most pure strategy equilibria, abandoning the equilibrium makes you strictly worse off. (There are exceptions, and these create their own problems, but the normal case is that deviation has positive cost.) In a mixed equilibrium nothing pushes you towards the equilibrium mixture. The only thing your mixture accomplishes is keeping the other player indifferent, which is an odd thing for a self-interested agent to be aiming at.

So the usual story about equilibrium, that each player is doing the best they can for themselves, does not explain why anyone would mix in the proportions the equilibrium specifies. So what does explain it? The different interpretations suggest different answers to this question. We'll look at the three most important interpretations, though note that there are others, and that each of these three admits some internal variation.

3.8.1 Objective Chance

On the first of our three interpretations, playing a mixed strategy is using a randomising device, and the probabilities are objective chances.

This is the original reading. Neumann and Morgenstern (1944) introduced mixtures precisely so that an opponent could not find out what you were going to do, and they meant you're using a randomising device. This is the natural reading for games where one side is trying to keep their strategy hidden from the other. (Think about poker, or sports where deception is important, or a tax authority deciding who to audit.) In those cases, being predictable is what you most want to avoid. Professional athletes do come close to the predicted mixtures; there is a small empirical literature on tennis serves and penalty kicks.

There are a few of reasons to not like this interpretation. One is that it's a wide open empirical question whether there really are any randomising devices. The most common interpretations of modern physics are indeterministic, but some are deterministic. Another problem is that if there is indeterminism at the quantum level in physics, most people we describe as playing a mixed strategy are not actually using devices which are indeterministic in this quantum sense. And a third is that it doesn't actually explain the puzzle we started with. Even if you have a randomising device, and you have reason to use it, you need a further reason to set it to the equilibrium probabilities. In equilibrium, you don't have such a reason.

There is a weaker version of the same thought that avoids the first two of these problems. What matters, one might say, is not that I actually use a randomiser, but that you cannot tell the difference between what I am doing and a chance process. On this version I might be running something perfectly deterministic; if you cannot model it, then as far as the game is concerned it is a mixture. Notice that this has quietly relocated the probability from my machinery to your ignorance, which makes it a step towards the epistemic reading rather than a version of this one. And it still doesn't solve the explanatory problem. But it least it doesn't make game theory sensitive to how quantum mechanics ends up.

3.8.2 Frequentist

On the second interpretation the mixture is not in anyone's head. It is a long-run frequency. This is similar to the *frequentist* interpretation of probability, most associated with Hans Reichenbach (1949). Playing a mixture of 1/3 U and 2/3 D just means that if the game is repeated enough times, you will average towards 1/3 of your plays being U, and 2/3 being D. (There need to be some extra clauses to, to prevent playing U exactly every third play. Just what those constraints should be, and why they are justified inside the frequentist framework, are quite delicate questions, but not ones we'll deal with here.)

The immediate problem is that you might not be playing the game repeatedly. We know from other games that when a game is repeated, sometimes that changes the strategy which one should play. A completely orthodox view is that one should defect in a one-shot Prisoners' Dilemma, but cooperate (until the other party defects) in a repeated Prisoners' Dilemma. The frequentist interpretation simply assumes that what we say about repeated games applies to the one-shot game, and that assumption is not, in general, true.

There is a variant that avoids the problem by moving from one player to many. On the **population** reading, the mixture describes a population rather than a person: some fraction of players are U-types who always play U, the rest are D-types, and nobody randomises at all. The equilibrium becomes a fact about how the population is composed, and the interesting question is which compositions are stable. This is the entry point to evolutionary game theory, which we are not doing in this course. Skyrms (1996) is where to go if you want to follow it up.

There is also a variant which is a more radical departure from the textbooks, but I think not uncommon if you talk to economists informally. It's that the presupposition of the question is wrong: just like the puzzle says, there is *zero* reason to mix in a one-shot game. On this view, the frequentist is right about the repeated game, and there is nothing systematic to say about the one-shot game. (Perhaps the rationalizability view, which

we'll come back to in Section 4.1, is right.) Let's set that aside, and stick with the assumption that the orthodox game theoretic account is right, and there is a reason to mix. This pushes us towards the last of the interpretations.

3.8.3 Epistemic

The third interpretation gives up on the idea that anyone randomises. It says the probabilities in mixed strategies are all about beliefs not about actions.

Aumann and Brandenburger (1995) develop the strongest version of this idea. Row plays some pure strategy. The mixture attributed to Row is Col's *belief* about what Row will do. Row's indifference is then not a puzzle about Row's motivation at all; it is a fact about Col's uncertainty.

This doesn't just change what a mixed strategy is. It changes what an equilibrium is. On this view, an equilibrium is a series of beliefs about the parties that have the following two features. First, they are all true. Second, on the assumption that everyone shares these beliefs, everyone is maximizing expected ytility. On this view, the theorem to prove is that mutual knowledge of rationality and of the conjectures about what people are believing forces those conjectures into the pattern that Nash equilibrium describes. It also dissolves the indifference puzzle rather than answering it: there was never a decision to explain, because nobody was choosing a mixture.

To the extent that contemporary philosophers talk about rational choice game theory (as distinct from evolutionary game theory), this seems to be the most popular interpretation. It's certainly the interpretation which minimizes the gap between game theory and decision theory. But while this view is not without adherents in economics, indeed, Aumann won a Nobel Prize in part for his work on it, it is far from orthodoxy in economics.

3.8.4 Where This Leaves Us

The three readings disagree about where the probability lives: in a device, in a sequence of plays, or in someone's beliefs. They also disagree about what an equilibrium is a claim about. The first makes it a claim about individual behaviour, the second about a series or a population, the third about states of information.

There is a fourth answer, due to Harsanyi, which says the randomness is real but sits in the other player's ignorance of your exact payoffs. We cannot state it properly yet, because it needs machinery we have not built. We will come back to it in Chapter 6.

You do not have to choose between these interpretations. You do have to notice that "the players mix 50-50" is not yet a claim with a settled meaning.

4 Other Equilibrium Concepts

The primary concept we'll use in this course is Nash Equilibrium. But there are many many other solution concepts for games. Many of them have practical applications. Many of them have a case to be the one true constraint on rational agents playing games.

4.1 Rationalizability

We looked at this briefly in Section 3.3, under the label of 'Iterated Best Response'. But it's worth stating the idea more carefully, and saying a bit about why it is an important concept.

A class of strategies is rationalizable if:

- A class of strategies is rationalizable if every member of the class is a best response, assuming that the probability that every other player plays some strategy from the class is 1;
- A strategy is rationalizable if it is a member of some class of rationalizable strategies.

This is strictly weaker than the notion of a Nash equilibrium. Note that for any Nash equilibrium, the class consisting of just the strategies that form the equilibrium will be rationalizable, so every member of the class will be rationalizable. But some strategies that are not part of any Nash equilibrium are rationalizable. Consider, for example, Table 4.1 (which is similar to some we've looked at in the previous chapter).

Table 4.1: A game with no pure strategy equilibria

		Col	
		a	*b*
Row	*A*	5,0	0, 1
	B	0,1	1, 0

There is a unique Nash equilibrium for the game. Row plays $\langle A, 0.5; B, 0.5\rangle$, and Col plays $\langle a, \frac{1}{6}; b, \frac{5}{6}\rangle$. But any strategy whatsoever, including all mixtures, is rationalizable. For any strategy you like, for either player, there is some opposing strategy it is a best response to.

Now compare the following two claims about rationality and game playing.

1. Rational players always play some part of a Nash equilibrium strategy.
2. Rational players always play rationalizable strategies.

These two theories make different assessments of Table 4.1. Principle 1 says that rationality rules out all but 1 strategy for each player. Principle 2 says anything goes. Which of them get it right?

If the game was being repeated many times, the answer is easy. The pure strategy *A* will lead to disaster for Row, since after a while, Col will pick up what's going on and respond by playing *b*. In a repeated game, randomizing as Principle 1 requires is the right thing to do.

But in a one shot game it is harder to say what is correct. There is a nice argument that Principle 1 is self-undermining. Assume that the only rational thing to do is to play the Nash Equilibrium strategy. Then Row will flip a coin to decide what to play. So Col has the same expected return either way, given Row's strategy. Moreover, since Col knows what rationality is, Col knows that Row will flip a coin. So now Col is just facing a decision problem; not a game. And in that decision problem, all responses have equal expected utility. So all responses are equally rational. So Col can rationally do anything. But that's just what Principle 2 says.

The argument of the last paragraph is perfectly general. In any game where the only Nash equilibrium is a mixed strategy equilibrium, the following principles cannot all be true.

A. Only strategies that are part of Nash equilibria are rational, and all players know this. B. All players are rational, and all players know this. C. Any strategy that is known to have the same expected return as a rational strategy can be rationally played.

By (A) and (B), the other player will play a mixed strategy. But as we've seen, this will mean that every strategy in the support for one's own mixed strategy will have equal expected return. And by (C), that means that the pure strategies are just as rational as the mixed strategies. But that contradicts the assumption in (A) that only mixed strategies are rational.

Conventional wisdom in game theory texts is that (C) fails. Principle 2 says that (A) fails. I'm not going to take a stand on this, just to note that there is a clash between the idea that Nash equilibria have a distinctive role to play in rational strategy choice, and the principle that rational agents maximize expected utility.

In games where the players only have two choices to pick from, the difference between the Principles comes down to whether it is obligatory to randomize, or whether it is permissible to play one of the pure strategies from the support of the mixed strategy Nash equilibrium. This not a trivial question to even have intuitions about, because it is tied up with the metaphysics of what a mixed strategy is. But in three way choices there can be more separation between the two principles. Consider either Table 4.2 or Table 4.3.

Table 4.2: A three-way game with a poor pure strategy equilibrium

		Col		
		a	*b*	*c*
	A	3,0	0,3	0,2
Row	*B*	0,3	3,0	0,2
	C	2,0	2,0	1,1

Table 4.3: A version of Table 4.2 with the pure strategy equilibrium raised to 2

		Col		
		a	*b*	*c*
	A	3,0	0,3	0,2
Row	*B*	0,3	3,0	0,2
	C	2,0	2,0	2,2

In both games, there is a unique Nash equilibrium, *Cc*. But all strategies are rationalizable. Note that *A* is a best response to *a*, which is a best response to *B*, which is a best response to *b*, which is a best response to *A*. Once you have a cycle like this, every member of the cycle is rationalizable. But there is no mixed strategy Nash equilibrium with those pure strategies as supports. For reasons that should be familiar by now, the only mixed strategy equilibrium that would be available is for each player to play a 50/50 split between their first two options. But if one player plays that mixed strategy, the other player has a return of 1.5 if they reciprocate, and a return of 2 if they play the third option. So there is only one equilibrium, pure or mixed, for the game, while all options are rationalizable.

Ideally, we would have clear intuitions here about what strategies are rationally permissible, and this would let us decide between Principle 1 and Principle 2. Unfortunately, I don't think the intuitions about the cases are at all clear. For what it is worth, I think intuitively that playing one of *A* or *B* is rational when the alternative is the equilibrium that returns 1 to each player, but not when the alternative is a guaranteed 2. So these cases don't ultimately settle the question. But they do make more vivid what is at stake. Could it ever be rational, under conditions of common knowledge of rationality, to play *A* or *B* in either of these games?

4.2 Correlated Equilibrium

To get Nash equilibrium, we have to assume that every player has a way of letting their play be chosen by a random process. But we assume that these random choices are independent. If Row has a 40% chance of playing Up, and Column has a 30% chance of playing Left, then the probability of the Up-Left outcome is 12%. (That is, 40% times 30%.) The idea behind correlated equilibrium is that the players may have a way to coordinate on a random strategy.

Start with a game, Table 4.4 where we might want such a coordination. Imagine that Row and Col are drivers, approaching an intersection at right angles. Each of them has a choice to go (written as G/g), or stop (written as S/s). They would each prefer to get through the intersection faster. But the other person going and them stopping isn't terrible; there is no crash and the intersection is cleared. And both going is a disaster. So we can approximate the game table as this. (In reality it's more like a many move game, but this is a first approximation.)

Table 4.4: A version of Chicken

		Col	
		g	*s*
Row	*G*	0,0	10, 9
	S	9,10	8, 8

Table 4.4 has two pure strategy equilibria, Gs and Sg. It also has a rather annoying mixed strategy equilibria, where each player plays goes with probability $2/11$, and stays otherwise, and they crash about 3% of the time.

Those are all the mixed strategy equilibria. Two of them are asymmetric, and intuitively unfair, and it isn't easy to see how they would get chosen. The only symmetric equilibrium is much worse than either pure strategy equilibria; it even allows for the possibility of crashes.

The correlated equilibria are designed to avoid those problems. Imagine that instead of choosing their own strategy at random, each player could

see a common randomising device. Maybe we could have a coin that was flipped in public, and Row goes if it lands heads, and Col goes if it lands tails. That would give each player an expected return of 9.5. And, crucially, once each player sees the (common) signal, each player has no incentive to defect. If Row knows that Col will go if the coin comes up tails, and Row sees that the coin comes up tails, and that Col sees this, Row will prefer to Stay than Go.

This is not completely unlike how we actually solve these problems in real life. Traffic lights are ways of correlating our actions without having one party always getting what they want. I'm simplifying over a lot of details here, but the main thing to stress is that a traffic light (a very public signal) is a way of introducing a new kind of solution to the game drivers face.

In this case, the coin flip is the best symmetric strategy that we can have. But that is to some extent a feature of the values. I won't try to do a just so story to make the following table make sense, but imagine that players faced Table 4.5. (Perhaps it comes about if they really dislike waiting at lights while the other person goes through.)

Table 4.5: A traffic game that isn't quite Chicken

		Col	
		g	*s*
Row	*G*	0,0	10, 7
	S	7,10	9, 9

The Nash equilibria don't change much with this change. And the correlated equilibrium of the previous problem still works. That is, the solution: flip a coin in public, if Heads, Row goes, if Tails, Col goes, is still an equilibrium. Each player has an incentive to do their part in the equilibrium once they see the coin. But there is a yet better equilibrium available.

Imagine an outside judge makes three cards: *Gs*; *Sg* and *Ss*. Each card has a strategy for Row (in Capital letters) and a strategy for Col (in small letters). The judge selects a card, and then tells each player the strategy that is

recommended for them, but not what's recommended for the other player. Each player still has an incentive to follow the judge's recommendations. If the judge says go, they know the other will stop, so they will want to go and get the best result. If the judge says stop, it is 50/50 whether the judge has told the other player go or stop. And in that circumstance, going is a bad gamble. It has an expected return of 5; stopping has an expected return of 8. So both parties will do what the judge says.

And if the parties agree in advance to this process, which is fair and symmetric, their expected return is even higher than if they agree to the coin flip. If they flip a coin to decide who will go, their expected return is 8.5. Under this strategy, their expected return is $8\frac{2}{3}$. That's because they have equal chances of getting 7, 9 or 10 as their return.

Let's take stock. A **correlated equilibrium** is a way of randomly choosing instructions for each player to follow such that each player is motivated to follow the instructions assuming (a) the probabilities of different instructions being chosen are common knowledge, and (b) each other player will follow the instructions. Every correlated equilibrium will consist entirely of rationalizable strategies, but some will not be Nash equilibria. Every Nash equilibria will be a correlated equilibrium.

Given a correlated equilibrium, we can work out the expected return to each player. We do this, as always, by working out their probability of getting each outcome, multiplying the probability by the utility of that outcome, and adding up. We can also find the correlated equilibrium that maximises expected returns, but that takes some more time. Let's look again at Table 4.5.

First, let's work out the mixed strategy equilibrium for the game. We want to find a probability for Row to Go that makes Col indifferent between going and staying. Let that probability be x. Then we have all the following lines being equivalent.

$$
\begin{aligned}
E(g) &= E(s) \\
10(1-x) &= 7x + 9(1-x) \\
10 - 10x &= 7x + 9 - 9x \\
10 - 10x &= 9 - 2x \\
10 &= 9 + 8x \\
1 &= 8x \\
x &= \frac{1}{8}
\end{aligned}
$$

So the mixed strategy equilibrium is that each player goes with probability 1/8. That leaves them in the crash outcome in $\frac{1}{64}$ of cases. Despite that, the strategies have a surprisingly high expected return of 8.75.

Note one other consequence of the calculation we just did. If the probability of the other player going is 1/8, the other player will be indifferent between their options. If it is higher than 1/8, the other player will maximise expected return by stopping; if it is lower, the other player will maximise expected return by going.

Now look at the simplest correlated equilibrium. A coin is flipped, and if it is Heads, Row Goes, and Col stops, while if it is Tails, Row Stops and Col goes. It is easy to confirm that each player will conform to this instruction; if you know what the other player is doing, you do the opposite. But this isn't a great equilibrium. Each player gets 10 with probability 0.5, and 7 with probability 0.5, for an expected return of 8.5. That's worse even than the mixed strategy equilibrium.

If the judge selects one of the three cards *Gs*; *Sg* and *Ss* at random, then each player will follow the instructions. That's obvious if they are told to go. If they are told to stop, the probability that the other player was told to go is 0.5. Just to confirm that,

$$
\begin{aligned}
&Pr(\text{Other player told to go}|\text{I'm told to stop}) \\
=&\frac{Pr(\text{Other player told to go and I'm told to stop})}{Pr(\text{I'm told to stop})} \\
=&\frac{Pr(\text{Other player told to go and I'm told to stop})}{Pr(\text{They told go and I'm told stop}) + Pr(\text{They told go and I'm told stop})} \\
=&\frac{\frac{1}{3}}{\frac{1}{3} + \frac{1}{3}} \\
=&0.5
\end{aligned}
$$

And if the probability that the other player goes is greater than 1/8, a player will stop. But as we noticed above, the expected return of this strategy is 8 2/3, which is still lower than the expected return of the mixed strategy. Can we do better?

It turns out that we can. The expected return of each player is just half the sum of the expected returns of each player (in any symmetric equilibrium). And the sum of the expected returns is maximised when we maximise the probability of ending up in the bottom left cell, where the players combine for 18 utils, compared to just 17 utils in the other two non-crash cells. So we want to make the probability of ending up there as high as possible. But we can't just tell the players to go to that cell; they won't have an incentive to cooperate.

What we want to do is to maximise the probability of getting the *Ss* instruction, consistent with each player being motivated to follow that instruction. And that will happen as long as each player thinks the probability the other player goes (conditional on them being told to stop) is at least 1/8. So let's assume that the judge makes the *Gs* instruction with probability x, the *Sg* instruction with probability x, and the *Ss* instruction with probability $1 - 2x$. Now do the above conditional probability calculation with those values.

$$
\begin{aligned}
&Pr(\text{Other player told to go} | \text{I'm told to stop}) \\
=&\frac{Pr(\text{Other player told to go and I'm told to stop})}{Pr(\text{I'm told to stop})} \\
=&\frac{Pr(\text{Other player told to go and I'm told to stop})}{Pr(\text{They told go and I'm told stop}) + Pr(\text{They told go and I'm told stop})} \\
=&\frac{x}{x + (1 - 2x)}
\end{aligned}
$$

And we want that last value to be 1/8. A little calculation shows that this will be true if $x = 1/9$. So consider the following instructions:

- The judge makes the Gs instruction with probability 1/9.
- The judge makes the Sg instruction with probability 1/9.
- The judge makes the Ss instruction with probability 7/9.

If a player is told to go, they will know the other player is stopping and will go. If a player is told to stop, they will think the probability the other player is going is 1/8, so they will be indifferent between stopping and going, so they will not have an incentive to deviate from judge's advice.

If the equilibrium is followed, each player expects the following probability distribution over their returns:

- They get 10 with probability 1/9;
- They get 7 with probability 1/9; and
- They get 9 with probability 7/9

That means their expected return is $8\frac{8}{9}$, which is a little larger than the mixed strategy equilibrium. And that's the highest expected return of any correlated equilibrium.

The possibility of attractive correlated equilibria shows us something about the value of **cheap talk**. Cheap talk is the general term we use for any kind of costless signalling. There are all sorts of interesting results that we can derive about costly signalling, and we'll look at that later in the course.

But it is somewhat intuitive that cheap talk is never valuable. After all, it is of no use in games like Prisoners Dilemma.

The existence of games with nice correlated equilibria shows that this intuitive finding is actually false. Players who are allowed cheap talk can work out a correlated equilibrium that they both prefer to strategies that are available without cheap talk.

4.3 Coordination Games

Many games have multiple equilibria. In such cases, it is interesting to work through the means by which one equilibrium rather than another may end up being chosen. Consider, for instance, the following three games (two of which we've seen before).

Table 4.6: Simple coordination

		Col	
		a	*b*
Row	*A*	1,1	0,0
	B	0,0	1,1

Table 4.7: Asymmetric coordination

		Col	
		a	*b*
Row	*A*	2,1	0,0
	B	0,0	1,2

Table 4.8: Stag Hunt

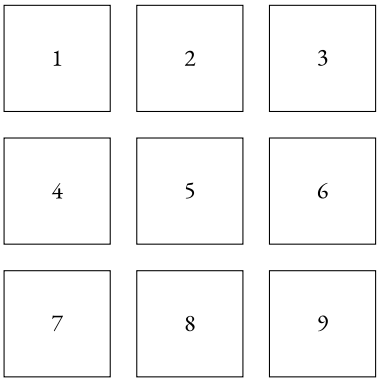
		Col	
		a	*b*
Row	*A*	5,5	0,4
	B	4,0	2,2

In each case, both *Aa* and *Bb* are Nash equilibria.

The top game is a purely cooperative game; the players have exactly the same payoffs in every outcome. It is a model for some real-world games. The two players, Row and Col, have to meet up, and it doesn't matter where. They could either go to location *A* or *B*. If they go to the same location, they are happy; if not, not.

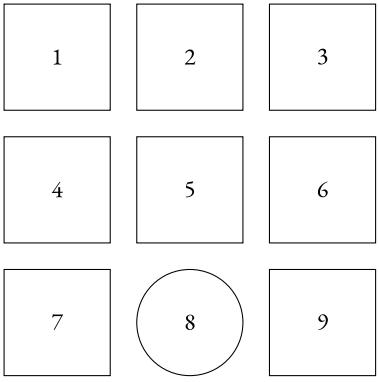
In such an abstract presentation of game, it is hard to see how we could select an equilibrium out of the two possibilities. In practice, it often turns out not to be so hard. Thomas Schelling (1960) noted that a lot of real-life versions of this game have what he called 'focal points'. (These are sometimes called Schelling points now, in his honour.) A focal point is a point that stands out from the other options in some salient respect, and it can be expected that other players will notice that it stands out. Here's a nice example devised by the linguist Christopher Potts (2008).

> *A* and *B* have to select, without communicating, one of the following nine figures. They each get a reward iff they select the same figure.



It's not obvious what they should do there, but compare the situation our players face in the following game.

> *A* and *B* have to select, without communicating, one of the following nine figures. They each get a reward iff they select the same figure.



We could run experiments to test this, but intuitively, players will do better at the second game than the first. That's because in the second game, they have a focal point to select; one of the options stands out from the crowd.

Schelling tested this by asking people where they would go if they had to meet a stranger in a strange city, and had no way to communicate. The answers suggested that meetups would be much more common than you might expect. People in Paris would go to the Eiffel Tower; people in New York would go to (the information booth at) Grand Central Station, and so on. (This is all very contingent. I don't think Grand Central would still work as a focal point in New York. Or maybe it would work as a focal point for people who regularly caught east-bound regional trains from New York.)

The second game we started this section with is often called 'Battle of the Sexes'. The real world example of it that is usually used is that Row and Col are a married couple trying to coordinate on a night out. But for whatever reason, they are at the stage where they simply have to go to one of two venues. Row would prefer that they both go to A, while Col would prefer that they both go to B. But the worst case scenario is that they go to different things. I won't go over that game much more, because the previous section told you most of what you need to know about it. The game has two asymmetric pure strategy Nash equilibria. It has a symmetric, but quite unfortunate, mixed strategy equilibria. And it has a nice correlated equilibrium, where they flip a coin to decide what to do. That's probably what real couples would do, demonstrating again the importance of correlated equilibria.

The third coordination game I started the section with is called Stag Hunt. The idea behind the game goes back to Rousseau ([1755] 2004), although you can find similar examples in Hume ([1739] 2003). The general idea is that A is some kind of cooperative action, and B is a 'defecting' action. If players make the same selection, then they do well. If they both cooperate, they do better than if they both defect. But if they don't cooperate, it is better to defect than to cooperate.

Rousseau's example was of a group of hunters out hunting a stag, each of whom sees a hare going by. If they leave the group, they will catch the hare for sure, and have something to eat, but the group hunting the stag will be left with a much lower probability of stag catching. Hume

was more explicit than Rousseau that these games come up with both two player and many player versions. Here is a nice many player version. I've adapted it from a game Ben Polak (2007) plays with his own class; the threshold is his, the payoffs are simplified.

> **The Investment Game**
>
> Everyone in the group has a choice between Investing and Not Investing. The payoff to anyone who doesn't invest is 0. If 90% or more of the group Invests, then everyone who Invests gains \$1. If less than 90% of the group Invests, then everyone who invests loses \$1.

Again, there are two Nash equilibria: everyone invests, and everyone does not invest. Hume's hypothesis was that in games like this, the cooperative move (in this case Investing) would be more likely in small groups than large groups. This is basically what happened in Polak's classes at Yale.

As well as the two pure strategy Nash equilibria in Stag Hunt, there is a mixed strategy Nash equilibria. We find it, as always, by finding the mixed strategy that makes the other player indifferent between their choices. So assume that Row plays A with probability x. Then the expected return for Col from her two options is

$$\begin{aligned} E(a) &= x \times 5 + (1 - x) \times 0 \\ &= 5x \\ E(b) &= x \times 4 + (1 - x) \times 2 \\ &= 4x + 2 - 2x \\ &= 2 + 2x \end{aligned}$$

So there will be equality only if $5x = 2 + 2x$, i.e., only if $x = 2/3$. If that happens, C's expected return from any strategy will be $10/3$. Since the game is completely symmetric, a very similar argument shows that if C plays the

mixed strategy of a with probability $2/3$, and b with probability $1/3$, then Row will be indifferent between her choices. So each player cooperating with probability $2/3$ is a Nash equilibrium.

Unlike it Meeting or Battle of the Sexes, the mixed strategy equilibrium is not worse for each player than either pure strategy equilibria. But it isn't a great outcome either. In Stag Hunt, the mixed equilibrium is better than the defecting equilibrium, but worse than the cooperating equilibrium.

Say an equilibrium is **Pareto-preferred** to another equilibrium iff every player would prefer the first equilibrium to the second. An equilibrium is **Pareto-dominant** iff it is Pareto-preferred to all other equilibria. The cooperative equilibrium is Pareto-dominant in Stag Hunt; neither of the other games have Pareto-dominant equiilbria.

Consider each of the above games from the perspective of a player who does not know what strategy their partner will play. Given a probability distribution over the other player's moves, we can work out which strategy has the higher expected return, or whether the various strategies have the same expected returns as each other. For any strategy S, and possible strategy S' of the other player, let f be a function that maps S into the set of probabilities x such that if the other player plays S' with probability x, S has the highest expected return. In two strategy games, we can remove the relativisation to S', since for the purposes we'll go on to, it doesn't matter which S' we use. In somewhat formal language, $f(S)$ is the **basin of attraction** for S; it is the range of probability functions that points towards S being played.

Let m be the usual (Lesbegue) measure over intervals; all that you need to know about this measure is that for any interval $[x, y]$, $m([x, y]) = y - x$; i.e., it maps a continuous interval onto its length. Say $m(f(S))$ is the risk-attractiveness of S. Intuitively, this is a measure of how big a range of probability distributions over the other person's play is compatible with S being the best thing to play.

Say that a strategy S is risk-preferred iff $m(f(S))$ is larger than $m(f(S^*))$ for any alternative strategy S^* available to that agent. Say that an equilib-

rium is **risk-dominant** iff it consists of risk-preferred strategies for all players.

For simple two-player, two-option games like we've been looking at, all of this can be simplified a lot. An equilibrium is risk-dominant iff each of the moves in it are moves the players would make if they assigned probability 1/2 to each of the possible moves the other player could make.

Neither the simple Meeting game nor Battle of the Sexes have a risk-dominant equilibrium. An asymmetric Meeting game, however, does. Table 4.9 is one example.

Table 4.9: A coordination game with a favored outcome

		Col	
		a	*b*
Row	*A*	2,2	0,0
	B	0,0	1,1

In this case *Aa* is both Pareto-dominant and risk-dominant. Given that, we might expect that both players would choose *A*. (What if an option is Pareto-dominated, and risk-dominated, but focal? Should it be chosen then? Perhaps; the Eiffel Tower isn't easy to get to, or particularly pleasant once you're there, but seems to be chosen in the Paris version of Schelling's meetup game because of its focality.)

The real interest of risk-dominance comes from Stag Hunt. In that game, cooperating is Pareto-dominant, but defecting is risk-dominant. The general class of Stag Hunt games is sometimes defined as the class of two-player, two-option games with two equilibria, one of which is Pareto-dominant, and the other of which is risk-dominant. In those games, players have a choice between risk and reward. And at the same time they have to choose between a socially useful choice, and a socially harmful choice. The risky choice is the most socially useful one. But note that the socially useful choice is not necessarily a self-sacrifice. Assuming

other players are being cooperative, each player does better by being cooperative than by defecting.

It's at this point that the analogy to society formation becomes, for some philosophers, quite vivid. The thought is that we don't require coercion to get people to stay living in civilised society. Assuming that a civilisation is up and running, each member of the society has a preference for staying in the society to leaving it. But some combination of coercion and public spiritedness might be required to get a society running. If the players are unsure whether the others will behave socially or anti-socially, the rational thing to do is to be anti-social. And in the long run, that's a tragic outcome. I'll leave it to other classes, however, to figure out how much social interactions look like iterated Prisoners Dilemma, and how much they look like iterated Stag Hunt.

5 Games and Time

So far we've looked just at games where each player makes just one move, and they make it simultaneously. That might not feel like most games that you know about. It isn't how we play, for instance, chess. Instead, most games involve players making multiple moves, and these moves taking place in time. Here is one simple such game. It's not very interesting; you won't have fun games nights playing it. But it is useful to study.

> **Five**
> There are two players, who we'll call A and B. First A moves, then B, then finally A moves again. Each move involves announcing a number, 1 or 2. A wins if after the three moves, the numbers announced sum to 5. B wins otherwise.

This is a simple zero-sum game. The payoff is either 1 to A and 0 to B, or 1 to B and 0 to A. For simplicity, we'll describe this as A winning or B winning. We'll soon be interested in games with draws, which are a payoff of to each player. But for now we're looking at games that someone wins.

Before we go on, it's worth thinking about how you would play this game from each player's perspective. The formal approach we'll eventually take is pretty similar, I think, to the way one thinks about the game.

5.1 Normal Form and Extensive Form

Figure 5.1 is the **extensive form** representation of **Five**. We will use W to denote that A wins, and L to denote that B wins.

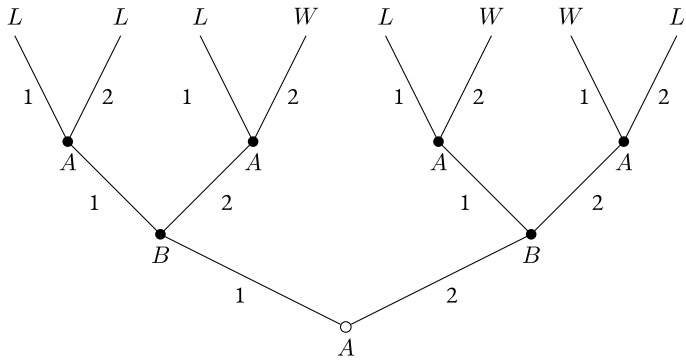


Figure 5.1: Five

To read what's going on here, start at the node that isn't filled in, which in this case is the node at the bottom of the tree. The first move is made by *A*. She chooses to play either 1 or 2. If she plays 1, we go to the left of the chart, and now *B* has a choice. She plays either 1 or 2, and now again *A* has a choice. As it turns out, *A* has the same choice to make whatever *B* does, though there is nothing essential to this. This is just a two-player game, though there is also nothing essential to this. We could easily have let it be the case that a third player moved at this point. Or we could have made it that who moved at this point depended on which move *B* had made. The extensive form representation allows for a lot of flexibility in this respect.

At the top of the diagram is a record of who won. In the more general form, we would put the payoffs to each player here. But when we have a simple zero-sum game, it is easier to just record the payoffs to one player. You should check that the description of who wins and loses in each situation is right. In each case, *A* wins if 2 is played twice, and 1 once.

The extensive form representation form is very convenient for turn taking games. But you might think that it is much less convenient for games where players move simultaneously, as in Prisoners' Dilemma. But it turns out that we can represent that as well, through a small notational innovation. Figure 5.2 is the game tree for Prisoners' Dilemma.

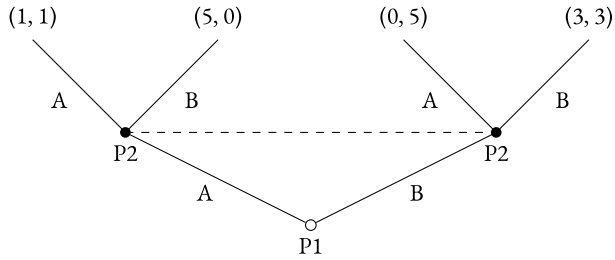


Figure 5.2: Prisoners' Dilemma

The crucial thing here is the dashed line between the two nodes where P2 (short for Player 2) moves. What this means is that P2 doesn't know which of these nodes she is at when she moves. We normally assume that a player knows what moves are available to her. So we normally only put this kind of dashed line in when it connects two nodes at which the same player moves, and the available moves are the same. When we put in notation like this, we say that the nodes that are connected form an **information set**. If we don't mark anything, we assume that a node is in a degenerate information set, one that only contains itself.

Strictly speaking, you could regard this tree as a representation of a game where Player 1 goes first, then Player 2 moves, but Player 2 does not know Player 1's move when she moves. But that would be reading, I think, too much into the symbolic representation. What's crucial is that we can represent simultaneous move games in extensive form.

There is a philosophical assumption in this notational convention that I just want to mention. We won't have time to cover it further here, though it is worth thinking about. It is usual to use dashed lines, like I've done, or circles or ovals to represent that for all Player 2 knows, she is at one of the connected nodes. But drawing circles around a set of possibilities is only a good way to represent Player 2's uncertainty if we assume quite a lot about knowledge. In particular, it is only good if we assume that which nodes are epistemically open for Player 2 is independent of which node, within that group, she is at. In formal terms, it amounts to assuming that each player

knows exactly what they know, and what they don't know.[7] This is not, in general, a true assumption. Normal people have some false beliefs. Those false beliefs are not pieces of knowledge, but the holders of the false belief do not in general know that they do not amount to knowledge. A more philosophically defensible game theory would not assume that everyone knows what it is that they know. But for now we'll follow orthodoxy and assume we can use representations like these dashed lines to represent uncertainty, even though that requires a lot more self-knowledge than most people have.

These representations using graphs are knows as **extensive form** representations of games. The representations using tables that we've used previously are known as **normal form** or **strategic form** representations. The idea is that any game really can be represented as game of one simultaneous move. The 'moves' the players make are the selections of **strategies**. A strategy in this sense is a plan for what to do in any circumstance whatsoever.

Let's do this for the **Five**. A strategy for A has three variables. It must specify what she does at the first move, what she does at the second move if B plays 1, and what she does at the second move if B plays 2. (We're assuming here that A will play what she decides to play at the first move. It's possible to drop that assumption, but it results in much more complexity.) So we'll describe A's strategy as $\alpha\beta\gamma$, where α is her move to begin with, β is what she does if B plays 1, and γ is what she does if B plays 2. A strategy for B needs to only have two variables: what to do if A plays 1, and what to do if A plays 2. So we'll notate her strategy as $\delta\epsilon$, where δ is what she does if A plays 1, and ϵ is what she does if A plays 2. So A has 8 possible strategies, and B has 4 possible strategies. Let's record the giant table listing the outcomes if they play each of those strategies.

[7] The assumption that if someone knows something, they know that they know it, is famously criticised by Timothy Williamson (2000). This has been incredibly controversial in philosophy. The assumption that if someone does not know something, they know that they do not know it, is much more widely rejected in philosophy. The most careful statement of why we should reject it that I know is by Lloyd Humberstone (2015, 380–401).

		B			
		11	12	21	22
A	111	*L*	*L*	*L*	*L*
	112	*L*	*L*	*W*	*W*
	121	*L*	*L*	*L*	*L*
	122	*L*	*L*	*W*	*W*
	211	*L*	*W*	*L*	*W*
	212	*L*	*L*	*L*	*L*
	221	*W*	*W*	*W*	*W*
	222	*W*	*L*	*W*	*L*

There is something quite dramatic about this representation. We can see what A should play. If her strategy is 221, then whatever strategy B plays, A wins. So she should play that; it is a (weakly) dominant strategy. This isn't completely obvious from the extended form graph.

Here's a related fact. Note that there are only 8 outcomes of the extended form game, but 32 cells in the table. Each outcome on the tree is represented by multiple cells of the table. Let's say we changed the game so that it finishes in a draw, represented by D, if the numbers picked sum to 3. That just requires changing one thing on the graph; the L in the top-left corner has to be changed to a D. But it requires making many changes to the table.

		B			
		11	12	21	22
A	111	*D*	*D*	*L*	*L*
	112	*D*	*D*	*W*	*W*
	121	*L*	*L*	*L*	*L*
	122	*L*	*L*	*W*	*W*
	211	*L*	*W*	*L*	*W*
	212	*L*	*L*	*L*	*L*
	221	*W*	*W*	*W*	*W*
	222	*W*	*L*	*W*	*L*

In part because of this fact, i.e., because every change to the value assignment in the extensive form requires making many changes in the values on the normal form, it isn't a coincidence that there's a row containing nothing but *W*. The following claim about our game can be proved.

> Assign *W* and *L* in any way you like to the eight outcomes of the extended form game. Then draw table that is the normal form representation of the game. It will either have a row containing nothing but *W*, i.e., a winning strategy for *A*, or a column containing nothing but *L*, i.e., a winning strategy for *B*.

We will prove this in a little bit, but first we will look at how to 'solve' games like **Five**.

5.2 Backwards Induction

The way to think through games like **Five** is by working from top to bottom. *A* moves last. Once we get to the point of the last move, there is no tactical decision making needed. *A* knows what payoff she gets from each move, and she simply will take the highest payoff (assuming rationality).

So let's assume she does that. Let's assume, that is, that *A* does play her best strategy. Then we know three things about what will happen in the game.

- If *A* plays 1, and *B* plays 2, *A* will follow with 2.
- If *A* plays 2, and *B* plays 1, *A* will follow with 2.
- If *A* plays 2, and *B* plays 2, *A* will follow with 1.

Moreover, once we make this assumption there is, in effect, one fewer step in the game. Once *B* moves, the outcome is determined. So let's redraw the game using that assumption, and just listing payoffs after the second move. This will be Figure 5.3.

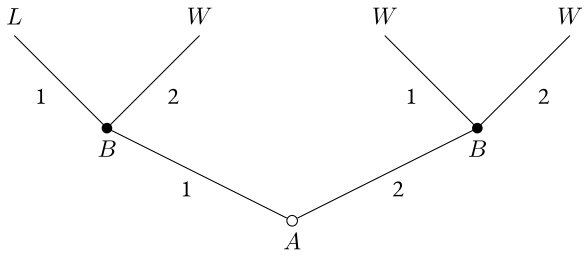


Figure 5.3: Five with last move assumed

Now we can make the same assumption about *B*. Assume that *B* will simply make her best move in this (reduced) game. Since *B* wins if *A* loses, *B*'s best move is to get to *L*. This assumption then, gives us just one extra constraint.

- If *A* plays 1, *B* will follow with 1.

And, once again, we can replace *B*'s actual movement with a payoff of the game under the assumption that *B* makes the rational move. This gives us an even simpler representation of the game that we see in Figure 5.4.

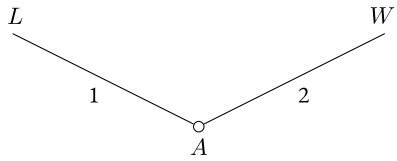


Figure 5.4: Five with last two moves assumed

And from this version of the game, we can draw two more conclusions, assuming A is rational.

- A will play 2 at the first move.
- A will win.

Let's put all of that together. We know A will start with 2, so her strategy will be of the form $2\beta\gamma$. We also know that B doesn't care which strategy she chooses at that point, so we can't make any further reductions. But we do know that if A plays 2 and B plays 1, A will follow with 2. So A's strategy will be of the form 22γ. And we know that if A plays 2 and B plays 2, then A will play 1. So A's strategy will be 221, as we saw on the table.

Note that the use of backwards induction here hides a multitude of assumptions. We have to assume each player is rational, and each player knows that, and each player knows that, and so on for at least as many iterations as there are steps in the game. If we didn't have those assumptions, it wouldn't be right to simply replace a huge tree with a single outcome. Moreover, we have to make those assumptions be very modally robust.

We can see this with a slight variant of the game. Let's say this time that the left two outcomes are D. So the graph looks like Figure 5.5.

Now we assume that A makes the optimal move at the last step, so we can replace the top row of outcomes with the outcome that would happen if A moves optimally. This gets us Figure 5.6.

Now assume that A plays 1 on the first round, then B has to move. From Figure 5.6 it looks like B has an easy choice to make. If she plays 1, she gets

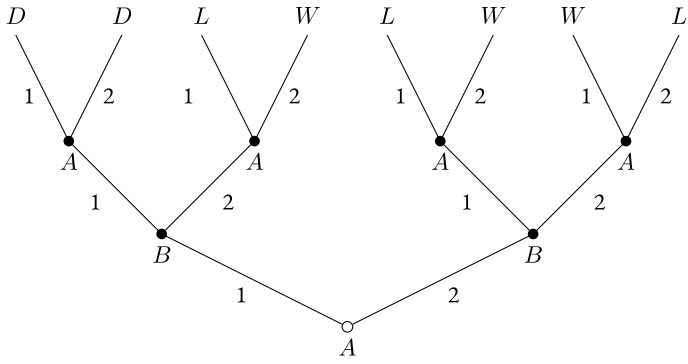


Figure 5.5: Five' '

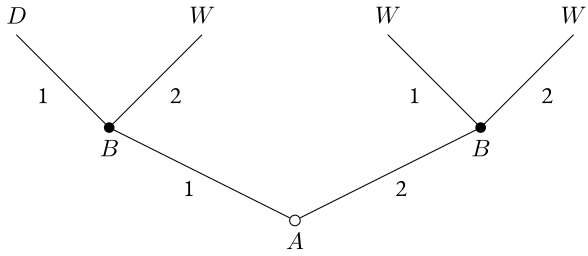


Figure 5.6: Five' ' with last move assumed

a draw, if she plays 2, then *A* wins, i.e., she loses. Since drawing is better than losing, she should play 1 and take the draw.

But why think that playing 2 will lead to *A* winning? The argument that it did depending on assuming *A* is perfectly rational. And assuming *B* is in a position to make this choice, that seems like an unlikely assumption. After all, if *A* were perfectly rational, she'd have chosen 2, and given herself a chance to force a win.

Now you might think that even if *A* isn't perfectly rational, it still is crazy to leave her with an easy winning move. And that's probably a sufficient reason for *B* to accept the draw, i.e., play 1. But the argument that *B* should regard playing 2 as equivalent to choosing defeat seems mistaken. *B* knows that *A* isn't perfectly rational, and she shouldn't assume perfect rationality from here on out.

We will come back to this point, a lot, in subsequent discussions of backwards induction. Note that it is a point that doesn't really arise in the context of normal form games. There we might wonder about whether common knowledge of rationality is a legitimate assumption at the *start* of the game. But once we've settled that, we don't have a further issue to decide about whether it is still a legitimate assumption at later stages of the game.

i Exercise: Last Counter

Here is another game to run backwards induction on. Some counters sit on a table. *A* and *B* take turns removing them, with *A* moving first. On your turn you must take either one counter or two. Whoever takes the **last** counter loses.

1. Draw the extensive form for the game that starts with four counters. Note that, unlike **Five**, the branches are not all the same length, and some nodes offer only one move.
2. Work out the normal form. Each player has four strategies, so it is a 4×4 table. How many distinct outcomes does the tree

have, and how many cells does the table have?
3. Solve the game by backwards induction. Who wins?
4. Now do the same for five counters, and for six. What is the general pattern?
5. Suppose the game starts with six counters and A opens badly. B now finds herself at a node that a perfectly rational A would never have created. Does that matter for how B should play from here? Compare the case of **Five'** above.

5.3 Value of Games

Consider games with the following characteristics.

- A and B take turns making moves. We will call each point at which they make a move, or at which the game ends, a **node** of the game.
- At each move, each player knows what moves have been previously made.
- At each move, the players have only finitely many possible moves open.
- The players' preferences over the outcomes are opposite to one another. So if A prefers outcome o_1 to o_2, then B prefers o_2 to o_1, and if A is indifferent between o_1 and o_2, then B is indifferent between them as well.
- A's preferences over the outcomes are complete; for any two outcomes, she either prefers the first, or prefers the second, or is indifferent between them.
- There is a finite limit to the total number of possible moves in the game.

The finiteness assumptions entail that there are only finitely many possible outcomes. So we can order the outcomes by A's preferences. (Strictly speaking, we want to order sets of outcomes that are equivalence classes

with respect to the relation that A is indifferent between them.) Assign the outcome A least prefers the value 0, the next outcome the value 1, and so on.

Now we recursively define the **value** of a node as follows.

- The value of a **terminal node**, i.e., a node at which the game ends, is the payoff at that node.
- The value of any node at which A makes a choice is the greatest value of the nodes between which A is choosing to move to.
- The value of any node at which B makes a choice is the least value of the nodes between which B is choosing to move to.

Finally, we say that the value of the game is the value of the initial node, i.e., the node at which the first choice is made. We can prove a number of things about the value of games. The proof will make crucial use of the notion of a **subgame**. In any extensive form game, a **subgame** of a game is the game we get by treating any perfect information node as the initial node of a game, and including the rest of the game 'downstream' from there.

By a perfect information node, I mean a node such that when it is reached, it is common knowledge that it is reached. That's true of all the nodes in most of the games we're looking at, but it isn't true in, for instance, the extensive form version of Prisoners' Dilemma we looked at. Nodes that are in non-degenerate information sets, i.e., information sets that contain other nodes, can't trigger subgames. That's because we typically assume that to play a game, players have to know what game they are playing.

Note that a subgame is really a game, just like a subset is a set. Once we're in the subgame, it doesn't matter a lot how we got there. Indeed, any game we represent is the consequence of some choices by the agent; they are all subgames of the game of life.

- The value of a game is the value of one of the terminal nodes.

We prove this by induction on the length of games. (The length of a game is the *maximum* number of moves needed to reach a terminal node. We've only looked so far at games where every path takes the same number of moves to reach a conclusion, but that's not a compulsory feature of games.)

If the game has zero moves, then it has just one node, and its value is the value of that node. And that node is a terminal node, so the value is the value of a terminal node.

Now assume that the claim is true for any game of length k or less, and consider an arbitrary game of length $k + 1$ The first node of the game consists of a choice about which path to go down. So the value of the initial node is the value of one of the subsequent nodes. Once that choice is made, we are in a subgame of length k, no matter which choice is made. By the inductive hypothesis, the value of that subgame is the value of one of its terminal nodes. So the value of the game, which is the value of one of the immediate subsequent nodes to the initial node, is the value of one of its terminal nodes.

- A can guarantee that the outcome of the game is at least the value of the game.

Again, we prove this by induction on the length of games. It is trivial for games of length 0. So assume it is true for all games of length at most k, and we'll prove it for games of length $k+1$. The initial node of such a game is either a move by A or a move by B. We will consider these two cases separately.

Assume that A makes the first move. Then the value of the initial node is the maximum value of any immediate successor node. So A can select to go to the node with the same value as the value of the game. Then we're in a subgame of length k. By the inductive assumption, in that game A can guarantee that the outcome is at least the value of the subgame. And since the value of that node is the value of the subgame, so it is also the value of the initial node, i.e., the value of the initial game. So by choosing that

node, and starting that subgame, A can guarantee that the outcome is at least the value of the game.

Now assume that B makes the first move. B can choose the node with the least value of the available choices. Then, as above, we'll be in a subgame in which (by the inductive hypothesis) B can guarantee has an outcome which is at most its value. That is, B can guarantee the outcome of the game is at most the value of the initial node of the subgame. And since B can guarantee that that subgame is played, B can guarantee that the game has an outcome of at most its value.

- B can guarantee that the outcome of the game is at most the value of the game.

The proof of this exactly parallels the previous proof, and the details are left as an exercise.

Let's note a couple of consequences of these theorems.

First, assume that the rationality of each player is common knowledge, and that it is also common knowledge that this will persist throughout the game. Then the kind of backwards induction argument we used is discussing **Five** will show that the outcome of the game will be the value of the game. That's because if A is rational and knows this much about B, the outcome won't be lower than the value, and if B is rational and knows this much about A, the outcome won't be greater than the value.

Second, these theorems have many applications to real-life games. Both chess and checkers, for instance, satisfy the conditions we listed. The only condition that is not immediately obvious in each case is that the game ends in finite time. But the rules for draws in each game guarantee that is true. (Exercise: Prove this!) Since these games end with White win, Black win or draw, the value of the game must be one of those three outcomes.

In the case of checkers, we know what the value of the game is. It is a draw. This was proved by the Chinook project at the University of Alberta (Schaeffer et al. 2007). We don't yet know what the value of chess is, but it is probably also a draw. Given how many possible moves there are in

chess, and in principle how long games can go on, it is hard to believe that chess will be 'solved' any time soon. But advances in chess computers may be able to justify to everyone's satisfaction a particular solution, even if we can't prove that is the value of chess.

5.4 Subgame Perfect Equilibrium

Consider the following little game. The two players, call them Player I and Player II, have a choice between two options, call them Good and Bad. The game is a sequential move game; first Player I moves then Player II moves. Each player gets 1 if they choose Good and 0 if they choose Bad. We will refer to this game as **Easy Game**. Its game tree is Figure 5.7.

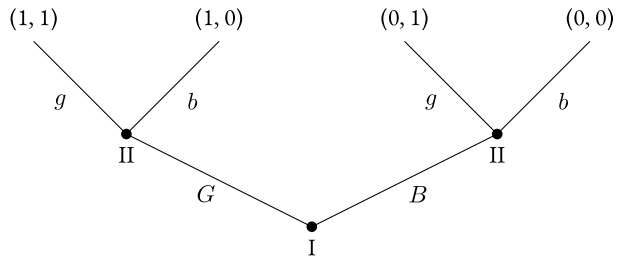


Figure 5.7: Easy Game

A strategy for Player I in **Easy Game** is just a choice of one option, Good or Bad. A strategy for Player II is a little more complicated. She has to choose both what to do if Player I chooses Good, and what to do if Player II chooses Bad. We'll write her strategy as $\alpha\beta$, where α is what she does if Player I chooses Good, and β is what she does if Player II chooses Bad. (We will often use this kind of notation in what follows. Wherever it is potentially ambiguous, I'll try to explain it. But the notation is very common in works on game theory, and it is worth knowing.)

The most obvious Nash equilibrium of the game is that Player I chooses Good, and Player II chooses Good whatever Player I does. But there is

another Nash equilibrium, as looking at the strategic form of the game reveals. We'll put Player I on the row and Player II on the column. (We'll also do this from now on unless there is a reason to do otherwise.)

		Player II			
		gg	gb	bg	bb
Player I	G	1, 1	**1, 1**	1, 0	1, 0
	B	0, 1	0, 0	0, 1	0, 0

Look at the cell that I've bolded, where Player I plays Good, and Player II plays Good, Bad. That's a Nash equilibrium. Neither player can improve their outcome, given what the other player plays. But it is a very odd Nash equilibrium. It would be very odd for Player II to play this, since it risks getting 0 when they can guarantee getting 1.

It's true that Good, Bad is weakly dominated by Good, Good. But as we've already seen, and as we'll see in very similar examples to this soon, there are dangers in throwing out *all* weakly dominated strategies. Many people think that there is something else wrong with what Player II does here.

Consider the sub-game that starts with the right-hand decision node for Player II. That isn't a very interesting game; Player I has no choices, and Player II simply has a choice between 1 and 0. But it is a game. And note that Good, Bad is not a Nash equilibrium of that game. Indeed, it is a *strictly* dominated strategy in that game, since it involves taking 0 when 1 is freely available.

Say that a strategy pair is a **subgame perfect equilibrium** when it is a Nash equilibrium, and it is a Nash equilibrium of every sub-game of the game. The pair Good and Good, Bad is not subgame perfect, since it is not a Nash equilibrium of the right-hand subgame.

When we solve extensive form games by backwards induction, we not only find Nash equilibria, but subgame perfect equilibria. Solving this game by backwards induction would reveal that Player II would choose Good wherever she ends up, and then Player I will play Good at the first

move. And the only subgame perfect equilibrium of the game is that Player I plays Good, and Player II plays Good, Good.

The stakes here don't seem to high, since the ruled out Nash equilibrium has the same payout as the subgame perfect equilibrium. But that need not be the case. Sometimes the equilibrium that is not subgame perfect is an **incredible threat**. We mean *incredible* here in the literal sense of not credible. In the spirit of this, let's consider a game I'll call **Incredible**. Its game tree is Figure 5.8.

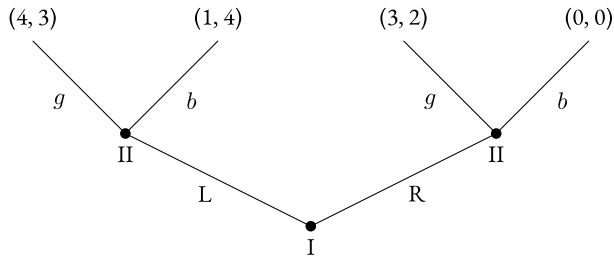


Figure 5.8: Incredible

Using backwards induction, we can see that the only subgame perfect equilibrium of the game is *Rbg*. That is, Player I plays *R*, and Player II is disposed to play *b* if Player I plays Left, and *g* if Player plays *R*. But writing out the able reveals another Nash equilibria.

		Player II			
		gg	*gb*	*bg*	*bb*
Player I	*L*	4,3	4,3	1,4	**1,4**
	R	3,2	0,0	**3,2**	0,0

I've bolded both the Nash equilibria. And note the one in the top right corner of the table. If Player II is committed to playing *bb*, then the best response Player I can make is to play *L*.

But look how implausible this is in real time. It means that if we get to the right hand node, Player II will choose to get 0, when 2 is available.

Of course, we won't actually get to that node. But it requires Player II to have dispositions that make no sense. This is the intuition behind subgame perfect equilibrium. We require that players not just make choices which make sense, but are disposed to do so at all times.

In the rest of this chapter, we'll look at reasons why this might be too strong a constraint on rational players.

5.5 Problems with Backwards Induction

When each player only moves once, it is very plausible that each player should play their part of a subgame perfect equilibrium solution to the game. But this is less compelling when players have multiple moves. We can start to see why by looking at a game that the philosopher Robert Stalnaker (1996, 1998) has used in a few places to argue against backwards induction reasoning. Again, it is an extensive form game. We will call it **Cooperative Caterpillar**.

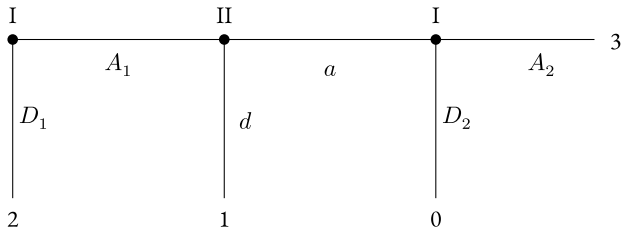


Figure 5.9: Cooperative Caterpillar

The game starts in the upper-left corner. There are up to three moves, each of them Across or Down. As soon as one player moves Down, the game ends. It is a common interest game; each player gets the payout at the terminal node.

Since there is only one path to each decision node, a strategy merely has to consist of a set of plans for what to play if that node is reached. Alice's strategy will consist of two capitalised moves, and Bob's strategy will consist of one lower-case move.

If we apply backwards induction, we get that the unique solution of the game is $\langle A_1 A_2, a \rangle$. Alice would play A_2 if it gets that far. Given that, Bob would be better off playing a than d if he gets to play. And given that, Alice is better off playing A_1 than D_1.

But there are many other Nash equilibria of the game. One of them is $\langle D_1 A_2, d \rangle$. Given that Player I is playing D_1, Player II can play anything without changing her payout; it will be 2 come what may. Given that Player II is playing d, Player I is best off playing D_1 and taking 2, rather than just getting the 1 that comes from leaving Player II to make a play.

Could this equilibrium be one that rational players, who know each other to be rational, reach? Stalnaker argues that it could. Assume each player knows that the other player is rational, and is playing that strategy. Given what Player I knows, her strategy choice is clearly rational. She takes 2 rather than the 1 that she would get by playing A_1, and she is disposed to take 3 rather than 0 if she gets to the final decision node. So her actual choice is rational, and her disposition is to make another rational choice if we reach the end of the game.

Things are a little trickier for Player II. You might think it is impossible for rational Player II, who knows Player I to be rational, to move d. After all, if Player I is rational, then she'll play A_2, not D_2. And if she plays A_2, it is better to play a than d. So it looks hard to justify Player II's move. But looks can be deceiving. In fact there isn't anything wrong with Player II's move, as long as he has the right beliefs to justify it. It's very important to distinguish the following two conditionals.

- If Player I has the choice, she chooses A_2 over D_2.
- If Player I were to have the choice, she would choose A_2 over D_2.

Player II knows that the first, indicative, conditional is true. And indeed it is true. But he doesn't know that the second, subjunctive, conditional is true. After all, if Player I were to have the choice between A_2 and D_2, she would have, *irrationally*, chosen A_1 over D_1. And if she had chosen irrationally once, it's possible that she would choose irrationally again.

Here's an analogy that may help explain what's going on. The following set of beliefs is consistent.

- Any perfectly rational being gets all of the questions on their algebra exam right.
- Alice is perfectly rational.
- If Alice had got the second-hardest question on the algebra exam wrong, she would have got the hardest question on the algebra exam wrong as well.

Player II's beliefs are like that. He believes Player I is perfectly rational. He also believes that if Player I were to make an irrational move, she would continue to make irrational moves. That's consistent with belief in perfect rationality, and nothing about the game setup rules out such a belief. He also believes that playing A_1 would be irrational. That's correct, given what Player I knows about Player II. Given all those beliefs, playing d, if he had the chance, would be rational.

Stalnaker argues that many game theorists have tacitly confused indicative and subjunctive conditionals in reasoning about games like **Cooperative Caterpillar**. There are other games where similar confusions arguably take place.

5.6 The Centipede

In **Cooperative Caterpillar**, backwards induction reasoning gets the players to the outcome that many people would say is intuitive. Stalnaker's objection is that the reasoning doesn't go through in theory, but at least you might say it works in practice. Once we make the game less cooperative, backwards induction reasoning ceases to be either intuitive, or to match what we see in experimental settings.

The game is due to Rosenthal (1981), and is usually called the **centipede**, because that is what the diagram looks like once you draw enough legs on it.

Two players alternate. At each turn a player can take the larger share of a growing pot, ending the game, or pass, in which case the pot grows and the other player faces the same choice. Table 5.1 is a short version, with four decision nodes.

Table 5.1: A short version of centipede

Node	1	2	3	4	end
Mover	Alice	Bob	Alice	Bob	
Take	2, 0	1, 3	4, 2	3, 5	
Pass	on	on	on	on	6, 4

Read that as follows. If Alice takes at node 1, she gets 2 and Bob gets 0. If she passes, Bob chooses at node 2, where taking gets him 3 and gets Alice 1. And so on. If everyone passes at every node, they end at 6, 4.

Notice two things about the payoffs. First, the total grows: 2 at the first node, 4 at the second, 6 at the third, 8 at the fourth, 10 at the end. Passing always makes the pie bigger. Second, at each node the mover does better by taking now than by taking one round later. Alice gets 2 by taking at node 1 and only 1 if Bob takes at node 2.

Now look what happens when we step through the backwards induction reasoning. At node 4, Bob compares taking, which gets him 5, with passing, which gets him 4. So Bob takes. Given that, at node 3 Alice compares taking, which gets her 4, with passing, which leads to Bob taking and gets her 3. So Alice takes. Given that, at node 2 Bob compares 3 with the 2 he gets when Alice takes at node 3. So Bob takes. And given that, at node 1 Alice compares 2 with the 1 she gets when Bob takes at node 2. So Alice takes at node 1.

The unique subgame perfect equilibrium is that Alice takes immediately, and the players walk away with 2 and 0. They never reach the part of the tree where the money is. Moreover, we could have let the game go on arbitrarily long, and this would have been the same result. As long as there's a final node, where one player knows the game will end no matter

what they do, backwards induction reasoning can get started, and it results in Alice taking at node 1.

This is really strange. No individual step is strange. Bob really should take rather than pass at the last step. It is fairly plausible that Alice should take rather than pass at the second last step. But if we add up enough of these plausible steps, we get something quite implausible.

The implausibility is not just that they end up with a result that is worse, for both of them, than what they could have got. One-shot prisoners' dilemma has that structure, and we mostly think that it's tragic, rather than irrational, that the players end up in a Pareto-inferior outcome.

Rather, the implausibility is that it feels like there is a way for them to have done better. By passing in the early rounds, they can give each other good reason to think that they will keep on passing, and hence a reason to pass themselves. Think about what happens too the backwards induction reasoning if anyone deviates. (This will be just like what Stalnaker says goes wrong in **Cooperative Caterpillar**.) Suppose Alice passes at node 1. Bob now has to decide what to do at a node that backwards induction says should never be reached. The argument for taking at node 2 ran through the claim that Alice will take at node 3, and that claim was derived from the assumption that Alice is rational. But Alice has just done the thing that assumption ruled out.

So what should Bob believe? There are at least three answers, and each has been defended.

- Alice made a mistake, a slip of the hand, and is otherwise rational. Then Bob should still expect her to take at node 3, and should take now.
- Alice is not rational after all. Then Bob has no idea what she will do at node 3, and the backwards induction argument for taking at node 2 has lost its premise.
- Alice is rational, and is signalling that she intends to pass again, in the hope that Bob will pass too and they will both do better. This is

> forward induction reasoning of the kind we will see in the **Money Burning Game** (Section 5.7), and on this reading Bob should pass.

The centipede is therefore not just a puzzle about whether backwards induction gives the right answer. It is a puzzle about what the assumption of common knowledge of rationality is even saying, given that the reasoning it licenses runs through nodes that it says will never be reached. Stalnaker's discussion of **Cooperative Caterpillar** is designed to show that we can see the same mistake in backwards induction reasoning even in cases where it gets the intuitive right answer, and that hence the first option here is mistaken.

There is an empirical footnote. When people play centipede games in the laboratory, they mostly do not take at the first node (McKelvey and Palfrey 1992). They pass for a while and take somewhere in the middle. That does not settle anything about what is rational, but it does mean the theory is predicting something that does not happen, which is at least a reason to look at the theory again.

5.7 Money Burning Game

Elchanen Ben-Porath and Eddie Dekel (1992) suggested this variant to the Battle of the Sexes game. The variation is in two steps. First, we change the payouts for the basic game to those in Table 5.2. (Note that I'm using a lowercase letter for one of Row's options here; this is to distinguish the *d* of down from the *D* of Don't burn.)

Table 5.2: The version of Battle of the Sexes being used

		Player II	
		l	*r*
Player I	*u*	4, 1	0, 0
	d	0, 0	1, 4

Then we give Player I the option of publicly burning 2 utils before playing this game. We will use D for Don't burn, and B for burn. So actually each player has four choices. Player I has to choose both D or B, then u or d. Player 2 has to choose whether to play l or r in each of the two possibilities: first, when D is played, second, when B is played. We'll write lr for the strategy of playing l if D, and r if B, and so on for the other strategies.

Table 5.3: The strategy table for the money burning game

		Player II			
		ll	lr	rl	rr
Player I	Du	4, 1	4, 1	0, 0	0, 0
	Dd	0, 0	0, 0	1, 4	1, 4
	Bu	2, 1	-2, 0	2, 1	-2, 0
	Bd	-2, 0	-1, 4	-2, 0	-1, 4

Now we can analyse this game a couple of ways. First, we can go through eliminating weakly dominated strategies in Table 5.3. Note that Du strictly dominated Bd, so we can eliminate it. If Bd is out, then ll weakly dominates lr, and rl weakly dominates rr. So we can eliminate lr and rr. Now Bu weakly dominates Dd, relative to the remaining options, so we can eliminate it. Given that just Du and Bu remain, ll weakly dominates all options for Player II, so it is all that is left. And given that ll is all that remains, Du strongly dominates Bu. So the iterative deletion of weakly dominated strategies leaves us with $\langle Du, ll \rangle$ as the unique solution of the game.

Alternatively, we can think the players reason as follows. (The following reasoning is sometimes called 'forward induction' reasoning.) Playing Bd is obviously irrational for Player I, since its maximum return is -1, and playing Du or Dd guarantees getting at least 0. So any rational player who plays B must be playing Bu. That is, if Player I is rational and plays B, she will play Bu. Moreover, this is common knowledge among the players. So if Player I plays B, she will recognise that Player II will know that she is playing Bu. And if Player II knows Player I is playing u after burning, she will

play *l*, since that returns her 1 rather than 0. So Player I knows that playing *B* leads to the 2,1 state; i.e., it returns her 2. But now it is irrational to play *Dd*, since that gets at most 1, and playing *B* leads to a return of 2. So Player I will play *Bu* or *Du*. And since the reasoning that leads to this is common knowledge, Player II must play *ll*, since playing *l* is her best response to a play of *u*, and she knows Player I will play *u*. But if Player II is going to play *ll*, Player I doesn't need to burn; she can just play *Du*.

There is a crucial conditional in the middle of that argument; let's isolate it.

- If Player I is rational and plays *B*, she will play *Bu*.

But that's not what is needed to make the argument work. What Player I needs, and in fact needs to be common knowledge, is the following subjunctive.

- If Player I is rational then, if she were to play *B*, she would play *Bu*.

But in fact there's no reason to believe that. After all, if the forward induction argument is right, then it is *irrational* to burn the money. So if Player I were to play *B*, i.e., burn the money, then she would be doing something irrational. And the fact that Player I is actually rational is consistent with the counterfactual that if she were to do one irrational thing, it would be because she is following an irrational strategy. Note that on the assumption that *Du* is optimal, then if Player I finds herself having played *B*, it isn't clear that playing *d* is irrational; it isn't like it is dominated by *u*.

So it isn't true that if Player I were to burn the utils, Player II would know that she is playing *Bu*, and react accordingly by playing *l*. And if that's not true, then there's no reason to think that *Bu* is better than *Dd*. And if there's no reason to think that, there's no reason to be confident Player 2 will play *ll*.

Robert Stalnaker (1998), in his commentary on the game, goes further and provides a positive model where the players start with a common belief in rationality, and in what each other will do, but in which the players play

Bu and *rl*. I'll leave it as an exercise to work out what counterfactuals the agents have to believe to make this rational, but suffice to say that it is a situation where each side has true beliefs about the other, and maximises expected utility given their beliefs.

There is another, relatively simple, equilibrium to the game. Player I will play *Dd*, and Player II will play *rr*. Player II is certain that Player I will play *d*, and will keep that belief even if Player I irrationally burns 2 utils to start with. Given that certainty, it is rational for Player I to play *Dd*, so Player II's belief is consistent with believing Player I to be rational. And since Player I is playing *Dd*, playing *rr* is perfectly rational for Player II; indeed it results in her best possible outcome. Moreover, given that Player II is playing *rr*, it makes sense for Player I to play *d* even if they were, irrationally, to burn the utils. So even though playing *Bd* would be irrational, since it is strictly dominated, if they were to (irrationally) play *D*, they *should* follow that by playing *d*. There's an important point about the scope of the claim that playing *Bd* is irrational. It *doesn't* imply that if the player were to irrationally play *B*, it would be irrational to follow with a play of *d*. If Player I was convinced that Player II was playing *rr*, then it would be rational to follow *B* with *d*.

5.8 Iterated Prisoners' Dilemma

Let's return to Prisoners Dilemma, using *C* and *D* for cooperate and defect. We'll call the players I and II, and use uppercase letters for Player I's strategies, and lower case letters for Player II's strategies. We'll assume that each player knows that the other players are perfectly rational in Stalnaker's sense. (That is, they have true beliefs about the other, and they maximise expected utility given their beliefs.)

There are a couple of arguments that the players must end up at the equilibrium where every player defects on every move. One of these is an argument by backwards induction.

At the last move of the game, defecting dominates cooperating. Both players know this. At the second-last move, you might have thought antecedently that there was some benefit to cooperating. After all, it might induce cooperation in the other player. But the only benefit one could get from cooperating was cooperation at the next (i.e., last) move. And no rational player will cooperate on the last move. So, if you're playing with a rational player, there's no benefit to cooperating on the second-last move. But if that's right, and everyone knows it, then there is no benefit to cooperating on the third-last move. The only benefit would be if that would induce cooperation, and it couldn't, since we just proved that any rational player would defect on the second-last move. And so on for any finite length game that you like.

Alternatively, we could look at the game from a strategic perspective. As we've seen in games like **Burning**, sometimes there are Nash equilibria that don't appear in backwards induction reasoning. But this isn't the case in finitely iterated Prisoners' Dilemma. The only Nash equilibrium is that both players defect in every circumstance.

This is a little tricky to check since the strategic form of iterated Prisoners' Dilemma gets very complex very quickly. Let's just consider the three round version of the game. Already each player has 128 strategies to choose from. The choice of a strategy involves making 7 distinct choices:

1. What to do in the first round.
2. What to do in the second round if the other player cooperates in the first round.
3. What to do in the second round if the other player defects in the first round.
4. What to do in the third round if the other player cooperates in each of the first two rounds.
5. What to do in the third round if the other player cooperates in the first round then defects in the second round.
6. What to do in the third round if the other player defects in the first

round then cooperates in the second round.

7. What to do in the third round is the other player defects in each fo the first two rounds.

Since these 7 choices are distinct, and the player has 2 choices at each point, there are $2^7 = 128$ possible strategies. So the strategic form of the table involves $128 \times 128 = 16384$ cells. Needless to say, we *won't* be putting that table here. (Though it isn't too hard to get a computer to draw the table for you. The four round game, where each player has 2^{15} choices, and there are over one billion cells in the decision table, requires more computing power!)

We'll write a strategy as $\langle x_1 x_2 x_3 x_4 x_5 x_6 x_7 \rangle$, where x_i is 0 if the player's answer to the i'th question above is to cooperate, and 1 if it is to defect. So $\langle 0000000 \rangle$ is the strategy of always cooperating, $\langle 1111111 \rangle$ is the strategy of always defecting, $\langle 0010101 \rangle$ is 'tit-for-tat', the strategy of cooperating on the first move, then copying the other player's previous move, and so on.

Let $\langle x_1 x_2 x_3 x_4 x_5 x_6 x_7 \rangle$ be any strategy that doesn't always involve defection on the final round, i.e., a strategy where $x_4 + x_5 + x_6 + x_7 < 4$. It is easy enough to verify that such a strategy is weakly dominated by $\langle x_1 x_2 x_3 1111 \rangle$. In some plays of the game, the defection on the final round leads to getting a better outcome. In other plays of the game, $\langle x_1 x_2 x_3 x_4 x_5 x_6 x_7 \rangle$ and $\langle x_1 x_2 x_3 1111 \rangle$ have the same play. For instance, $\langle 0001000 \rangle$ will do just as well as $\langle 0001111 \rangle$ if the opponent plays any strategy of the form $\langle 000 x_4 x_5 x_6 x_7 \rangle$, but it can never do better than $\langle 0001111 \rangle$. So if each player is perfectly rational, we can assume their strategy ends with 1111.

That cuts each player's choices down to 8. Let's do that table. We'll use C and c rather than 0, and D and d rather than 1, so it is easier to distinguish the two players' strategies. When we label the table, we'll leave off the trailing Ds and d's, since we assume players are defecting on the last round. We'll also leave off the 3 units the players each get in the last round. (In other words, this will look a lot like the table for a *two* round iterated Prisoners' Dilemma, but it is crucial that it is actually a three-round game.)

		Player II							
		ccc	*ccd*	*cdc*	*cdd*	*dcc*	*dcd*	*ddc*	*ddd*
Player I	*CCC*	6, 6	6, 6	3, 8	3, 8	3, 8	3, 8	0, 10	0, 10
	CCD	6, 6	6, 6	3, 8	3, 8	5, 5	5, 5	1, 6	1, 6
	CDC	6, 6	8, 3	4, 4	4, 4	3, 8	3, 8	0, 10	0, 10
	CDD	6, 6	8, 3	4, 4	4, 4	5, 5	5, 5	1, 6	1, 6
	DCC	6, 6	5, 5	8, 3	5, 5	4, 4	1, 6	4, 4	1, 6
	DCD	6, 6	5, 5	8, 3	5, 5	6, 1	2, 2	6, 1	2, 2
	DDC	6, 6	6, 1	10, 0	6, 1	4, 4	1, 6	4, 4	1, 6
	DDD	6, 6	6, 1	10, 0	6, 1	6, 1	2, 2	6, 1	2, 2

In this game *CCC* is *strongly* dominated by *CDD*, and *DCC* is strongly dominated by *DDD*. Similarly *ccc* and *dcc* are strongly dominated. So let's delete them.

		Player II					
		ccd	*cdc*	*cdd*	*dcd*	*ddc*	*ddd*
Player I	*CCD*	6, 6	3, 8	3, 8	5, 5	1, 6	1, 6
	CDC	8, 3	4, 4	4, 4	3, 8	0, 10	0, 10
	CDD	8, 3	4, 4	4, 4	5, 5	1, 6	1, 6
	DCD	5, 5	8, 3	5, 5	2, 2	6, 1	2, 2
	DDC	6, 1	10, 0	6, 1	1, 6	4, 4	1, 6
	DDD	6, 1	10, 0	6, 1	2, 2	6, 1	2, 2

Note that each of these is a best response. In particular,

- *CCD* is a best response to *dcd*.
- *CDC* is a best response to *ccd*.
- *CDD* is a best response to *ccd* and *dcd*.
- *DCD* is a best response to *ddc*.
- *DDC* is a best response to *cdc* and *cdd*.
- *DDD* is a best response to *cdc*, *cdd*, *ddc* and *ddd*.

Now the only Nash equilibrium of that table is the bottom-right corner. But there are plenty of strategies that, in a strategic version of the game,

would be consistent with common belief in perfect rationality. The players could, for instance, play *CDC* and *cdc*, thinking that the other players are playing *ccd* and *CCD* respectively. Those beliefs would be false, but they wouldn't be signs that the other players are irrational, or that they are thinking the other players are irrational.

But you might suspect that the strategic form of the game and the extensive form are crucially different. The very fact that there are Nash equilibria that are not subgame perfect equilibria suggests that there are tighter constraints on what can be played in an extensive game consistent with rationality and belief in rationality. Stalnaker argues, however, that this isn't right. In particular, any strategy that is (given the right beliefs) perfectly rational in the strategic form of the game is also (given the right beliefs and belief updating dispositions) perfectly rational in the extensive form. We'll illustrate this by working more slowly through the argument that the game play $\langle CDC, cdc \rangle$ is consistent with common belief in perfect rationality.

The first thing you might worry about is that it isn't clear that *CDC* is perfectly rational, since it is looks to be weakly dominated by *CDD*, and perfect rationality is inconsistent with playing weakly dominated strategies. But in fact *CDC* isn't weakly dominated by *CDD*. It's true that on the six columns represented here, *CDC* never does better than *CDD*. But remember that each of these strategies is short for a *three*-round strategy; they are really short for *CDCDDDD* and *CDDDDDD*. And *CDCDDDD* is not weakly dominated by *CDDDDDD*; it does better, for example, against *dcccddd*. That is the strategy is defecting the first round, cooperating the second, then defecting on the third round unless the other player has cooperated on each of the first two rounds. Since *CDC* does cooperate each of the first two rounds, it gets the advantage of defecting against a cooperator on the final round, and ends up with 8 points, whereas *CDD* merely ends up with 6.

But why would we think Player II might play *dcccddd*? After all, it is itself a weakly dominated strategy, and perfectly rational beings (like Player II) don't play weakly dominated strategies. But we're not actually thinking Player II *will* play that. Remember, Player I's assumption is that Player II

will play *ccddddd*, which is not a weakly dominated strategy. (Indeed, it is tit-for-tat-minus-one, which is a well-known strategy.) What Player I also thinks is that if she's wrong about what Player II will play, then it is possible that Player II is not actually perfectly rational. That's consistent with believing Player II is actually perfectly rational. The assumption of common belief in perfect rationality is not sufficient for the assumption that one should believe that the other player is perfectly rational *no matter what surprises happen*. Indeed, that assumption is barely coherent; some assumptions are inconsistent with perfect rationality.

One might be tempted by a weaker assumption. Perhaps a player should hold on to the belief that the other player is perfectly rational unless they get evidence that is *inconsistent* with that belief. But it isn't clear what could motivate that. In general, when we are surprised, we have to give up something that isn't required by the surprise. If we antecedently believe $p \wedge q$, and learn $\neg(p \wedge q)$, then what we've learned is inconsistent with neither p nor q, but we must give up one of those beliefs. Similarly here, it seems we must be prepared to give up a belief in the perfect rationality of the other player in some circumstances when that is not entailed by our surprise. (Note that even if you don't buy the argument of this paragraph, there still isn't a reason to think that perfectly rational players can't play *CDD*. But analysing that possibility is beyond the scope of these notes.)

What happens in the extensive form version of the game? Well, each player cooperates, thinking this will induce cooperation in the other, then each player is surprised by a defection at round two, then at the last round they both defect because it is a one-shot Prisoners' Dilemma. Nothing seems irrational there. We might wonder why neither defected at round one. Well, if they believed that the other player was playing tit-for-tat-minus-one, then it is better to cooperate at round one (and collect 8 points over the first two rounds) than to defect (and collect at most 6). And playing tit-for-tat-minus-one is rational if the other person is going to defect on the first and third rounds, and play tit-for-tat on round two. So as long as Player I thinks that Player II thinks that Player I thinks that she is going to defect on the first and third rounds, and play tit-for-tat on round two,

then her cooperation at round one is rational, and consistent with believing that the other player is rational.

But note that we had to attribute an odd belief to Player II. We had to assume that Player II is *wrong* about what Player I will play. It turns out, at least for Prisoners' Dilemma, that this is crucial. If both players are certain that both players are perfectly rational, and have no false beliefs, then the only strategy that can be rationalised is permanent defection. The proof (which I'm leaving out) is in Stalnaker (1996).

This *isn't* because the no false beliefs principle suffices for backwards induction reasoning in general. In **Cooperative Caterpillar**, we can have a model of the game where both players are perfectly rational, and have correct beliefs about what the other player will do, and both those things are common belief, and yet the backwards induction solution is not played. In that game A believes, truly, that B will play d, and B believes, truly, that A will play A_1D_2. And each of these moves is optimal given (true!) beliefs about the other player's strategy.

But Prisoners' Dilemma is special. It isn't that in a game played between players who know each other to have no false beliefs and be perfectly rational that we must end up at the bottom-right corner of the strategic table. But we must end up in a game where every player defects every time. It could be that Player I thinks Player II is playing either dcd or ddd, with ddd being much more, and on that basis she decides to play dcd. And Player II could have the converse beliefs. So we'll end up with the play being $\langle DCD, dcd \rangle$. But of course that means each player will defect on the first two rounds, and then again on the last round since they are perfectly rational. In such a case the requirement that each player have no false beliefs won't even be enough to get us to Nash equilibrium since $\langle DCD, dcd \rangle$ is not a Nash equilibrium. But it is enough to get permanent defection.

6 Bayesian Games

6.1 Two Kinds of Ignorance

In the games of Chapter 5, players were sometimes ignorant about what had already happened. When a player has to move without knowing which node of the tree they are at, we represent that with an information set, and we say the game is one of **imperfect information**. Prisoners' Dilemma in extensive form is like this. The second player moves knowing the rules, knowing the payoffs, knowing everything about the game, and not knowing only what the first player just did.

There is a different thing a player might not know. They might not know what the game is. They might not know what the other player's payoffs are, and so not know what the other player is trying to achieve. Games like this are games of **incomplete information**.

The distinction matters and it is easy to lose, so it is worth putting the two side by side. In a game of imperfect information you know the whole payoff table and you don't know where you are in the tree. In a game of incomplete information you don't know the payoff table.

Almost every interesting application of game theory outside the classroom involves incomplete information. When you bargain over a used car you do not know how much the seller values it. When a firm decides whether to enter a market it does not know the incumbent's costs. When two states approach a crisis, neither knows how much the other is willing to lose. Writing down a payoff table and assuming both players can see it is a serious idealisation.

But we don't obviously have a theory for these cases. Our solution concepts all take a game as input, and a game is a payoff table. If the players

don't know the payoff table, we cannot even say what game they are playing, and if we cannot say that, we cannot say what an equilibrium of it would be. Worse, there is a regress waiting. If Row doesn't know Col's payoffs, Row has beliefs about them. Col then has beliefs about Row's beliefs about Col's payoffs. And Row has beliefs about those. It looks like modelling ignorance about payoffs requires an infinite hierarchy of beliefs about beliefs, which is not something we can write on a page.

6.2 Types

John Harsanyi (1967) found a clever way through this. Games of incomplete information can be turned into games of imperfect information, which we already know how to handle.

The trick is to let the ignorance be about a move rather than about the game. We add a new player, **Nature**, who moves first, and whose move settles what the payoffs are. Nature's move is not a choice; it is a draw from a probability distribution that is common knowledge. The players then observe part of what Nature did, and not all of it.

What each player observes about Nature's move is called their **type**. A type settles everything about a player that the other players might be unsure of. In most of our examples the type just is the player's payoff function, but nothing turns on that. A type could encode how patient a player is, or how good they are at chess, or what they believe about the other player. It could encode something completely distinct; in a scenario where one player can observe whether it is raining and the other can't, it's sometimes useful to classify the first player's type as *rain* or *sun*. (If you insist, you could call their type *sees rain* or *sees sun*, but we often use the externally individuated types.)

So a **Bayesian game** has:

- a set of players;
- for each player, a set of actions;
- for each player, a set of possible types;
- a probability distribution over the profiles of types, called the **prior**;
- for each player, a payoff function that depends on the actions of everyone and on their own type.

We assume the prior is common knowledge. This is the **common prior assumption**, and without it we need a more complicated theory. One consequence of it is that the players start out agreeing about how likely each type profile is, and disagree only because they have seen different draws. That assumption is not always plausible in the cases these tools are applied to, and it's often worth questioning on an application. For now, notice one useful theoretical simplification that this assumption gets us. Instead of thinking about a potentially infinite hierarchy of beliefs about beliefs, we can just use the conditional probabilities a player gets from the prior, given their own type, and everyone can compute those.

6.3 Bayesian Nash Equilibrium

Once the game has been transformed this way, we can reuse Nash equilibrium almost unchanged. There is one adjustment to make first.

In a Bayesian game, a player knows their own type before acting. So a strategy cannot just be an action; it has to say what the player does for each type they might turn out to have. A **strategy** in a Bayesian game is a function from that player's types to that player's actions.

This is a common mistake, so it's worth belaboring the point. If Col has two types, and Col has two moves regardless of their type, then Col has *four* strategies, not two: do X whatever happens, do Y whatever happens, do X if the first type and Y if the second, or do Y if the first and X if the second. Col will only ever perform one action in any play of the game. But the

strategy has to be defined for the type Col didn't turn out to have, because Row's reasoning depends on what Col would have done in that case.

A **Bayesian Nash equilibrium** is then a profile of strategies, one for each player, such that each type of each player is choosing an action that maximises their expected payoff, given the strategies of the others and given the beliefs about the others' types that they get by conditioning the prior on their own type.

Equivalently, and this is the cleaner way to say it: a Bayesian Nash equilibrium is a Nash equilibrium of the imperfect information game that Harsanyi's transformation produces. In a sense, this isn't a new solution concept; it's a new term for an old concept.

6.4 An Example

Take Battle of the Sexes from Section 1.6, and make one change. Row still wants to meet Col. But Row is no longer sure that Col wants to meet Row.

Col has two types. With probability 1/2 Col is *sociable*, and wants to meet, with the payoffs we saw before. With probability 1/2 Col is *elusive*, and would rather be wherever Row is not. Col knows which type Col is. Row does not.

If Col is sociable, the game (shown in Table 6.1) is the one we already know.

Table 6.1: Familiar Battle of the Sexes

		Col	
		X	Y
Row	X	2, 1	0, 0
	Y	0, 0	1, 2

If Col is elusive, Row's payoffs are unchanged, but Col's change.

Table 6.2: A variant of Table 6.1 where Col does not want to meet

		Col	
		X	Y
Row	X	2, 0	0, 2
	Y	0, 1	1, 0

In Table 6.2, look first at Col, who has it easy, because Col knows which table is 'real'.

Sociable Col wants to match Row. If Row plays X, sociable Col gets 1 from X and 0 from Y, so plays X. If Row plays Y, sociable Col gets 0 from X and 2 from Y, so plays Y.

Elusive Col wants to mismatch. If Row plays X, elusive Col gets 0 from X and 2 from Y, so plays Y. If Row plays Y, elusive Col gets 1 from X and 0 from Y, so plays X.

Now Row, who has to average over the two. Suppose Col's strategy is *play X if sociable, Y if elusive*. Then Col plays X half the time and Y half the time, from Row's point of view, and Row's expected payoffs are

$$E(X) = \frac{1}{2} \times 2 + \frac{1}{2} \times 0 = 1$$

$$E(Y) = \frac{1}{2} \times 0 + \frac{1}{2} \times 1 = \frac{1}{2}$$

So Row plays X. And we should check this is consistent: given that Row plays X, sociable Col does want to play X, and elusive Col does want to play Y. Nothing unravels, so we have a Bayesian Nash equilibrium.

Two things about this are worth spelling out in more detail.

The first is that Col's two types do different things. The possibility of this happening is why the type system is interesting. Externally, Row is facing one person. But from the inside, from Row's perspective, they are facing a distribution over opponents, and has to pick an action that is good on average against it.

The second is that Row's behaviour looks, from the outside, exactly like behaviour in an ordinary game. Row picks X and sticks with it. The incomplete information is not visible in what Row does. But it is visible in why Row does it, and in what Row would have done had the probabilities been different. Suppose Col were sociable with probability p rather than $1/2$. Then Row's expected payoff from X is $2p$ and from Y is $1 - p$, so the equilibrium we just found survives exactly when $p \geqslant \frac{1}{3}$. Below that it falls apart, though something strange starts to happen. There is no pure strategy equilibrium when p is below $\frac{1}{3}$. Whatever Row does, if Col knows it, they will avoid Row, which Row doesn't want. So Row has to mix, playing X a third of the time, and Row and Col still find each other sometimes, by luck rather than design.

6.5 Purification

We can now settle a debt from Section 3.8.

The problem there was that mixed equilibria seem to require players to randomise, and it is not clear why anyone would. At a mixed equilibrium every pure strategy in the support does exactly as well as the mixture, so nothing recommends the mixture over any of them. The only thing a player's mixture accomplishes is keeping the *other* player indifferent, which is a strange thing for a self-interested agent to be aiming at.

Harsanyi's response is that the randomness is really in ignorance of the payoffs, not in what the other player does (Harsanyi 1973). Here's the idea. Imagine that we're in a game with a mixed strategy equilibrium and no pure strategy equilibrium. And each player has a small amount of private information. They know their own payoffs. For instance, in one outcome they might know that their payoff is 2, while the other player only knows that it is $2 + \delta$, where δ is randomly drawn from $[-\epsilon, \epsilon]$, for arbitrarily small ϵ. Now we're playing a Bayesian game; the player's type is the value of δ.

In the Bayesian equilibrium of the perturbed game, almost every type has a strict best reply. A player whose private draw tips them slightly towards one action simply takes it. Almost no one randomises; almost every type plays a pure strategy. But an opponent who cannot see the draw cannot predict the action, and from the opponent's point of view the player's behaviour is a probability distribution over actions.

Harsanyi's theorem (1973) is that as ϵ shrinks to zero, that sequence of pure strategy equilibria converges to the mixed equilibrium of the original game. The mixed equilibrium is the limit of pure strategy Bayesian Nash equilibria of nearby games of incomplete information.

If that is right, the interpretive problem dissolves. Nobody was ever randomising. Each player was doing the strictly best thing their own private situation called for, and the mixture was never a description of anyone's decision. It was a description of what the other player could work out.

Whether it is right is a further question, and there are two obvious worries. The first is that this all rests on reasoning from a sequence to what happens in the limit. You might worry that this is illegitimate. Relatedly, you might worry that we can't tell much about a game by looking at different games, even if they are neighbors. For what it's worth, the idea that we should look at nearby games has been really influential. We're not going to focus on it, but the formal statements of many of the concepts we'll somewhat informally introduce in the rest of this chapter uses the same idea.

6.6 Perfect Bayesian Equilibrium

The core idea behind subgame perfect equilibrium was that we wanted to eliminate equilibria that relied on 'incredible threats'. That is, there are some equilibria such that the first player makes a certain move only because if they make a different move, the second player to move would, on their current strategy, do something that makes things worse for both players.

But there's no reason to think that the second player would actually do that.

The same kind of consideration can arise in games where there aren't any subgames. For instance, consider the following game, which we'll call **Perfect**.

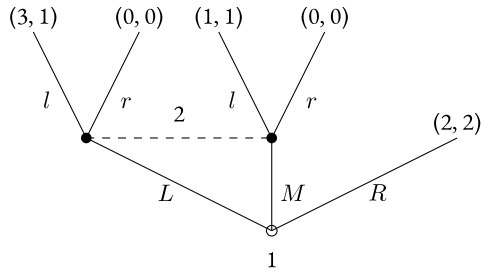


Figure 6.1: Perfect

The game here is a little different to one we've seen so far. First Player 1 makes a move, either L, M or R. If her move is R, the game ends, with the (2, 2) payout. If she moves L or M, the game continues with a move by Player 2. But crucially, Player 2 does not know what move Player 1 made, so she does not know which node she is at. So there isn't a subgame here to start. Player 2 chooses l or r, and then the game ends.

There are two Nash equilibria to **Perfect**. The obvious equilibrium is Ll. In that equilibrium Player 1 gets her maximal payout, and Player 2 gets as much as she can get given that the rightmost node is unobtainable. But there is another Nash equilibrium available, namely Rr. In that equilibrium, Player 2 gets her maximal payout, and Player 1 gets the most she can get given that Player 2 is playing r. But note what a strange equilibrium it is. It relies on the idea that Player 2 would play r were she to be given a move. But that is absurd. Once she gets a move, she has a straight choice between 1, if she plays l, and 0, if she plays r. So obviously she'll play l.

This is just like the examples that we used to motivate subgame perfect equilibrium, but that concept doesn't help us here. So we need a new con-

cept. The core idea is that each player should be modelled as a Bayesian expected utility maximiser. More formally, the following constraints are put on players.

1. At each point in the game, each player has a probability distribution over where they are in the game. These probability distributions are correct about the other players' actions. That is, if a player is playing a strategy S, everyone has probability 1 that they are playing S. If S is a mixed strategy, this might involve having probabilities between 0 and 1 in propositions about which move the other player will make, but players have correct beliefs about other players' strategies.
2. No matter which node is reach, each player is disposed to maximise expected utility on arrival at that node.
3. When a player had a positive probability of arriving at a node, on reaching that node they update by conditionalisation.
4. When a player gave 0 probability to reaching a node (e.g., because the equilibrium being played did not include that node), they have some disposition or other to form a set of consistent beliefs at that node.

The last constraint is very weak, but it does enough to eliminate the equilibrium Rr. The constraint implies that when Player 2 moves, she must have some probability distribution Pr such that there's an x such that $\Pr(L) = x$ and $\Pr(M) = 1 - x$. Whatever value x takes, the expected utility of l given Pr is 1, and the expected utility of r is 0. So being disposed to play r violates the second condition. So Rr is not a Perfect Bayesian equilibrium.

It's true, but I'm not going to prove it, that all Perfect Bayesian equilibria are Nash equilibria. It's also true that the converse does not hold, and this we have proved; **Perfect** is an example.

6.7 Signaling Games

Concepts like Perfect Bayesian equilibrium are useful for the broad class of games known as signaling games. In a signaling game, Player 1 gets some information that is hidden from player 2. Many applications of these games involve the information being *de se* information, so the information that Player 1 gets is referred to as her *type*. But that's inessential; what is essential is that only one player gets this information. Player 1 then has a choice of move to make. There is a small loss of generality here, but we'll restrict our attention to games where Player 1's choices are independent of her type, i.e., of the information she receives. Player 2 sees Player 1's move (but not, remember, her type) and then has a choice of her own to make. Again with a small loss of generality, we'll restrict attention to cases where Player 2's available moves are independent of what Player 1 does.

We'll start, in game **Simple Signal** with a signaling game where the parties' interests are perfectly aligned.

As above, we use an empty disc to signal the initial node of the game tree. In this case, it is the node in the centre of the tree. The first move is made by Nature, again denoted as N. Nature assigns a type to Player 1; that is, she makes some proposition true, and reveals it to Player 1. Call that proposition q. We'll say that Nature moves left if q is true, and right otherwise. We assume the probability (in some good sense of probability) of q is p, and this is known before the start of the game. After Nature moves, Player 1 has to choose *U*p or *D*own. Player 2 is shown Player 1's choice, but not Nature's move. That's the effect of the horizontal dashed lines. If the game reaches one of the upper nodes, Player 2 doesn't know which one it is, and if it reaches one of the lower nodes, again Player 2 doesn't know which it is. Then Player 2 has a make a choice, here simply denoted as *l*eft or *r*ight.

In any game of this form, each player has a choice of four strategies. Player 1 has to choose what to do if q is true, and what to do if q is false. We'll write $\alpha\beta$ for the strategy of doing α if q, and β if $\neg q$. Since α could be U or D, and β could be U or D, there are four possible strategies. Player 2

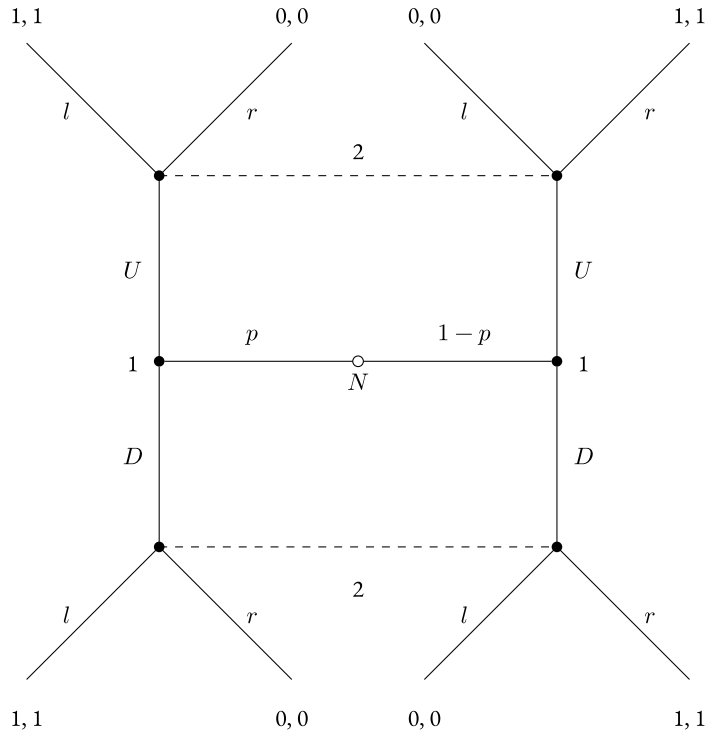


Figure 6.2: Simple Signal

has to choose what to do if U is played, and what to do if D is played. We'll write $\gamma\delta$ for the strategy of doing γ if U is played, and δ if D is played. Again, there are four possible choices. (If Player 2 knew what move Nature had made, there would be four degrees of freedom in her strategy choice, so she'd have 16 possible strategies. Fortunately, she doesn't have that much flexibility!)

Any game of this broad form is a signaling game. signaling games differ in (a) the interpretation of the moves, and (b) the payoffs. **Simple Signal** has a very simple payoff structure. Both players get 1 if Player 2 moves l iff q, and 0 otherwise. If we think of l as the formation of a belief that q, and r as the formation of the opposite belief, this becomes a simple communication game. The players get a payoff iff Player 2 ends up with a true belief about q, so Player 1 is trying to communicate to Player 2 whether q is true. This kind of simple communication game was used by David Lewis (1969) to show that game theoretic approaches could be fruitful in the study of meaning. The game is perfectly symmetric if $p = 1/2$; so as to introduce some asymmetries, I'll work through the case where $p = 3/5$.

Simple Signal has a dizzying array of Nash equilibria, even given this asymmetry introducing assumption. They include the following:

- There are two **separating** equilibria, where what Player 1 does depends on what Nature does. These are $\langle UD, lr\rangle$ and $\langle DU, rl\rangle$. These are rather nice equilibria; both players are guaranteed to get their maximal payout.
- There are two **pooling** equilibria, where what Player 1 does is independent of what Nature does. These are $\langle UU, ll\rangle$ and $\langle DD, ll\rangle$. Given that she gets no information from Player 1, Player 2 may as well guess. And since $\Pr(q) > 1/2$, she should guess that q is true; i.e., she should play l. And given that Player 2 is going to guess, Player 1 may as well play anything. So these are also equilibria.
- And there are some **babbling** equilibria. For instance, there is the equilibrium where Player 1 plays either UU or DD with some probability r, and Player 2 plays ll.

Unfortunately, these are all Perfect Bayesian equilibria too. For the separating and babbling equilibria, it is easy to see that conditionalising on what Player 1 plays leads to Player 2 maximising expected utility by playing her part of the equilibrium. And for the pooling equilibria, as long as the probability of q stays above $1/2$ in any 'off-the-equilibrium-path' play (e.g., conditional on D in the $\langle UU, ll \rangle$ equilibrium), Player 2 maximises expected utility at every node.

That doesn't mean we have nothing to say. For one thing, the separating equilibria are Pareto-dominant in the sense that both players do better on those equilibria than they do on any other. So that's a non-coincidental reason to think that they will be the equilibria that arise. There are other refinements on Perfect Bayesian equilibria that are more narrowly tailored to signaling games. We'll introduce them by looking at a couple of famous signaling games.

6.8 Signaling without Cooperation

Economists have been interested for several decades in game-theoretic representations of college that give make going to college essentially a signal in a game. For example, consider the following variant of the signaling game, **College Signal**. It has the following intended interpretation:

- Player 1 is a student and potential worker, Player 2 is an employer.
- The student is either type high or low[8] with probability p of being high. Nature reveals the type to the *student*, but only the probability to the employer, so p is that the student is high.
- The student has the choice of going to college (U) or the beach (D).
- The employer has the choice of hiring the student (l) or rejecting them (r).

[8] For some purposes we interpret 'high' as high intelligence, other times as 'high' labor market value, other times as something else. I'll use use 'high' and 'low', and leave off what this might mean until we get to the question of whether this is really a model of anything in the real world

In **College Signal** we make the following extra assumptions about the payoffs.

- Being hired is worth 4 to the student.
- Going to college rather than the beach costs the high student 1, and the low student 5, since college is much harder work for low types.
- The employer gets no benefit from hiring college students as such.
- Hiring a high student pays the employer 1, hiring a low student costs the employer 1, and rejections have no cost or benefit.

The resulting game tree is shown in Figure 6.3. The first move, by nature, sends high types to the left, and low types to the right. In this game, low students never prefer to go to college, since even the lure of a job doesn't make up for the pain of actually having to study. So a rational strategy for Player 1 will never be of the form αU, since for low students, college is a dominated option, being dominated by αD. But whether high students should go to college is a trickier question. That is, it is trickier to say whether the right strategy for Player 1 is UD or DD, which are the two strategies consistent with eliminating the strictly dominated strategies. (Remember that strictly dominated strategies cannot be part of a Nash equilibrium.)

First, consider the case where $p < 1/2$. In that case, if Player 1 plays DD, then Player 2 gets $2p - 1$ from playing either ll or rl, and 0 from playing lr or rr. So she should play one of the latter two options. If she plays rr, then DD is a best response, since when employers aren't going to hire, students prefer to go to the beach. So there is one **pooling** equilibrium, namely $\langle DD, rr \rangle$. But what if Player 2 plays lr? Then Player 1's best response is UD, since high students prefer college conditional on it leading to a job. So there is also a **separating** equilibrium, namely $\langle UD, lr \rangle$. The employer prefers that equilibrium, since her payoff is now p rather than 0. And students prefer it, since their payoff is $3p$ rather than 0. So if we assume that people will end up at Pareto-dominant outcomes, we have reason to think that high students, and only high students, will go to college, and employers will hire college students, and only college students. And all this is true

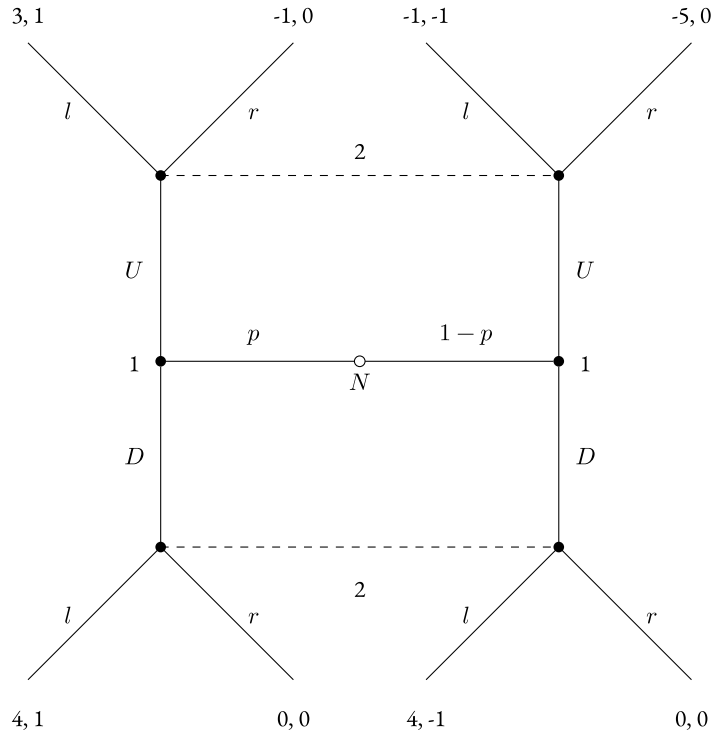


Figure 6.3: College Signal

despite there being no advantage whatsoever to going to college in terms of how good an employee one will be.

Especially in popular treatments of the case, the existence of this kind of model can be used to motivate the idea that college *only* plays a signaling role. That is, some people argue that in the real world college does not make students more economically valuable, and the fact that college graduates have better employment rates, and better pay, can be explained by the signaling function of college. We'll look in a bit at the detailed evidence for this, but first lets look at some variants of the model.

So far we assumed that $p < 1/2$. If we drop that assumption, and assume instead that $p > 1/2$, the case becomes more complicated. Now if Player 1 plays *DD*, i.e., goes to the beach no matter what, Player 2's best response is still to hire them. But note that now a very odd equilibrium becomes available. The pair $\langle DD, rl \rangle$ is a Nash equilibrium, and, with the right assumptions, a Perfect Bayesian equilibrium. This pair says that Player 1 goes to the beach whatever her type, and Player 2 hires only beach goers.

This is a very odd strategy for Player 2 to adopt, but it is a little hard to say just why it is odd. It is clearly a Nash equilibrium. Given that Player 2 is playing *rl*, then clearly beach-going dominates college-going. And given that Player 1 is playing *DD*, playing *rl* gets as good a return as is available to Player 2, i.e., $2p - 1$. Could it also be a Perfect Bayesian equilibrium? It could, provided Player 2 has a rather odd update policy. Say that Player 2 thinks that if someone goes to college, they are low type with probability greater than 1/2. That's consistent with what we've said; given that Player 1 is playing *DD*, the probability that a student goes to college is 0. So the conditional probability that a college-goer is high type is left open, and can be anything one likes in Perfect Bayesian equilibrium. So if Player 2 sets it to be, say, 0, then the rational reaction is to play *rl*.

But now note what an odd update strategy this is for Player 2. She has to assume that if someone deviates from the *DD* strategy, it is someone for whom the deviation is strictly dominated. A few economists and game theorists have thought that we can put more substantive constraints on probabilities conditional on 'off-the-equilibrium-path' behaviour. The simplest

such constraint is that you shouldn't read a deviation as coming from a player for whom that deviation is strictly dominated.

That constraint does rule out $\langle DD, rl \rangle$. Going to college is strictly dominated for the low type: the worst the beach can do for her is 0, and the best college can do is −1. It is not dominated for the high type, whose best outcome from college is 3, against a worst case of 0 from the beach. So Player 2, on seeing a degree, has to put probability 1 on the high type, and so has to hire. The same constraint does not rule out the similar pair $\langle DD, ll \rangle$, since there she hires anyway.

It is worth being careful about what did the work there, because the best known constraint in this area is a different one, and it does *not* rule out $\langle DD, rl \rangle$. That is the "**intuitive criterion**" of Cho and Kreps (1987), which we state in Section 6.9. Roughly, it restricts beliefs by asking which types could have gained by deviating, given what they get in the equilibrium. In $\langle DD, rl \rangle$, neither type could have. Both are getting 4, which is the most either type can get anywhere in this game, while college gets the high type at most 3 and the low type at most −1. So there is no type left to attribute a deviation to, and the criterion says nothing at all.

That looks like a weakness. $\langle DD, rl \rangle$ is just the sort of equilibrium a refinement ought to reject, and the most famous refinement cannot reject it. The reason is structural. The criterion works by comparing what a type gets in equilibrium with what it could get by deviating, and once $p > 1/2$ the employer is willing to hire an unsorted pool, so both types are already doing as well as they possibly could. A criterion that works by comparing payoffs has nothing to compare.

The criterion has a great deal more to say when the pooling equilibrium leaves somebody short of what they could get, which is what happens when $p < 1/2$. We'll close the discussion of the math of signaling games with an example where it does real work, and a more careful statement of it.

6.9 The Intuitive Criterion

The tree in Figure 6.4 represents **Canadian Signal**, which is also a guessing game. The usual statement of it involves all sorts of unfortunate stereotypes about the type of men who have quiche and/or beer for breakfast, and I'm underwhelmed by it. So I'll run with an example that relies on different stereotypes.

A North American tourist is in a bar. 60% of the North Americans who pass through that bar are from Canada, the other 40% are from the US. (This is a little implausible for most parts of the world, but maybe it is very cold climate bar. In Cho and Kreps' version the split is 90/10 not 60/40, but all that matters is which side of 50/50 it is.) The tourist, call her Player 1, knows her nationality, although the barman doesn't. The tourist can ask for the bar TV to be turned to hockey or to baseball, and she knows once she does that the barman will guess at her nationality. (The barman might also try to rely on her accent, but she has a fairly neutral upper-Midwest/central Canada accent.) Here are the tourist's preferences.

- If she is American, she has a (weak) preference for watching baseball rather than hockey.
- If she is Canadian, she has a (weak) preference for watching hockey rather than baseball.
- Either way, she has a strong preference for being thought of as Canadian rather than American. This preference is considerably stronger than her preference over which sport to watch.

The barman's preferences are simpler; he prefers to make true guesses to false guesses about the tourist's nationality. All of this is common knowledge. So the decision tree is given in Figure 6.4. In this tree, B means asking for baseball, H means asking for hockey, a means guessing the tourist is from the USA, c means guessing she is from Canada.

There are two Nash equilibria for this game. One is $\langle HH, ca \rangle$; everyone asks for hockey, and the barman guesses Canadian if hockey, American if baseball. It isn't too hard to check this is an equilibrium. The Canadian

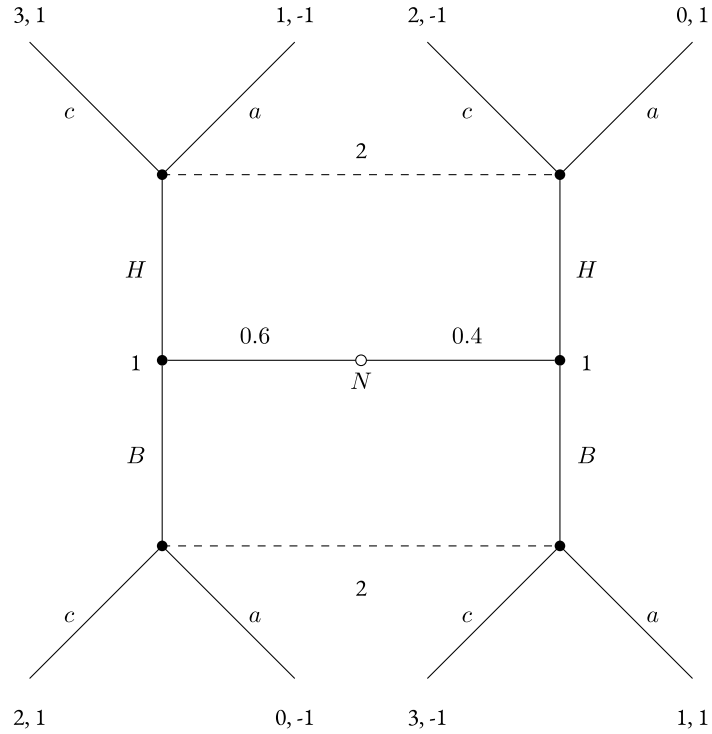


Figure 6.4: Canadian Signal

gets her best outcome, so it is clearly an equilibrium for her. The American gets the second-best outcome, but asking for baseball would lead to a worse outcome. And given that everyone asks for hockey, the best the barman can do is go with the prior probabilities, and that means guessing Canadian. It is easy to see how to extend this to a perfect Bayesian equilibrium; simply posit that conditional on baseball being asked for, the probability that the tourist is American is greater than 1/2.

The other equilibrium is rather odder. It is $\langle BB, ac \rangle$; everyone asks for baseball, and the barman guesses American if hockey, Canadian if baseball. Again, it isn't too hard to see how it is Nash equilibrium. The Canadian would rather have hockey, but not at the cost of being thought American, so she has no incentive to defect. The American gets her best outcome, so she has no incentive to defect. And the barman does as well as he can given that he gets no information out of the request, since everyone requests baseball.

Surprisingly, this could also be a perfect Bayesian equilibrium. This requires that the barman maximise utility at every node. The tricky thing is to ensure he maxmises utility at the node where hockey is chosen. This can be done provided that conditional on hockey being chosen, the probability of American rises to above 1/2. Well, nothing we have said rules that out, so there *exists* a perfect Bayesian equilibrium. But it doesn't seem very plausible. Why would the barman adopt just *this* updating disposition?

There were a bunch of suggestions for formal rules to rule out intuitively crazy updating dispositions like this. I'm just going to note one of the very early attempts in this area, Cho and Kreps' *Intuitive Criterion*. (Cho and Kreps 1987, 202)

Let an outcome o of the game be any combination of moves by nature and the players. Call the players' moves collectively P and nature's move N. So an outcome is a pair $\langle P, N \rangle$. Divide the players into three categories.

- The **happy** players in o are those such that given N, o maximises their possible returns. That is, for all possible P', their payoff in $\langle P, N \rangle$ is

at least as large as their payoff in $\langle P', N \rangle$.

- The **content** players are those who are not happy, but who are such that for no strategy s' which is an alternative to their current strategy s, would they do better given what nature and the other players would do if they played s'.
- The **unhappy** players are those who are neither happy nor content.

The standard Nash equilibrium condition is that no player is unhappy. So assume we have a Nash equilibrium with only happy and content players. And assume it is also a Perfect Bayesian equilibrium. That is, for any 'off-equilibrium' outcome, each player would have some credence function such that their play maximises utility given that function.

Now add one constraint to those credence functions:

> **Intuitive Criterion - Weak Version**
>
> Assume a player has probability 1 that a strategy combination P will be played. Consider their credence function conditional on another player playing s', which is different to the strategy s they play in P. If it is consistent with everything else the player believes that s' could be played by a player who is content with the actual outcome $\langle P, N \rangle$, then give probability 1 to s' being played by a content player.

That is, if some deviation from equilibrium happens, give probability 1 to it being one of the content players, not one of the happy players, who makes the deviation.

That's enough to rule out the odd equilibrium for the baseball-hockey game. The only way that is a Perfect Bayesian equilibrium is if the barman responds to a hockey request by *increasing* the probability that the tourist is American. But in the odd equilibrium, the American is happy, and the Canadian is merely content. So the Intuitive Criterion says that the barman should give credence 1 to the hockey requester, i.e., the deviator from equilibrium, being Canadian. And if so, the barman won't respond to a

hockey request by guessing the requestor is American, so the Canadian would prefer to request hockey. And that means the outcome is no longer an equilibrium, since the Canadian is no longer even content with requesting baseball.

The weak version is easier to apply than what Cho and Kreps actually offer, and it is sometimes weaker. The difference shows up as soon as two types differ over *which* deviations would have been worth making, even if neither type is happy overall. The official version is a little more complicated. First some terminology of mine, then their terminology next.

- A player is **happy to have played** s rather than s' if the payoff for s in o is greater than any possible outcome for s' given N.
- A player is **content to have played** s rather than s' if they are not happy to have played s rather than s', but the payoff for s in o is as great as the payoff for s' given N and the other players' strategies.
- A player is **unhappy to have played** s rather than s' if they are neither happy nor content to have played s rather than s'.

If a player is happy to have played s rather than s', and s is part of some equilibrium outcome o, then Cho and Kreps say that s' is an **equilibrium-dominated** strategy. The full version of the Intuitive Criterion is then:

> **Intuitive Criterion - Full Version**
>
> Assume a player has probability 1 that a strategy combination P will be played. Consider their credence function conditional on another player playing s', which is different to the strategy s they play in P. If it is consistent with everything else the player believes that s' could be played by a player who is merely content to have played s rather than s', then give probability 1 to s' being played by someone who is content to have played s, rather than someone who is happy to have played s.

This refinement matters whenever two types differ over which deviations are worth making, and that can happen with only two options each. Go

back to **College Signal** with $p < 1/2$, at the pooling equilibrium $\langle DD, rr \rangle$ where nobody goes to college and nobody is hired. Both types get 0, and both would do worse by going to college, so on my weak version both are merely content and nothing is ruled out. But the high type is *content* to have gone to the beach rather than to college, since college would have got her −1 here but could have got her 3, while the low type is *happy* to have gone to the beach rather than to college, since college could not have got her more than −1. So the full version says that a degree must be read as coming from a high type, the employer must hire, and the high type would then get 3 rather than 0. The pooling equilibrium fails, and the separating equilibrium $\langle UD, lr \rangle$ is what survives.

The difference also shows up with more options. It might be that a player's options are s_1, s_2 and s_3, and their type is t_1 or t_2. In the Nash equilibrium in question, they play s_1. If they are type t_1, they are happy to have played s_1 rather than s_2, but merely content to have played s_1 rather than s_3. If they are type t_2, they are happy to have played s_1 rather than s_3, but merely content to have played s_1 rather than s_2. In my terminology above, both players are content rather than happy with the outcome. So the weak version of the Intuitive Criterion wouldn't put any restrictions on what we can say about them conditional on them playing, say s_2. But the full version does say something; it says other players should assign probability 1 to the player being type t_2 conditional on them playing s_2, since the alternative is that s_2 is played by a player who is happy they played s_1 rather than s_2. Similarly, it says that conditional on the player playing s_3, other players should assign probability 1 to their being of type t_1, since the alternative is to assign positive probability to a player deviating to a strategy they are happy not to play.

The Intuitive Criterion has been the subject of an enormous literature. Google Scholar lists over 5000 citations for the Cho and Kreps paper alone, and similar rules are discussed in other papers. So there are arguments that it is too weak and arguments too strong, and refinements in all sorts of directions. But we'll leave those. What I most wanted to stress was the *form* a refinement of Perfect Bayesian equilibrium would take. It is a constraint

on the conditional credences of agents conditional on some probability 0 event happening. It's interesting that there are some plausible, even intuitive constraints; in general it seems the economic literature has investigated rationality under 0 probability evidence more than philosophers have.

6.10 Spence and the Value of a Useless Signal

Let's turn away from these very theoretical questions, and back to applications of these models to real world problems.

College Signal has an important history. It is a simplified version of the model Spence (1973) published as an account of the labor market, and which won him a share of a Nobel Prize. It triggered a huge debate, going to this day, about the economic significance of college. The debate is not about the mathematics of the model, which all check out. It's about whether the model is an accurate representation of the real world.

Let's look at those assumptions. In the separating equilibrium, high type students go to college and low type ones do not, employers hire college-goers and reject the rest, and everyone is behaving perfectly rationally. In the model, *no one learns anything* in college. A high type student is more valuable to employers, but they would have been just as valuable with no college. Indeed, them being hired without going to college would be a weak Pareto-improvement, since college is costly to the student. The student's productivity is settled by their type, which Nature fixed before the game started. College does not change it. All college does is cost less to a high type student than to a low type one.

That asymmetry in cost is why the model works. Because the high type student finds college 'cheaper', and we'll want to come back to what that means, there is a range of prices at which the high type student is willing to pay and the low type one is not, and paying it therefore reveals the type. The signal works because it is expensive, and because it is differentially expensive. It does not need to be useful.

Notice also how different this is from **Simple Signal**. There the players' interests were aligned, and a costless signal did the job. You can get the separating equilibrium in a case where it doesn't cost anything to leave one or the other signal.[9] In this case, where the interests diverge, a costless signal couldn't work. If college was costless, everyone would go. If it cost the *same amount* to everyone, it wouldn't work.

So we might worry what this says about what we do here. Is college just a sorting mechanism? Does it produce a private gain but no social gain? Is it a massively expensive system that just makes us collectively worse off?

Now note that the model only gives us a consistency proof. We know, and we'll go over this evidence more in Section 6.11, that college grads earn more than non grads. What Spence's model shows is that this fact alone is consistent with college not actually teaching anyone anything. But lots of things are consistent without being all true. To do more, we need to look more at the details.

6.11 The College Wage Premium

There is a key fact that needs explaining. In almost every rich country, people who finish a degree earn substantially more over a lifetime than people who do not. According to recent OECD (2025) figures (see especially Table A4.1), on average across a range of countries, degree holders earn 40% more over their lifetime than people with high school degrees who have not completed college. In the United States, this premium is even more. Here, grads earn 64% more than people whose highest qualification is either a high school degree *or a diploma*. That's a gap of well over a million dollars,

[9] The canonical example of this signalling is in David Lewis's book *Convention*, which grew out of his PhD Thesis (Lewis 1969). The patriots in Boston want to warn their soldiers out at Lexington which way the English are attacking. The sexton in the Old North Church can see what the English are doing, but he can't be seen talking to other patriots. So they have a plan; he'll put out one lantern if the English are marching up the highway, and two if they are sailing up the river. This works, even though it doesn't cost more to put up two lanterns.

probably closer to two million dollars. These are enormous numbers, and they need explaining.

There are three natural explanations which are worth distinguishing.

- **Human capital.** College teaches things that are worth money. The premium measures the skills. This explanation doesn't say *where* college teaches these skills; as far as this explanation goes it could be that you learn more from your fellow students in the dorms than in class.[10] What's central to this story is that people come out of college with more value to employers than they came in with.
- **Selection.** Colleges admit people who were already going to do well. Just knowing that one group of students had higher SAT scores on average would suggest they'd earn more money. College grads do, on average, have much higher SAT scores than the general populace. Maybe that's the whole story.
- **Signalling.** College is a hurdle that is cheaper to clear for the people employers want. The premium is a consequence of this fact.

These are not exclusive, and the full explanation is likely to involve all three. But they are different, and a story where one of them explains 80% of the premium, and the other two explain 10% each leaves us with a very different picture than if a different one was the 80%.

Still, let's see if we can make progress by thinking about what the world would look like if one of them were the whole story. Two of the explanations look fairly bad by that score.

If it were all human capital, then partial degrees should earn partial returns, and three years of a four-year degree should be worth roughly three quarters of the premium. If it were all selection, an admission letter should be worth almost as much as a diploma, since the letter is where the sorting happened.

[10] Prima facie this is plausible, but note that most students around the world don't live in dorms.

Neither prediction holds. Two or three years of college buys very little, and the premium mostly arrives when the degree does. This is the **sheepskin effect**, named for the old practice of printing diplomas on sheepskin. This is the single strongest piece of evidence for the signalling story. It is hard to see why the fourth year of skill acquisition should be worth several times the third. It is easy to see why clearing a hurdle should be worth more than nearly clearing it.

The effect is easiest to see in the United States, which has an unusually large population of people with some college and no degree. In most countries that group is too small to say much about. In the American data there are millions of them, and their advantage over high school graduates is small. While it would definitely be nicer to have more data, we shouldn't dismiss this American data: it does seem to tell against the purest version of the human capital story.

6.12 Does the Signalling Model Fit?

So that's the first bit of evidence that the signalling theory is right. It predicts the sheepskin effect, the other two, *if taken as comprehensive theories*, predict it won't exist. In fact, it does exist.

The signalling model also explains something else about public universities like this one: the doors to the lecture theatres are unlocked. Anyone can walk in to any lecture.[11] We are, on the whole, not overrun with people taking advantage of all these effectively free lectures. This is a little surprising on the human capital story.[12]

But there are also some difficulties.

The premium grows with age. Again, the Among workers in their late twenties the advantage is real but moderate; among workers in their

[11] Some of you may recall this fact plays an important part in the plot of *On the Calculation of Volume* (Balle [2021] 2025).

[12] Though a human capital story that focussed on what you learn in discussion sections and in discussions with fellow students will also predict this, and is frankly a little more plausible.

late forties it is substantially larger. That feels like the wrong way round for a signal. The whole apparatus rests on employers being unable to see a worker's type directly, and after twenty years of performance reviews, promotions and colleagues' opinions, the type is not hidden any more. The signal should be losing value, and it is gaining. The key thing is that the signalling model turns on the presence or absence of a degree being a key part of how employers judge the quality of a would-be-employee, and that's just not plausible after many years of work.

There are things the defender of the signalling model could say. The premium at forty-five may reflect twenty years of sorting inside firms rather than the degree itself, in which case the degree got someone through the first door and the rest is career. Or the wage structure may have been fixed early, by contracts or internal pay scales, and simply persist. Both are defensible, but we're back in a situation where one theory is explaining things away, and the other theory is readily predicting them.

The cost assumption may not hold. The model needs college to be more costly for the low type than the high type, and it needs the correlation to be tight; this is the single condition that makes the separating equilibrium work at all. It is worth asking whether it is true.

The 'cost' here mostly isn't literal dollar cost. Some people get more scholarships for academic success, and some people have to repeat courses they can't complete. But none of these are making college literally two million dollars more expensive for low type students.

Rather, the point is that college is more *pleasant* for high types than low types. OK, who actually finds college fun, who pays the least psychic cost for being here? The students who flourish, who enjoy the reading and the seminars and are untroubled by exams, are not obviously the students who will be the most valuable employees. The traits that make academic life comfortable are not the traits that make someone reliable under a deadline, or good with clients, or sound in practical judgement. When *I* was a student, the students who were paying the least psychic cost were the perpetual student types, and I would be very hesitant of having hired them. More generally, one can be an excellent at many jobs that require college

degrees while having found college genuinely painful, and one can be very happy indeed in a library and a poor fit for most workplaces.

This is a more serious objection than the first, because it attacks an assumption rather than a prediction. If there isn't a correlation between the cost of degree and the value to the employer, the model breaks. Plenty of low types, from the perspective of employers, will sail through college because they love it here.

College isn't expensive enough. The wage premium is *huge*. In the model, everyone is acting rationally. Some people ratioanlly turn down millions of dollars because the unpleasantness of those calculus classes is just too much. This beggars belief. In the modern university, with almost all courses being electives, not all courses will be that painful.

There is a technical point here. There is a variant on Spence's model where the difference between high and low is not how much college *costs*, but whether it is an option at all. In the biological literature, signalling models often start with a costly signal that other animals *cannot* send. That's still a signalling model, and a model where low types are simply incapable of finishing a degree, rather than able but unwilling, does not require thinking that there are all these people who will literally not take a calculus class even if they'll be paid a million dollars for it.

Changes in degree length provide natural experiments. If the hurdle is what matters, then changing its height should not change the premium *much*, because employers will recalibrate and a three-year degree will come to mean what a four-year degree used to mean. If skills are what matter, cutting the coursework should cut the premium.

Some of the evidence favours the signalling reading. In some countries bachelor's degrees take three years, and in other countries, like the US, they take four. On the whole, the premium doesn't obviously reflect this length difference. Though the US, which has four year degrees, does have a larger than average premium. In the EU, the Bologna reforms shortened degrees across much of the continent without eliminating the graduate premium, though the master's degree has become more expected since, which looks like the hurdle moving rather than disappearing.

But the cleanest social data we have points towards the human capital theory. Carolina Arteaga (2018) studied a reform at Universidad de los Andes in Colombia, which in 2006 cut the required coursework in economics by about 20% and in business by about 14%. The entering classes, their test scores and the graduation rates stayed put, and so did the university, the degree title and the diploma. The signal, in other words, was held fixed by construction. Earnings were not: graduates after the reform earned about 16% less in economics and about 13% less in business, and they did visibly worse in recruitment and ended up at weaker firms. Something in the missing courses was doing work that the diploma was not doing.

The reasonable reading is that both mechanisms operate. A degree is partly a hurdle and partly an education, and the proportions vary by institution and by labor market. What the model establishes is not that college teaches nothing. It is that a world in which college taught nothing could look, at least in broad outline, very much like ours.

6.13 Akerlof and the Market for Lemons

Spence (1973) shows what the informed side can do about asymmetric information. George Akerlof (1970) asks what happens when they cannot do anything about it, and the answer is that the market can break in unexpected ways.

For most of the twentieth century, the used car market had a striking feature. Of course the older a car is, and the more miles it has on it, the less value it has. And for most of the car's life, this is a *reasonably* linear decline. There was, however, a big discontinuity. When you bought a car, and drove it off the lot, it *immediately* lost a big chunk of its value, maybe as much as 20%. Everyone knew this, but no one had a good theory for why it would be so.[13]

[13] There were lots of bad theories, which I won't go into here. Suffice to say, they all made bad predictions.

Akerlof suggested this worked because of asymmetric information. Let's work with a really simple model. Most cars are good, but some are, in the terminology of the mid-twentieth century, *lemons*. They are just bad cars. You can drive them around the block on a test drive, but after a few days, weeks, or months at home, you'll realise you bought a dud. This is private information; a buyer, even one who kicks the tires and drives it around the block a few times, can't tell whether they are buying a lemon.

Let's work with a *really* simplified model. Assume that sellers are motivated to sell; that's to say, restrict attention to those people who have cars and would other things being equal like to sell them at market price. Assume that half the cars are lemons. That's absurd, but it keeps the numbers simple. The values of good and bad cars are in Table 6.3.

Table 6.3: An extreme version of the market for lemons

	To seller	**To buyer**
Good car	100	120
Lemon	40	60

Every car is worth more to a buyer than to its seller, so if everyone could see quality, every car would change hands, the market would clear, and everyone would be better off.

Now suppose buyers cannot tell the two apart. A buyer who thinks the odds are even values a random car at $\frac{120+60}{2} = 90$, and so is willing to pay up to 90.

But a good car is worth 100 to its owner. At 90, no owner of a good car sells. So the only cars on the market are lemons, and the buyer who works this out revises: a car actually on offer is worth 60. Buyers now offer at most 60. Lemon owners, who value their cars at 40, are happy to sell.

The market clears, in the sense that the cars put up for sale sell, but only lemons trade, at 60. Good cars never change hands, even though every good-car sale would have produced a surplus of 20.

In more realistic models, where cars have continuous quality, there aren't *that many* lemons, and some holders of good cars are *really* motivated to sell, you'll get more trade. But still, the fact that a car is up for sale will be evidence that it's bad, and the used-car discount is what buyers rationally demand in order to overcome this evidence.

The underlying structure here ends up being simmilar to the Spence model. Indeed, Spence was motivated to come up with his model after his advisors recommended reading Akerlof's paper.[14] The decision to put a car up for sale is a signal; a signal that you're an unfortunate owner of a lemon.

As well as explaining the used-car discount that was prevalent for the twentieth century, Akerlof's model also predicts the conditions under which the problem should ease. Those conditions have by now largely been met, and the discount is really small now. The model requires two things to hold. One is that there is enough variation in manufacturing quality that the possibility of lemons is a serious problem. That's just not as big a deal now; most car companies have excellent quality control and the cars rolling off the line are pretty similar to one another. The other condition is that there is an information asymmetry between buyers and sellers. That isn't nearly as dramatic now. Vehicle history services and cheap independent inspections, reduce the information gap, and for what's left manufacturer certification removes the downside of buying a lemon. Both of the model's assumptions were weakened, and the effect it was built to explain faded. That's a really neat result; it's one thing to explain a striking phenomenon, it's another altogether to predict when the phenomena will fade.

One last very striking point for the philosophy students. Akerlof's model was really unpopular with economists when it was first published. They thought he was abandoning rational choice explanation. This is odd because in his model, everyone is completely rational by the standards

[14]Spence tells this story in his Nobel lecture (Spence 2002). Those advisors included Thomas Schelling, who we've already met, and Kenneth Arrow, who we will discuss *at length* in the next part of the course.

that philosophers would use. They all have sensible priors, which they update by conditionalisation, and then they utility maximise. What's true is that in his model, not everyone is *fully informed*. Nowadays, I think economists have largely come around to philosophers' way of talking; ignorance is unfortunate, but it's not irrational. But there was something really sociologically puzzling about how economists reacted to it, and a different course would explore this more.

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